19. Troubleshooting Guide

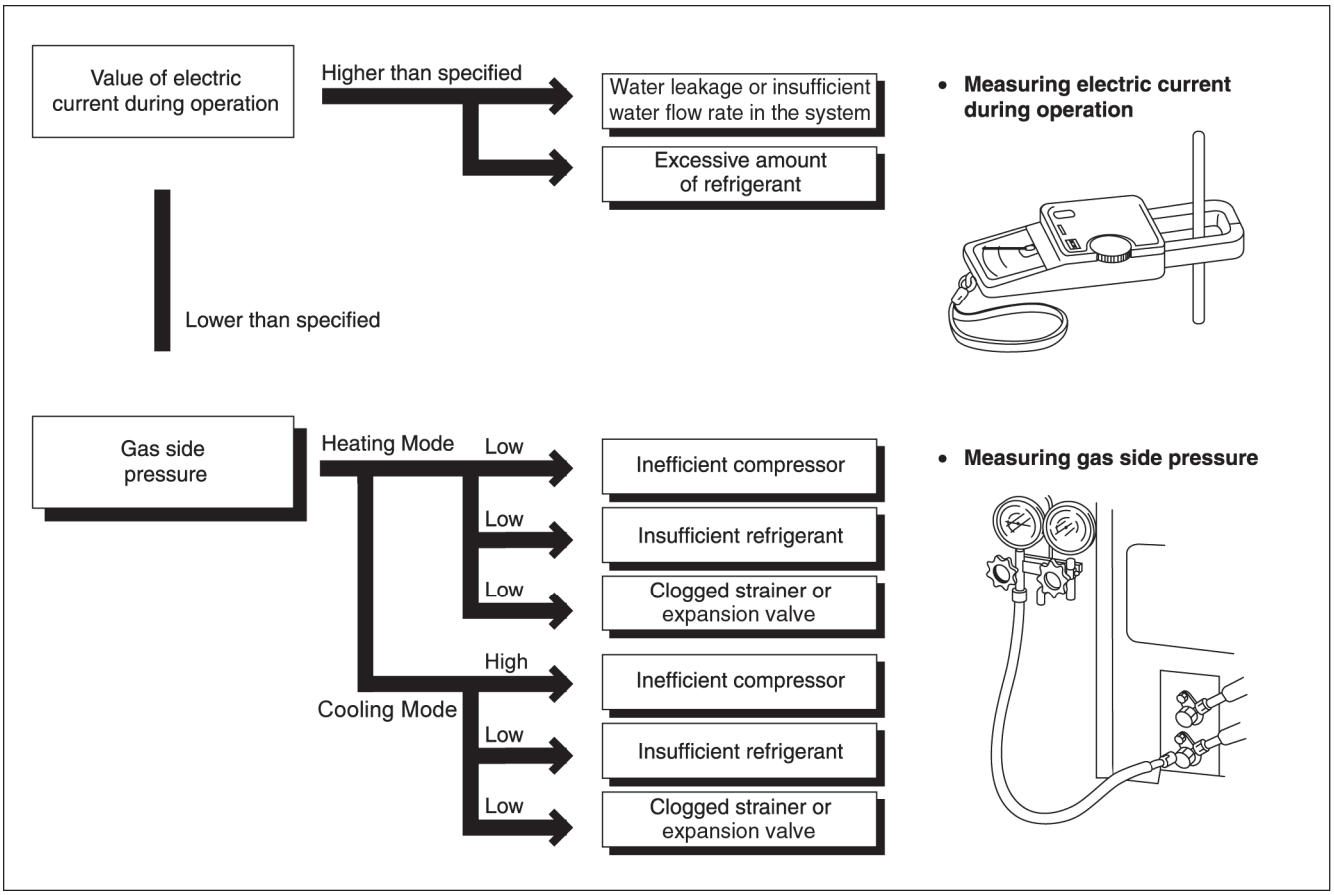
19.1 Refrigeration Cycle System

In order to diagnose malfunctions, make sure that there are no electrical problems before inspecting the refrigeration cycle.
Such problems include insufficient insulation, problem with the power source, malfunction of a compressor and a fan.
The normal pressure of the refrigeration cycle depends on various conditions, the standard values for them are shown in the table on the right.

Normal Pressure (Standard)

	Gas pressure MPa (kg/cm²G)
Heating Mode	2.3 ~ 2.9 (23 ~ 29)
Cooling Mode	0.9 ~ 1.2 (9 ~ 12)

★Condition:
- Outdoor temperature 7°C at heating mode and 35°C at cooling mode.
- Compressor operates at rated frequency.



19.2 Relationship between the Condition of the Air-to-Water Heatpump Indoor and Outdoor Units and Pressure and Electric Current

Condition of the Air-to-Water Heatpump indoor and outdoor units	Heating Mode			Cooling Mode		
	Low Pressure	High Pressure	Electric current during operation	Low Pressure	High Pressure	Electric current during operation
Water leakage or insufficient water flow rate in the system						
Excessive amount of refrigerant						
Inefficient compression						
Insufficient refrigerant (gas leakage)						
Outdoor heat exchange deficiency						
Clogged expansion valve or Strainer						

- Carry out the measurements of pressure, electric current, and temperature fifteen minutes after an operation is started.

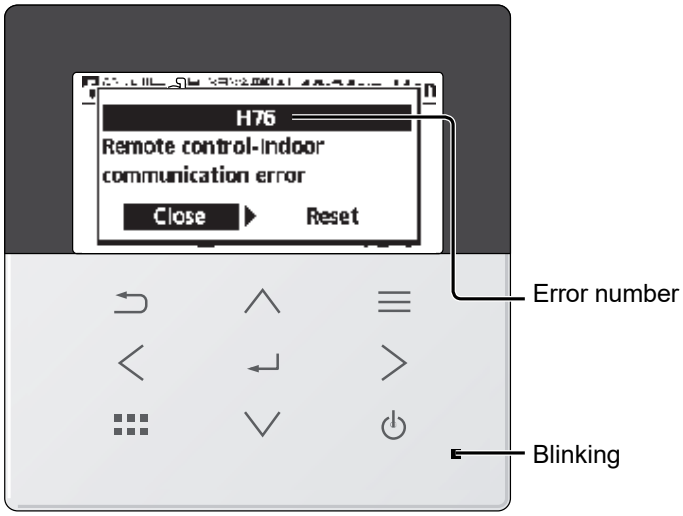
19.3 Breakdown Self Diagnosis Function

19.3.1 Self Diagnosis Function (Three Digits Alphanumeric Code)

- When abnormality occur during operation, the system will stop operation, and OFF/ON control panel LED will blink and error code will display on the control panel.
- Even error code is reset by turning OFF power supply or by selecting ERROR RESET, if the system abnormality is still unrepaired, system will again stop operation, and OFF/ON control panel LED will again blinks and error code will be display.
- The error code will store in IC memory.

- **To check the error code**
  1 When an abnormality occurs, system will stop operation and OFF/ON control panel LED will blink.
  2 Error code of the abnormality will be display on the control panel.
  3 To determine the abnormality description, the error code table needs to be referred.

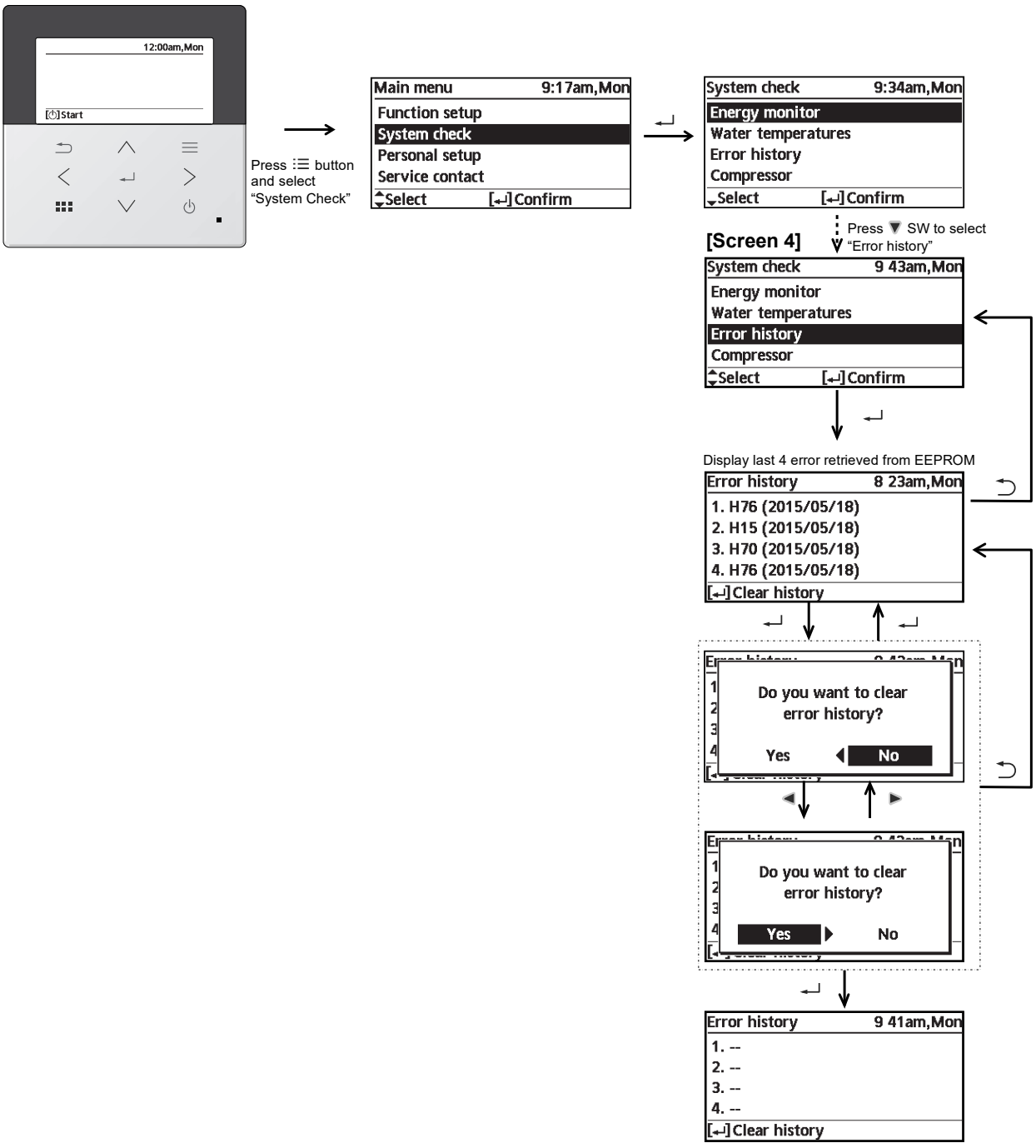
eg:



Press < > to select Close / Reset then press ↵

- **To display past/last error code**
  1 Turn ON power supply.
  2 Refer below procedure to retrieve the error code history.

- **To permanently delete error code from IC memory**
  1 Turn ON power supply.
  2 Refer below procedure to clear error history.


12:00am,Mon
Start
Press ☰ button and select "System Check"
Main menu 9:17am,Mon
Function setup
System check
Personal setup
Service contact
Select [↵]Confirm
System check 9:34am,Mon
Energy monitor
Water temperatures
Error history
Compressor
Select [↵]Confirm
[Screen 4]
Press ▼ SW to select "Error history"
System check 9 43am,Mon
Energy monitor
Water temperatures
Error history
Compressor
Select [↵]Confirm
Display last 4 error retrieved from EEPROM
Error history 8 23am,Mon
1. H76 (2015/05/18)
2. H15 (2015/05/18)
3. H70 (2015/05/18)
4. H76 (2015/05/18)
[↵]Clear history
Do you want to clear error history?
Yes No
Do you want to clear error history?
Yes No
Error history 9 41am,Mon
1. --
2. --
3. --
4. --
[↵]Clear history

19.4 Error Codes Table

Diagnosis display	Abnormality/Protection control	Abnormality judgement	Primary location to verify
H00	No abnormality detected	—	—
H12	Indoor/Outdoor capacity unmatched	90s after power supply	• Indoor/outdoor connection wire • Indoor/outdoor PCB • Specification and combination table in catalogue
H15	Outdoor compressor temperature sensor abnormality	Continue for 5 sec.	• Compressor temperature sensor (defective or disconnected)
H20	Water pump abnormality	Continue for 10 sec.	• Indoor PCB • Water pump (malfunction)
H23	Indoor refrigerant liquid temperature sensor abnormality	Continue for 5 sec.	• Refrigerant liquid temperature sensor (defective or disconnected)
H27	Service valve error	Continue for 5 minutes	• High pressure sensor (defective or disconnected)
H28	Abnormal solar sensor	Continue for 5 sec.	• Solar temperature sensor (defective or disconnected)
H31	Abnormal swimming pool sensor	Continue for 5 sec.	• Pool temperature sensor (defective or disconnected)
H36	Abnormal buffer tank sensor	Continue for 5 sec.	• Buffer tank sensor (defective or disconnected)
H38	Brand code not match	When indoor and outdoor brand code not same	—
H42	Compressor low pressure abnormality	—	• Outdoor pipe temperature sensor • Clogged expansion valve or strainer • Insufficient refrigerant • Outdoor PCB • Compressor
H43	Abnormal Zone 1 sensor	Continue for 5 sec.	• Water temperature Zone 1 sensor
H44	Abnormal Zone 2 sensor	Continue for 5 sec.	• Water temperature Zone 2 sensor
H62	Water flow switch abnormality	Continue for 1 min.	• Water flow switch
*H63	Refrigerant low pressure abnormality	Continue for 5 sec.	• Outdoor low pressure sensor (defective or disconnected)
H64	Refrigerant high pressure abnormality	Continue for 5 sec.	• Outdoor high pressure sensor (defective or disconnected)
*H65	Deice circulation error	Continue for 10 sec.	• Water flow switch sensor (defective or disconnected) • Water pump malfunction • Buffer tank (is used)
H67	Abnormal External Thermistor 1	Continue for 5 sec.	• Room temperature Zone 1 sensor
H68	Abnormal External Thermistor 2	Continue for 5 sec.	• Room temperature Zone 2 sensor
H70	Back-up heater OLP abnormality	Continue for 60 sec.	• Back-up heater OLP (Disconnection or activated)
H72	Tank sensor abnormal	Continue for 5 sec.	• Tank sensor
H74	PCB communication error	Communication or transfer error	• Indoor main PCB and Sub PCB
H75	Low water temperature control	Room heater disable and deice request to operate under low water temperature	• Heater operation must enable to increase water temperature
H76	Indoor - control panel communication abnormality	—	• Indoor - control panel (defective or disconnected)
H90	Indoor/outdoor abnormal communication	> 1 min after starting operation	• Internal/external cable connections • Indoor/Outdoor PCB
H91	Tank heater OLP abnormality	Continue for 60 sec.	• Tank heater OLP (Disconnection or activated)
H95	Indoor/Outdoor wrong connection	—	• Indoor/Outdoor supply voltage
H98 / F95	Outdoor high pressure overload protection Cooling high pressure overload protection	—	• Outdoor high pressure sensor • Water pump or water leakage • Clogged expansion valve or strainer • Excess refrigerant • Outdoor PCB
H99	Indoor heat exchanger freeze prevention	—	• Indoor heat exchanger • Refrigerant shortage
F12	Pressure switch activate	4 times occurrence within 20 minutes	• Pressure switch
F14	Outdoor compressor abnormal revolution	4 times occurrence within 20 minutes	• Outdoor compressor
F15	Outdoor fan motor lock abnormality	2 times occurrence within 30 minutes	• Outdoor PCB • Outdoor fan motor
F16	Total running current protection	3 times occurrence within 20 minutes	• Excess refrigerant • Outdoor PCB

Diagnosis display	Abnormality/Protection control	Abnormality judgement	Primary location to verify
F20	Outdoor compressor overheating protection	4 times occurrence within 30 minutes	• Compressor tank temperature sensor • Clogged expansion valve or strainer • Insufficient refrigerant • Outdoor PCB • Compressor
F22	IPM (power transistor) overheating protection	3 times occurrence within 30 minutes	• Improper heat exchange • IPM (Power transistor)
F23	Outdoor Direct Current (DC) peak detection	7 times occurrence continuously	• Outdoor PCB • Compressor
F24	Refrigeration cycle abnormality	2 times occurrence within 20 minutes	• Insufficient refrigerant • Outdoor PCB • Compressor low compression
F25	Cooling/Heating cycle changeover abnormality	4 times occurrence within 30 minutes	• 4-way valve • V-coil
F27	Pressure switch abnormality	Continue for 1 min.	• Pressure switch
F29	Low Discharge Superheat	1 times occurrence within 2550 minutes	• Discharge Temperature Sensor • Discharge Pressure Sensor • Pressure Switch • Outdoor PCB
F30	Water outlet sensor 2 abnormality	Continue for 5 sec.	• Water outlet sensor 2 (defective or disconnected)
F32	Abnormal Internal Thermostat	Continue for 5 sec.	• Control panel PCB thermostat
F36	Outdoor air temperature sensor abnormality	Continue for 5 sec.	• Outdoor air temperature sensor (defective or disconnected)
F37	Indoor water inlet temperature sensor abnormality	Continue for 5 sec.	• Water inlet temperature sensor (defective or disconnected)
F40	Outdoor discharge pipe temperature sensor abnormality	Continue for 5 sec.	• Outdoor discharge pipe temperature sensor (defective or disconnected)
F41	PFC control	4 times occurrence within 10 minutes	• Voltage at PFC
F42	Outdoor heat exchanger temperature sensor abnormality	Continue for 5 sec.	• Outdoor heat exchanger temperature sensor (defective or disconnected)
F43	Outdoor defrost sensor abnormality	Continue for 5 sec.	• Outdoor defrost sensor (defective or disconnected)
F45	Indoor water outlet temperature sensor abnormality	Continue for 5 sec.	• Water outlet temperature sensor (defective or disconnected)
F46	Outdoor Current Transformer open circuit	—	• Insufficient refrigerant • Outdoor PCB • Compressor low
F48	Outdoor EVA outlet temperature sensor abnormality	Continue for 5 sec.	• Outdoor EVA outlet temperature sensor (defective or disconnected)
F49	Outdoor bypass outlet temperature sensor abnormality	Continue for 5 sec.	• Outdoor bypass outlet temperature sensor (defective or disconnected)

Note: * This error code is not applicable for this system.

19.5 Self-diagnosis Method

19.5.1 Connection Capability Rank Abnormality (H12)

Malfunction Decision Conditions:

During startup operation of cooling and heating, the capability rank of indoor checked by the outdoor is used to determine connection capability rank abnormality.

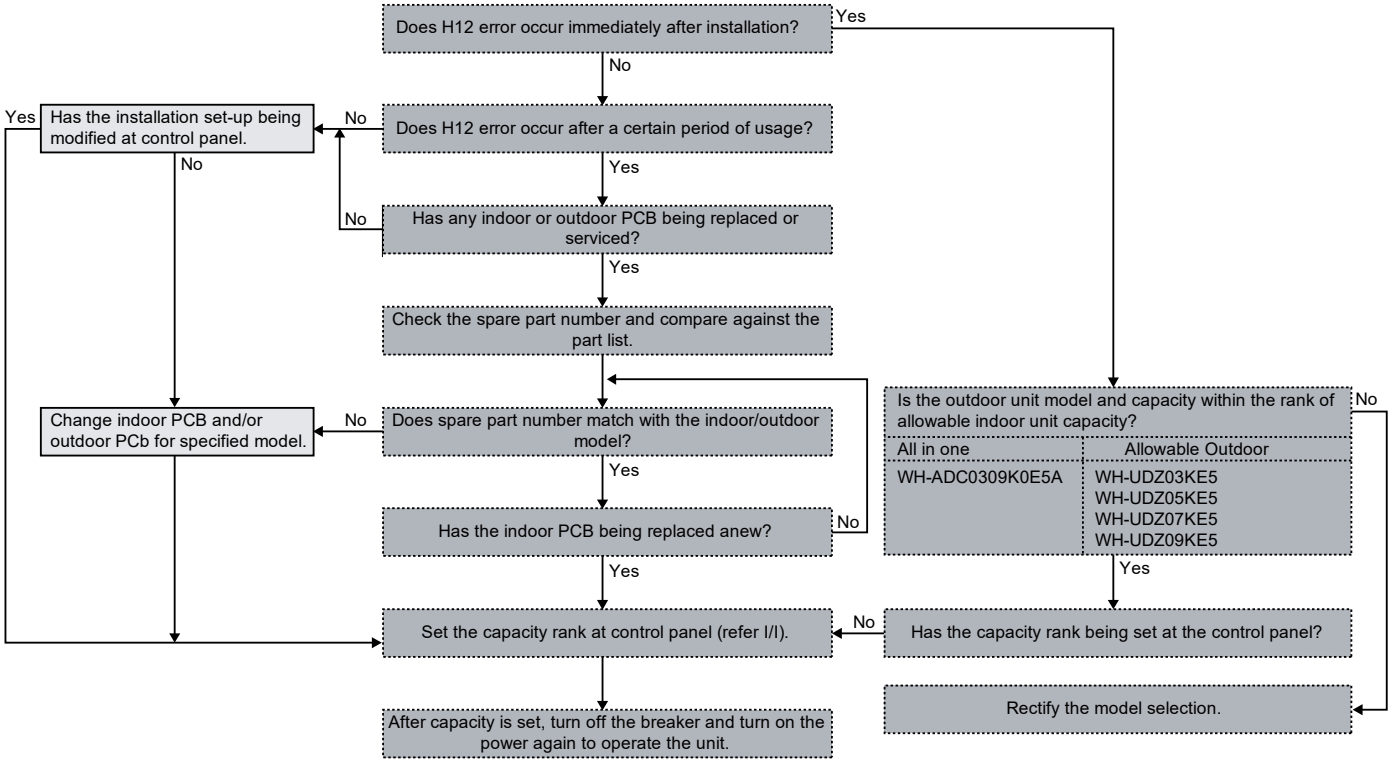
Malfunction Caused:

1 Wrong model interconnected.
2 Wrong indoor unit or outdoor unit PCB (main) used.
3 Faulty indoor unit or outdoor unit PCB (main).

Abnormality Judgment:

Continue for 90 seconds.

Troubleshooting:



19.5.2 Compressor Tank Temperature Sensor Abnormality (H15)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the compressor tank temperature sensor are used to determine sensor error.

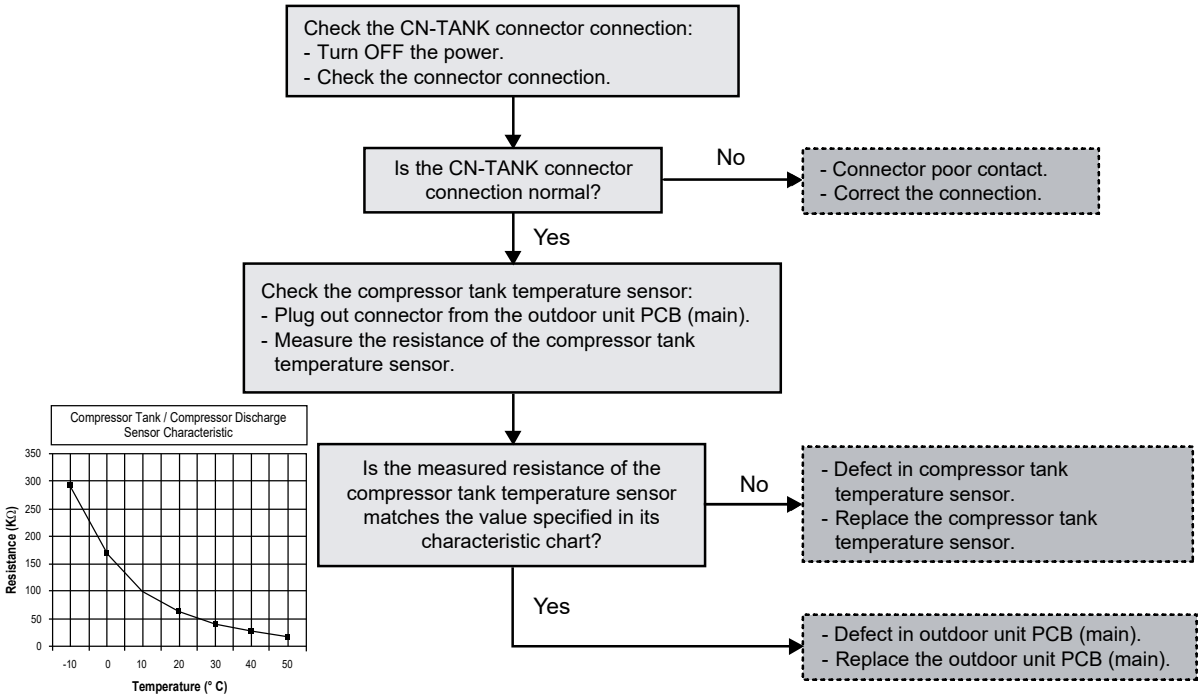
Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:



19.5.3 Water Pump Abnormality (H20)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the rotation speed detected by the IPM of water pump motor during water pump operation is used to determine abnormal water pump (feedback of rotation > 6,000rpm or < 1,000rpm).

Malfunction Caused:

1 Operation stop due to short circuit inside the water pump motor winding.
2 Operation stop due to breaking of wire inside the water pump motor.
3 Operation stop due to breaking of water pump lead wires.
4 Operation stop due to water pump motor IPM malfunction.
5 Operation error due to faulty indoor unit PCB.

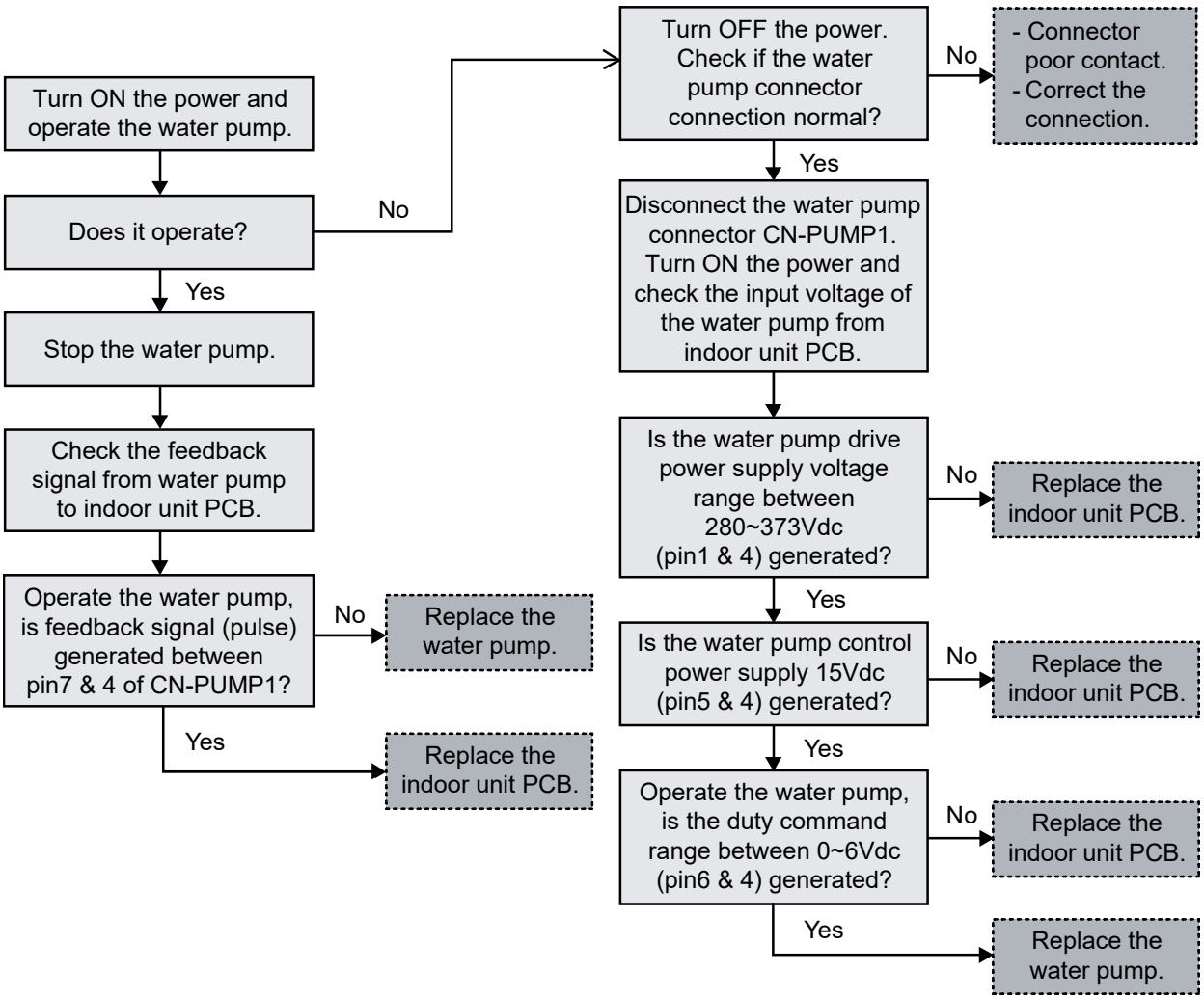
Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.4 Indoor Refrigerant Liquid Temperature Sensor Abnormality (H23)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the indoor refrigerant liquid temperature sensor are used to determine sensor error.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty indoor unit PCB (main).

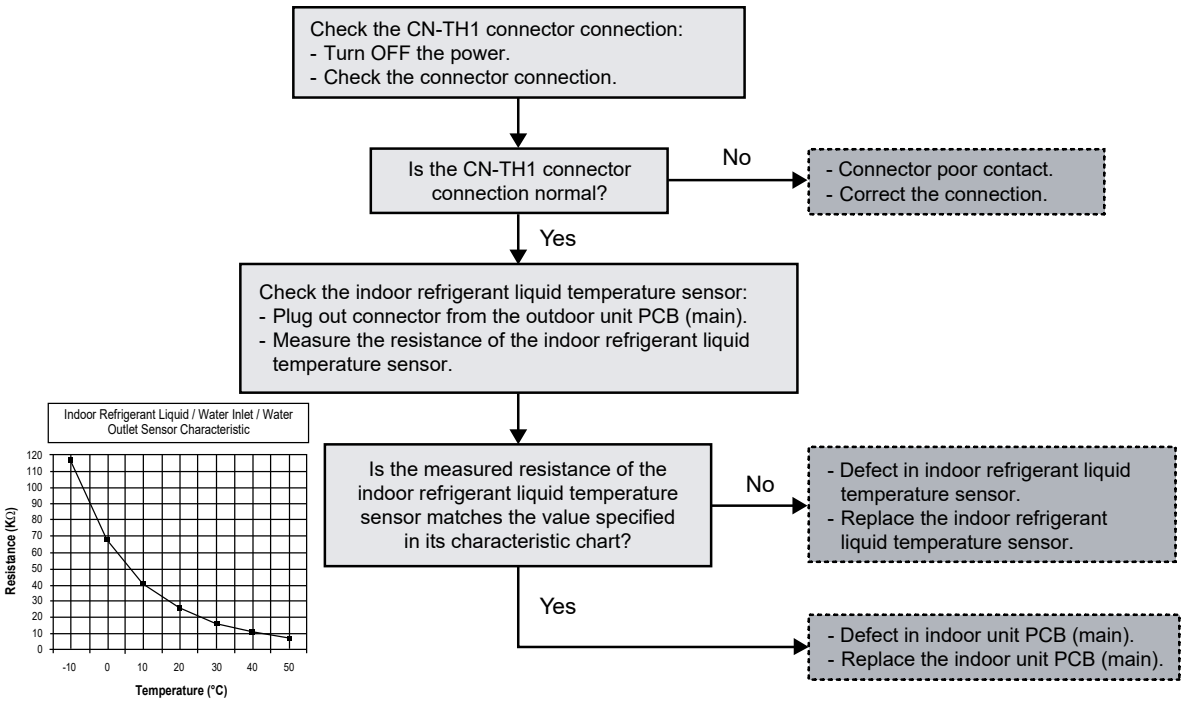
Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



19.5.5 Service Valve Error (H27)

Malfunction Decision Conditions:

During cooling operation, when:-

[a] Indoor refrigerant pipe temperature at compressor startup - present indoor refrigerant pipe temperature < 2°C

[b] Present high pressure - high pressure at compressor startup < 5kg/cm^2

**Judgment only for first time cooling operation and not during pump down operation.

Malfunction Caused:

1 3 way valves closed.
2 Faulty high pressure sensor.
3 Faulty indoor refrigerant pipe temperature sensor
4 Faulty outdoor unit PCB (main).

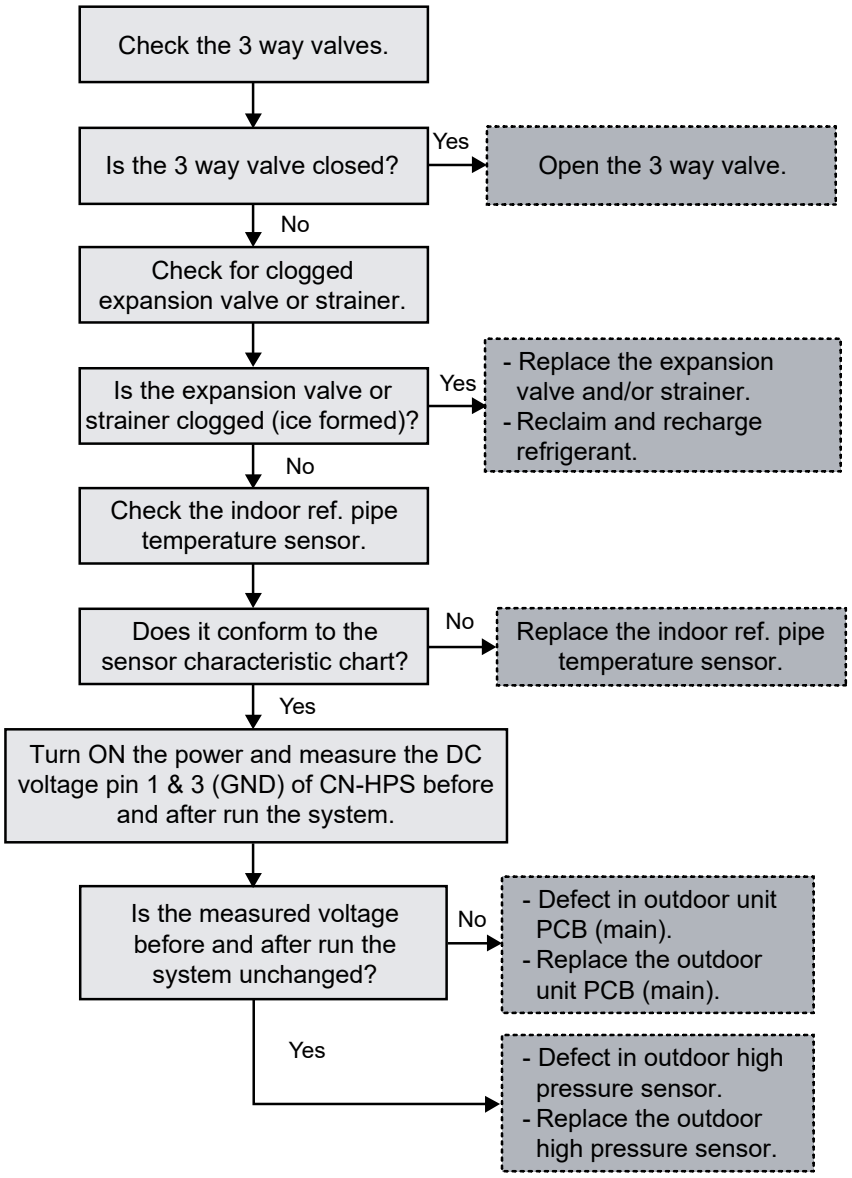
Abnormality Judgment:

Continue for 5 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



19.5.6 Abnormal Solar Sensor (H28)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty solar sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



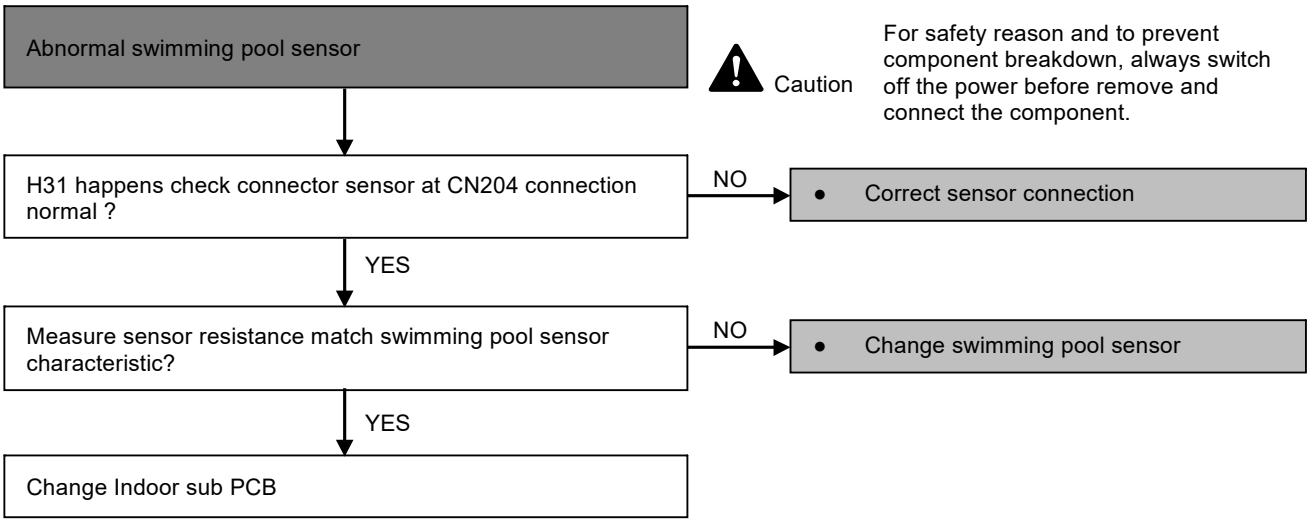
19.5.7 Abnormal Swimming Pool Sensor (H31)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty swimming pool sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



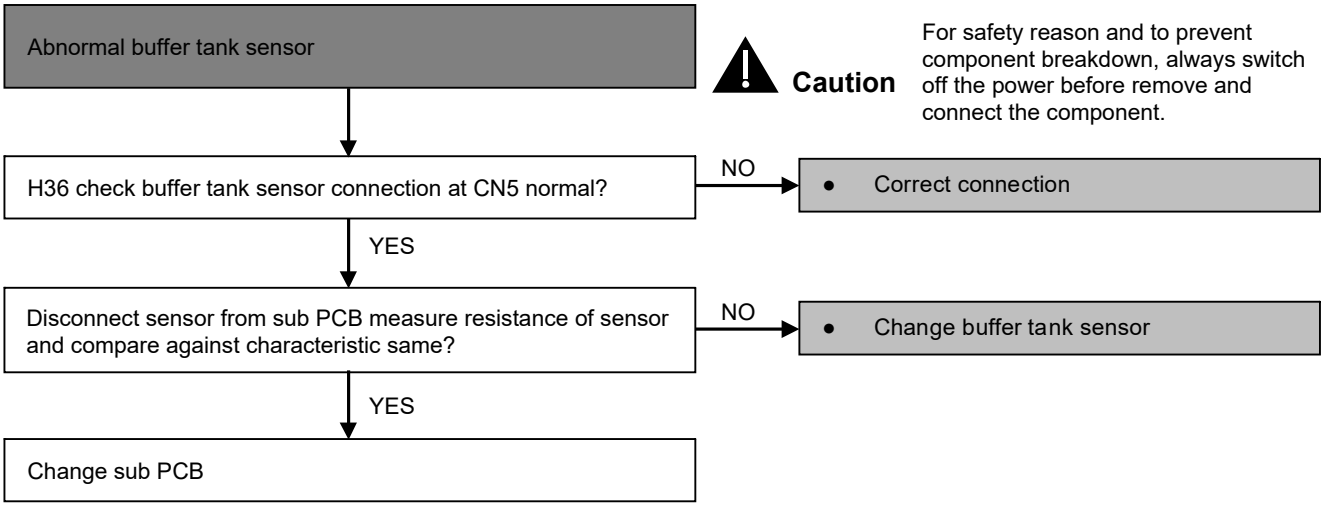
19.5.8 Abnormal Buffer Tank Sensor (H36)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty buffer tank sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

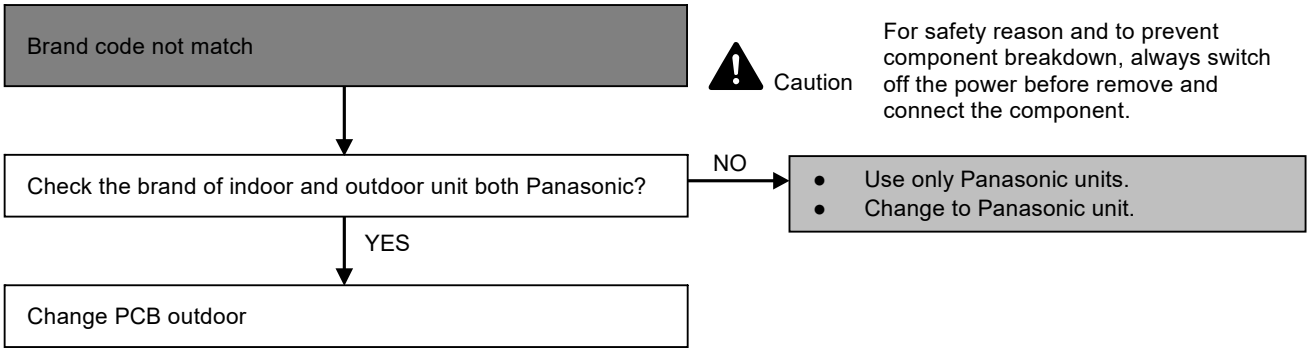
Continue for 5 seconds.



19.5.9 Brand Code Not Matching (H38)

Malfunction Caused:

1 Indoor and outdoor brand code not match.



19.5.10 Compressor Low Pressure Protection (H42)

Malfunction Decision Conditions:

During operation of heating and after 5 minutes compressor ON, when outdoor pipe temperature below -29°C or above 26°C is detected by the outdoor pipe temperature sensor.

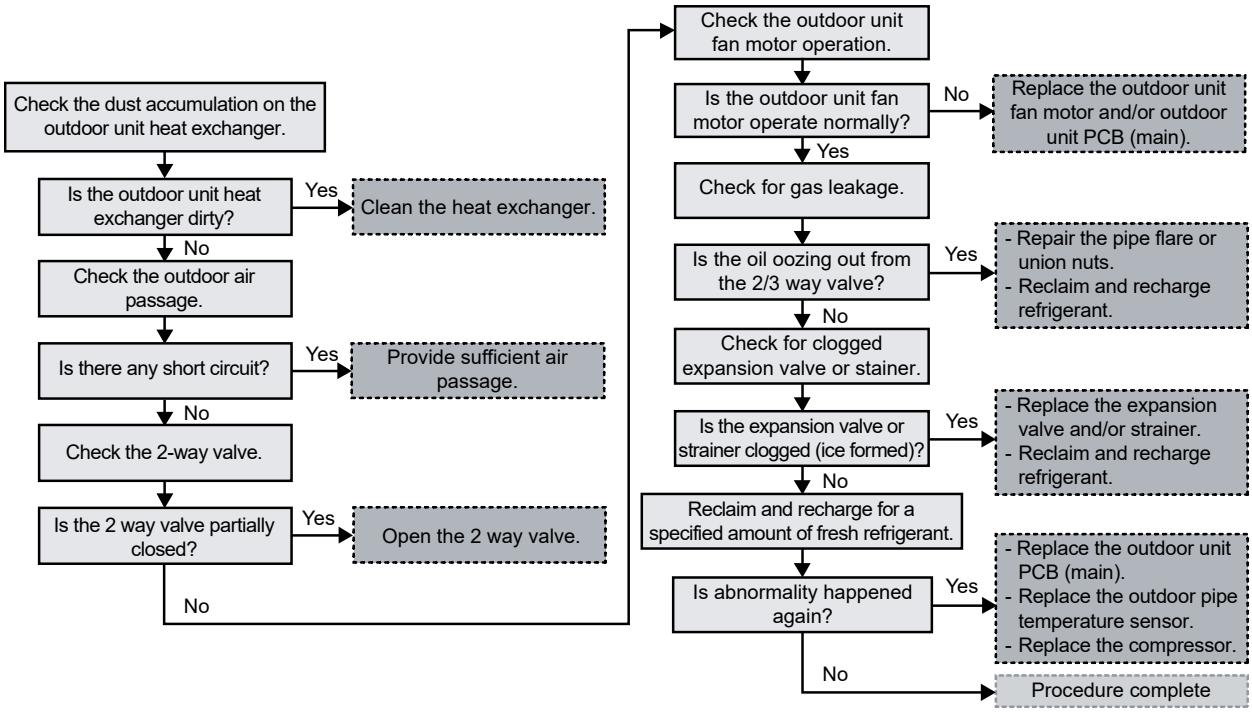
Malfunction Caused:

1 Dust accumulation on the outdoor unit heat exchanger.
2 Air short circuit at outdoor unit.
3 2 way valve partially closed.
4 Faulty outdoor unit fan motor.
5 Refrigerant shortage (refrigerant leakage).
6 Clogged expansion valve or strainer.
7 Faulty outdoor pipe temperature sensor.
8 Faulty outdoor unit main PCB (main).

Troubleshooting:

Caution

For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



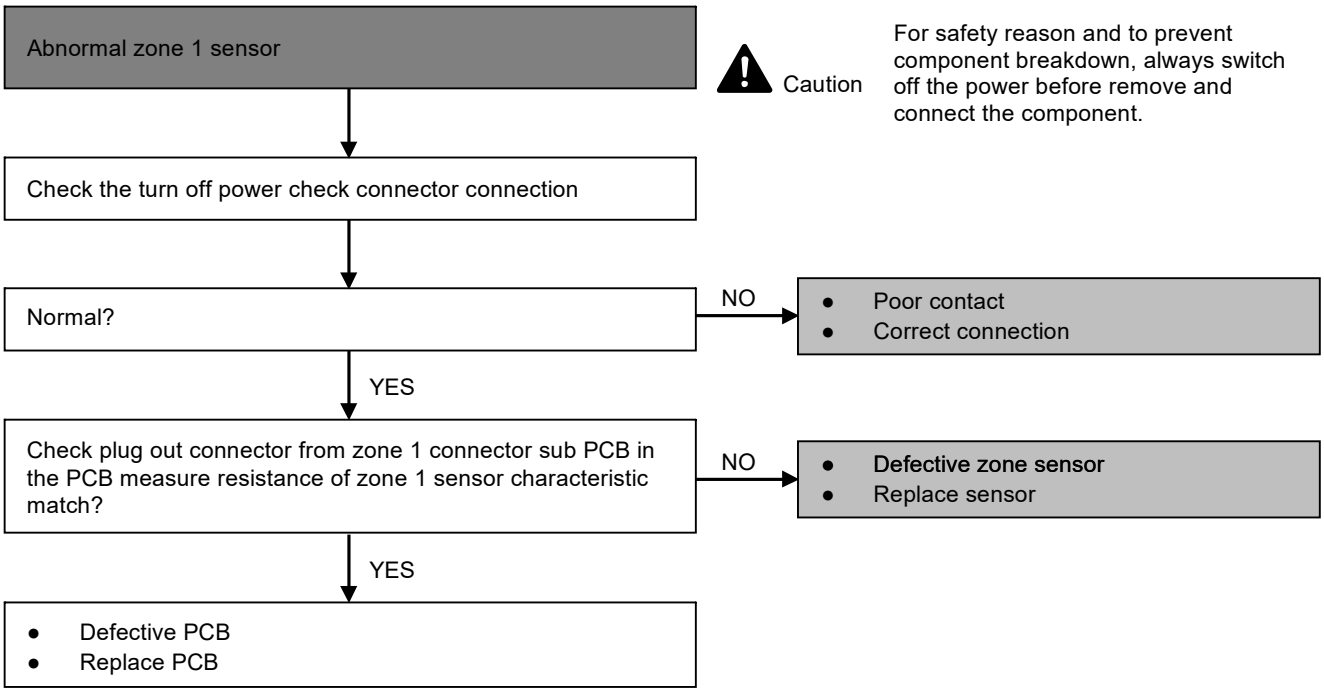
19.5.11 Abnormal Zone 1 Sensor (H43)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty buffer tank sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



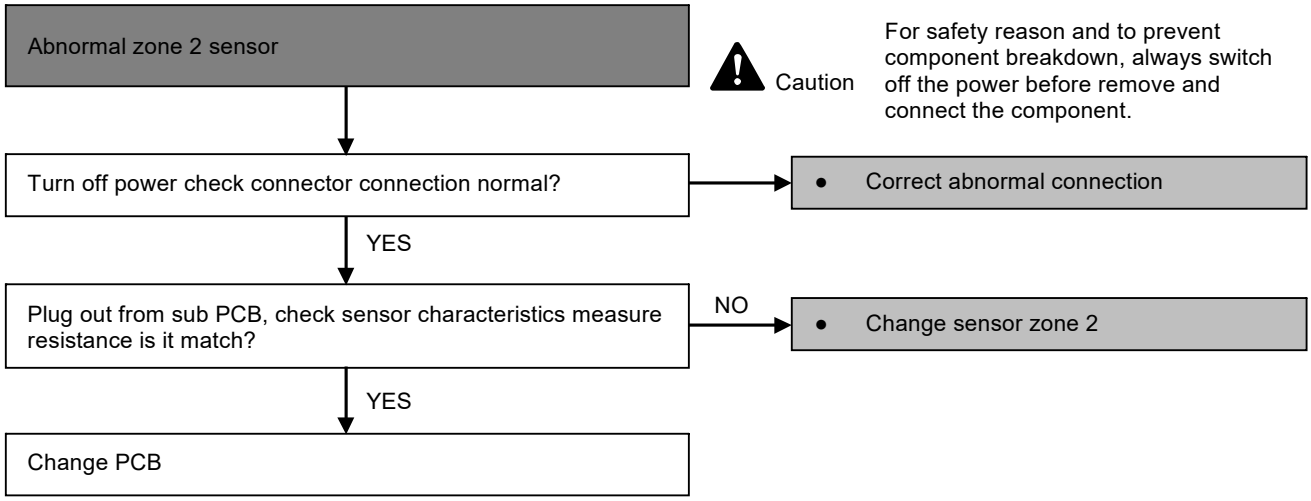
19.5.12 Abnormal Zone 2 Sensor (H44)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty buffer tank sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



19.5.13 Water Flow Switch Abnormality (H62)

Malfunction Decision Conditions:

During operation of cooling and heating, the water flow detected by the indoor water flow switch is used to determine water flow error.

Malfunction Caused:

1 Faulty water pump.
2 Water leak in system.
3 Faulty connector connection.
4 Faulty water flow switch.
5 Faulty indoor unit PCB (main).

Abnormality Judgment:

Continue for 10 seconds (but no judgment for 9 minutes after compressor startup/restart).

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.

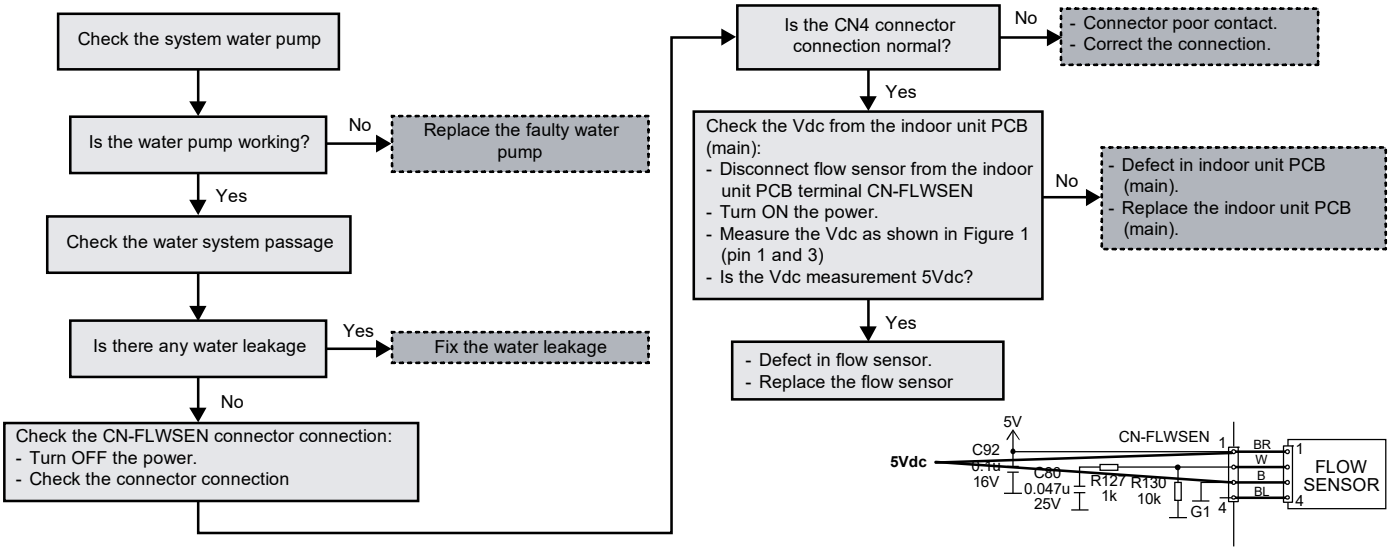


Figure 1

19.5.14 Outdoor High Pressure Abnormality (H64)

Malfunction Decision Conditions:

During operation of cooling and heating, when the outdoor high pressure sensor output signal is 0 Vdc or 5 Vdc.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

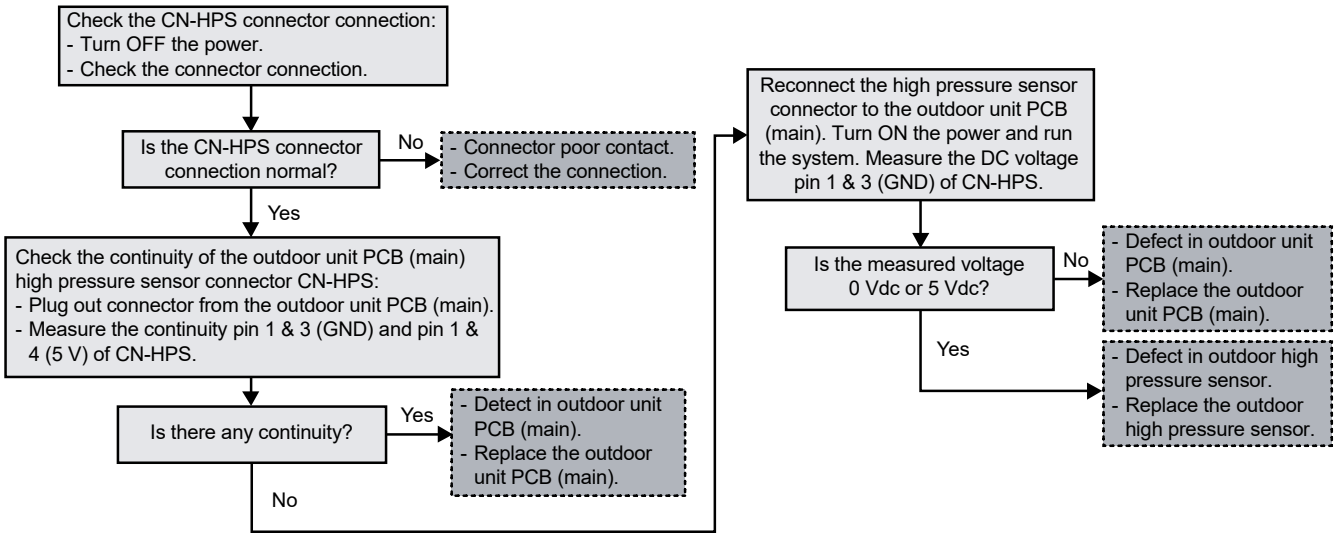
Abnormality Judgment:

Continue 4 times in 20 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



19.5.15 Deice Circulation Error (H65)

Malfunction Decision Conditions:

During startup and operation of deice (mode 2), the water flow (> 10l/min) detected by the water flow switch is used to determine deice circulation error.

Malfunction Caused:

1 Water flow in air-to-water heatpump indoor unit circuitry.
2 Faulty indoor unit water flow switch.
3 Faulty indoor unit water pump.
4 Faulty indoor unit PCB.

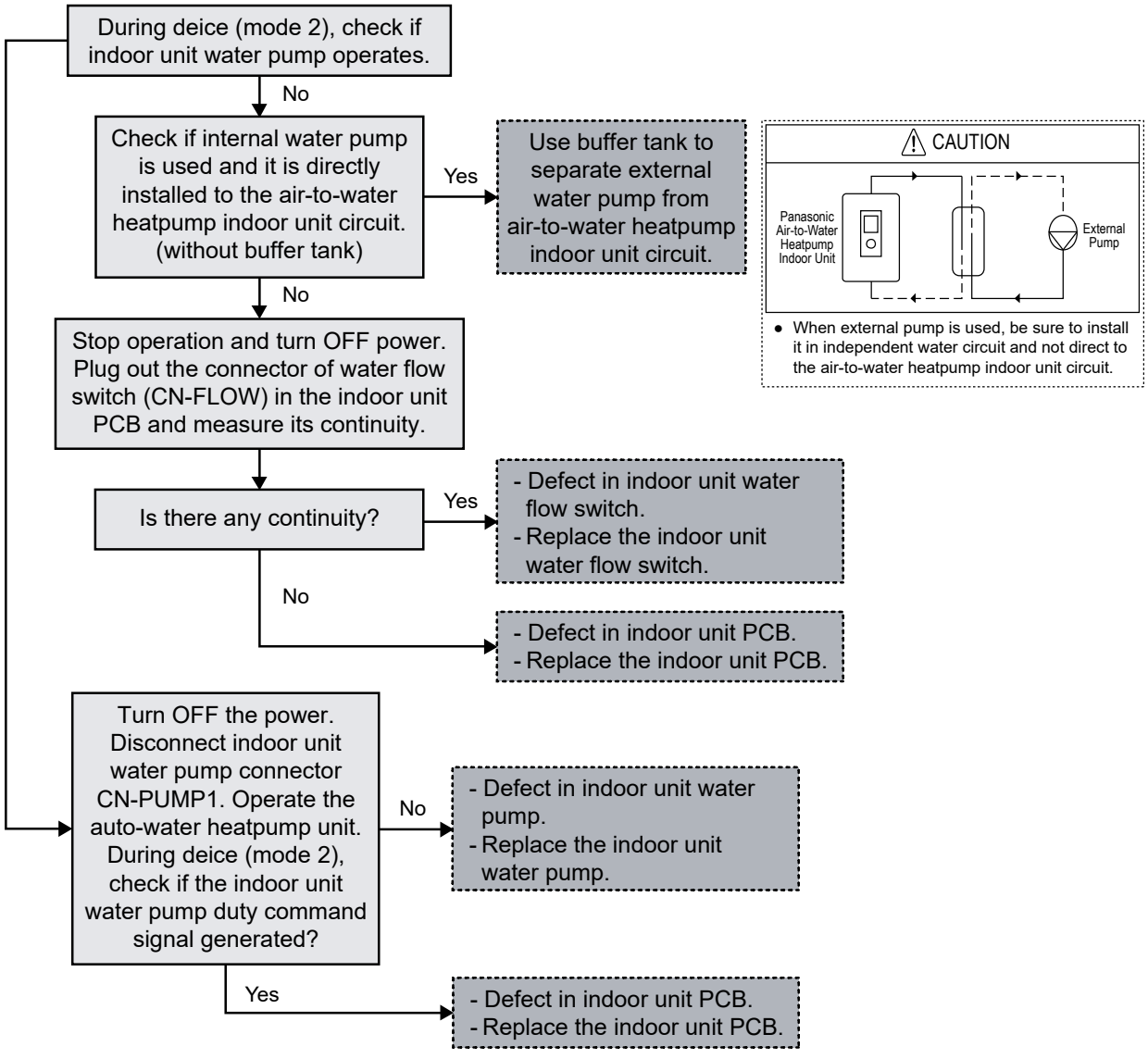
Abnormality Judgment:

Continue for 10 seconds.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



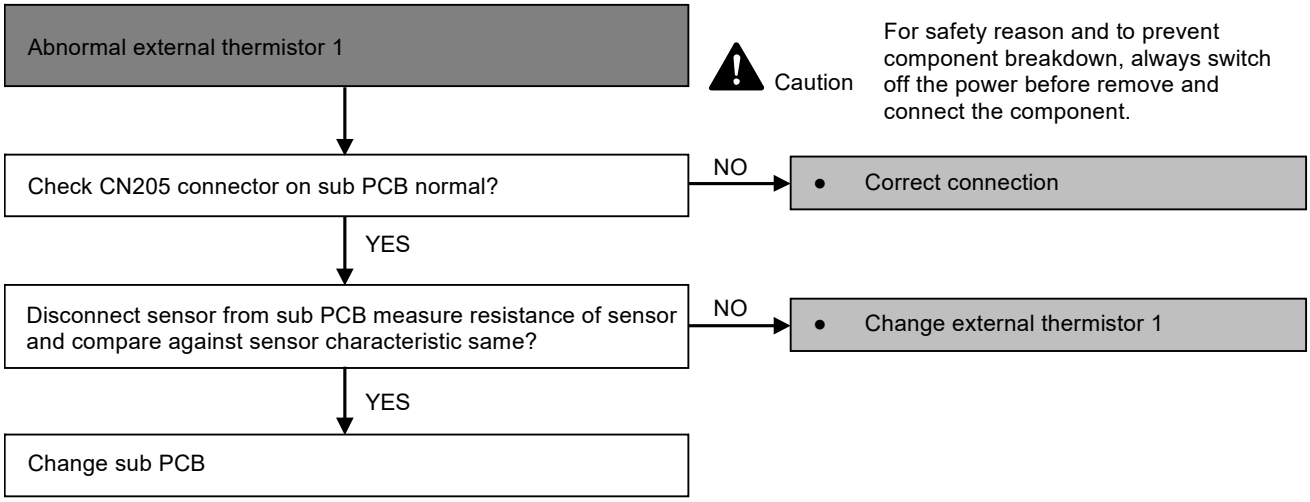
19.5.16 Abnormal External Thermistor 1 (H67)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty room temperature zone 1 sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



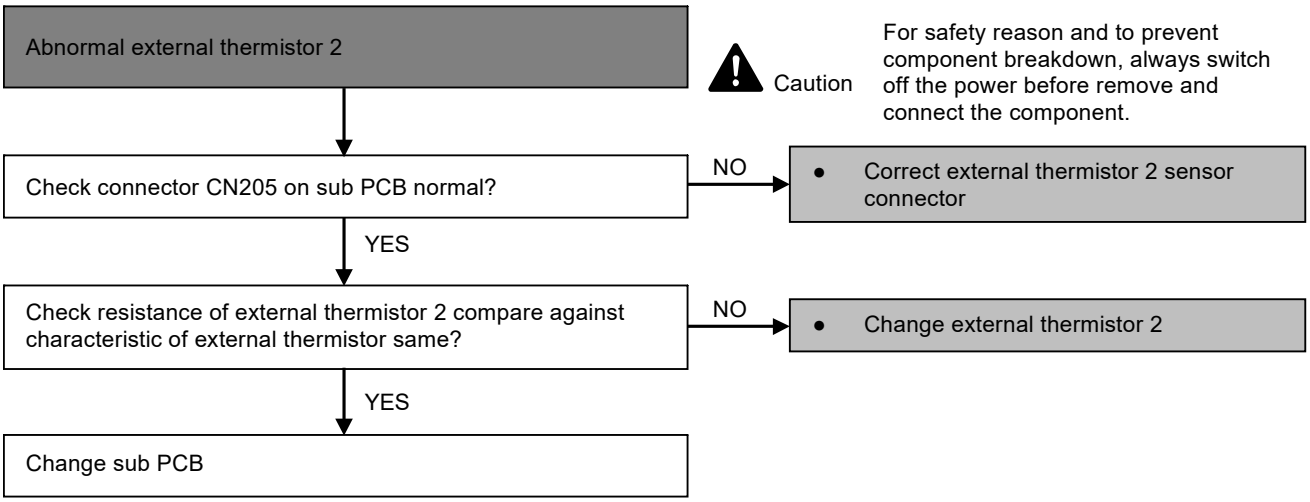
19.5.17 Abnormal External Thermistor 2 (H68)

Malfunction Caused:

1 Faulty connector connection.
2 Faulty room temperature zone 2 sensor.
3 Faulty indoor sub PCB.

Abnormality Judgment:

Continue for 5 seconds.



19.5.18 Indoor Backup Heater OLP Abnormality (H70)

Malfunction Decision Conditions:

During operation of indoor backup heater, when no power supplies to indoor backup heater or OLP open circuit.

Malfunction Caused:

1 Faulty power supply connector connection.
2 Faulty connector connection.
3 Faulty indoor backup heater overload protector (OLP).
4 Faulty indoor unit PCB (main).

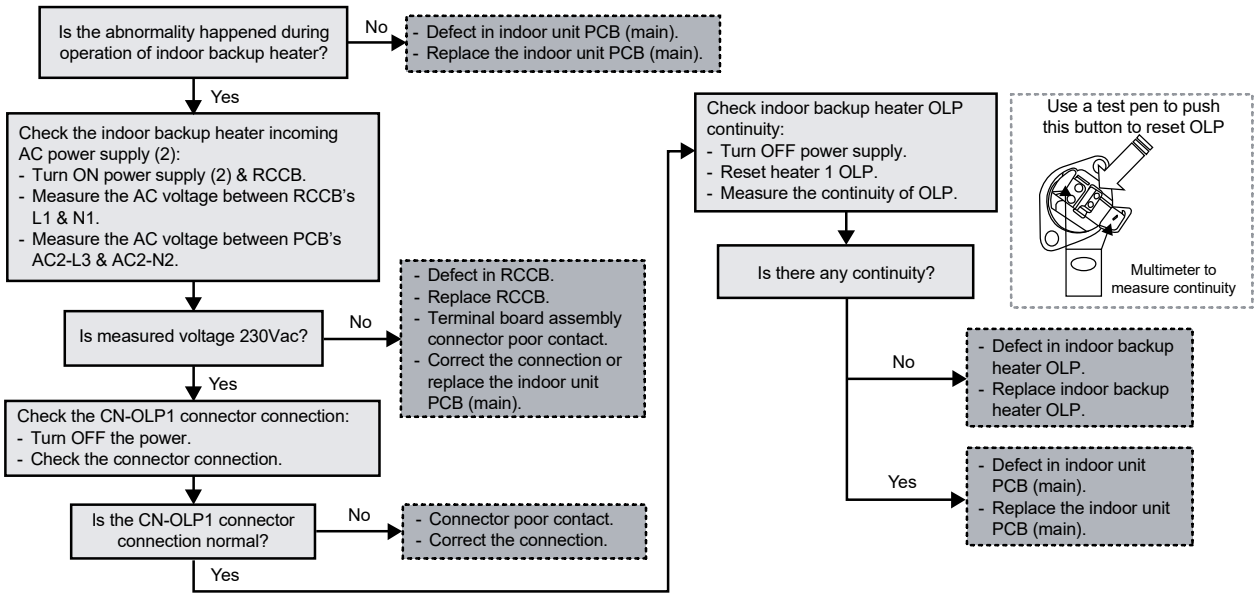
Abnormality Judgment:

Continue for 60 seconds.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



19.5.19 Tank Temperature Sensor Abnormality (H72)

Malfunction Decision Conditions:

When tank connection is set to ON, the temperatures detected by the tank temperature sensor are used to determine sensor error.

Malfunction Caused:

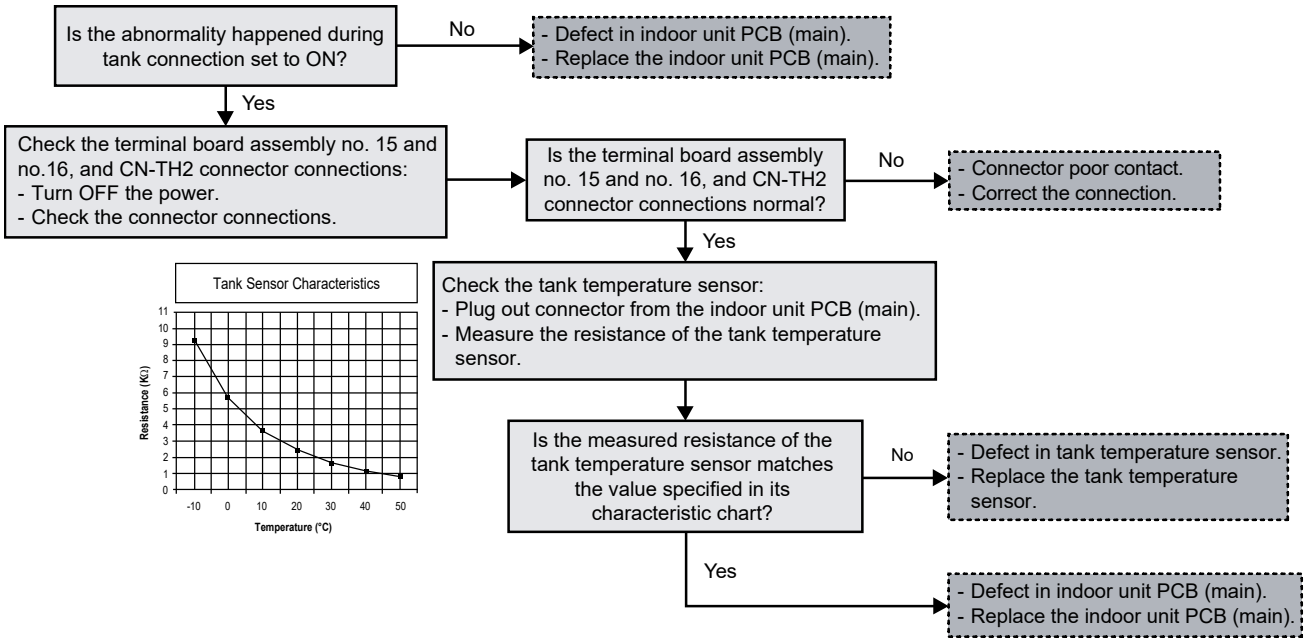
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty indoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.20 PCB Communication Error (H74)

Malfunction Decision Conditions:

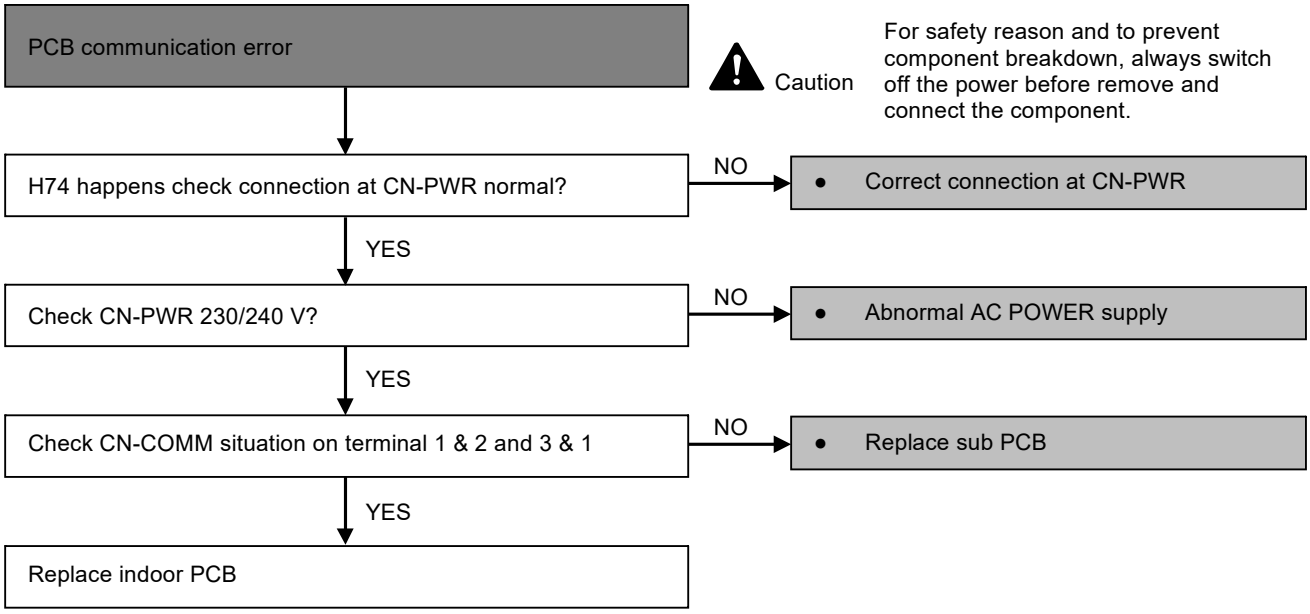
When External PCB connection is select “YES” and no communication with External PCB micon for 10 seconds and above.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty indoor PCB.
3 Faulty indoor sub PCB.

Abnormality Judgment:

After 1 minute operation started.



19.5.21 Indoor-Control Panel Communication Abnormality (H76)

Malfunction Decision Conditions:

During standby and operation of cooling and heating, indoor-control panel error occur.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty control panel.
3 Faulty indoor unit PCB (main).

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.

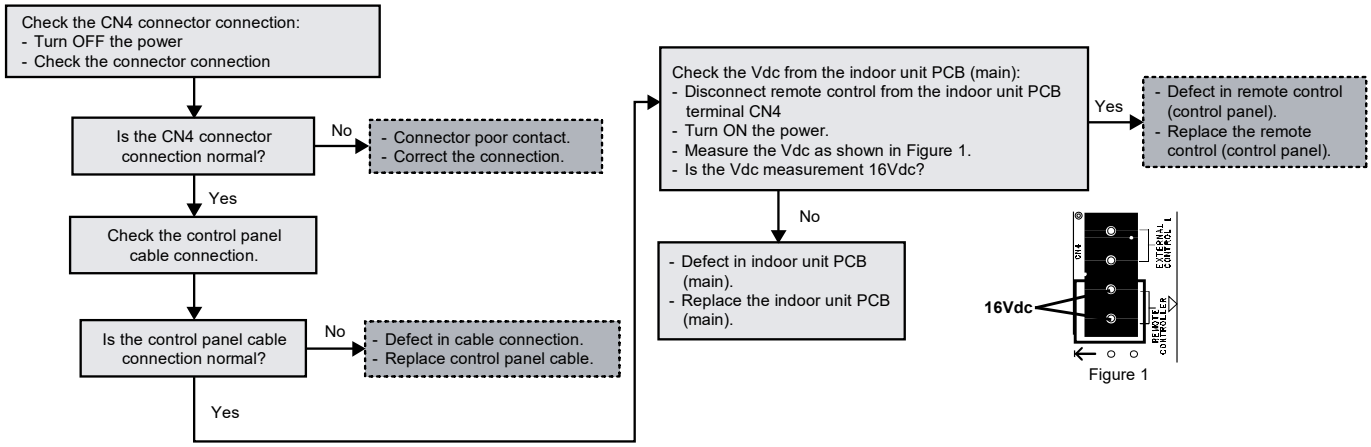


Figure 1

19.5.22 Indoor/Outdoor Abnormal Communication (H90)

Malfunction Decision Conditions:

During operation of cooling and heating, the data received from outdoor unit in indoor unit signal transmission is checked whether it is normal.

Malfunction Caused:

1 Faulty outdoor unit PCB (main).
2 Faulty indoor unit PCB (main).
3 Indoor-outdoor signal transmission error due to wrong wiring.
4 Indoor-outdoor signal transmission error due to breaking of wire in the connection wires between the indoor and outdoor units.
5 Indoor-outdoor signal transmission error due to disturbed power supply waveform.

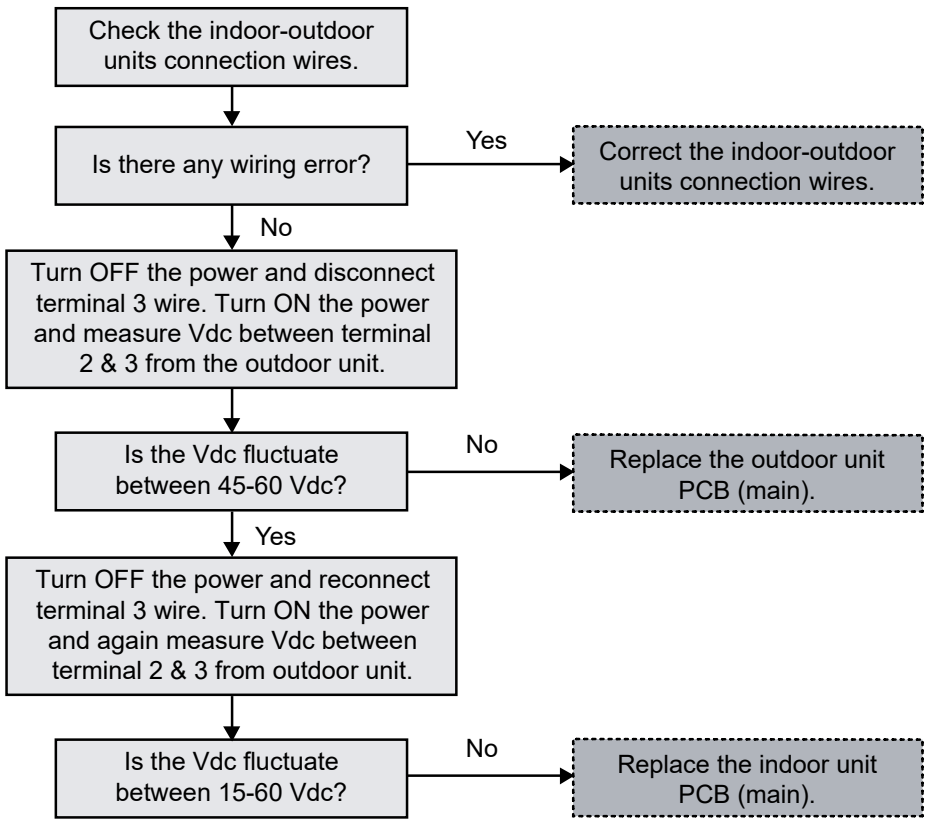
Abnormality Judgment:

Continue for 1 minute after operation.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



19.5.23 Tank Booster Heater OLP Abnormality (H91)

Malfunction Decision Conditions:

During operation of tank booster heater, and tank booster heater OLP open circuit.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty tank booster heater overload protector (OLP).
3 Faulty indoor unit PCB (main).

Abnormality Judgment:

Continue for 60 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.

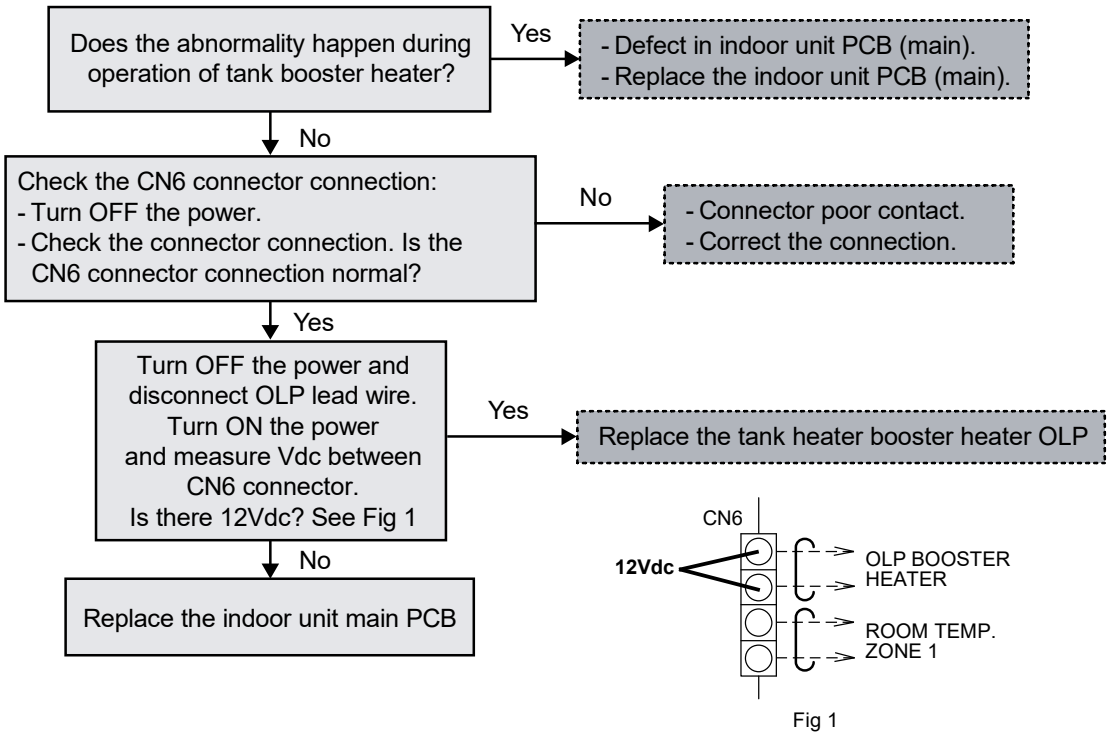


Fig 1

19.5.24 Unspecified Voltage between Indoor and Outdoor (H95)

Malfunction Decision Conditions:

The supply power is detected for its requirement by the indoor/outdoor transmission.

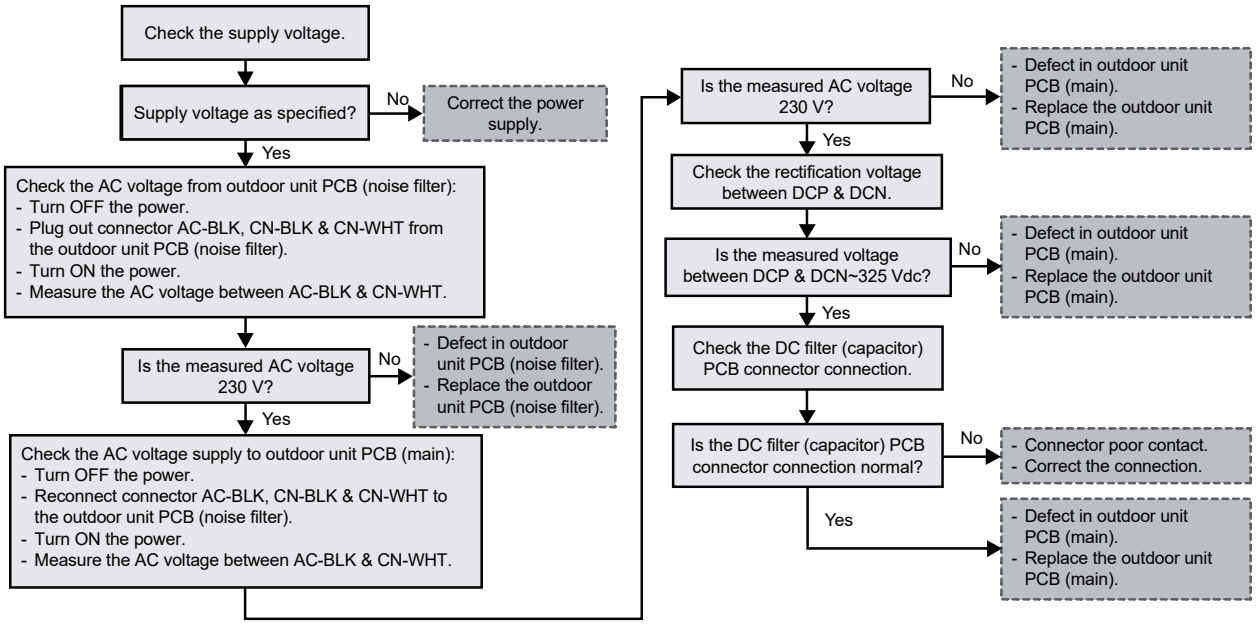
Malfunction Caused:

1 Insufficient power supply.
2 Faulty outdoor unit PCB (noise filter/main).

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



19.5.25 Outdoor High Pressure Protection (H98 / F95)

Malfunction Decision Conditions:

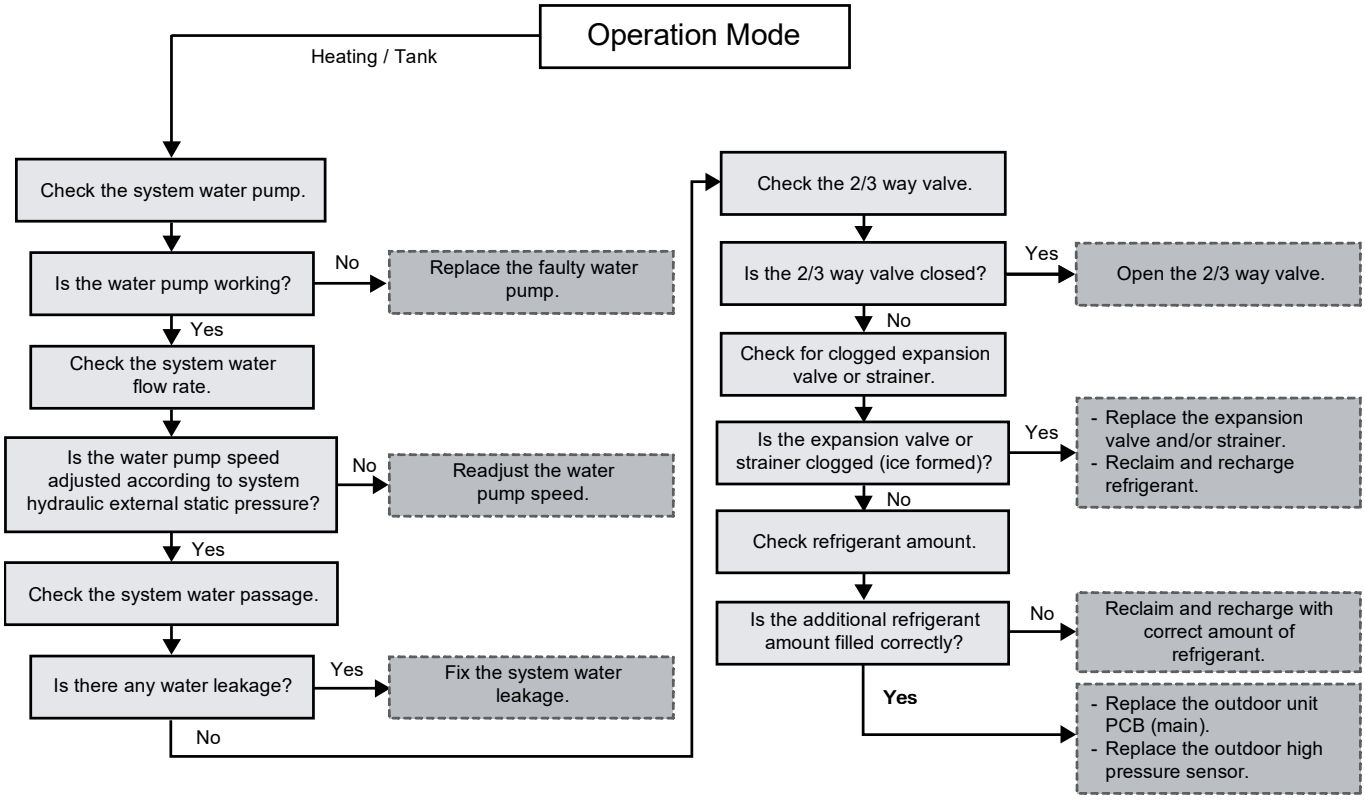
During operation of cooling / heating, when pressure 4.2 MPa and above is detected by outdoor high pressure sensor.

Malfunction Caused:

1 Faulty water pump.
2 Insufficient water flow rate in system.
3 Water leak in system.
4 Dust accumulation in the outdoor unit heat exchanger.
5 Air short circuit at outdoor.
6 Faulty outdoor unit fan motor.
7 2/3 way closed.
8 Clogged expansion valve or strainer.
9 Excessive refrigerant.
10 Faulty outdoor high pressure sensor.
11 Faulty outdoor unit PCB (main).

Troubleshooting:

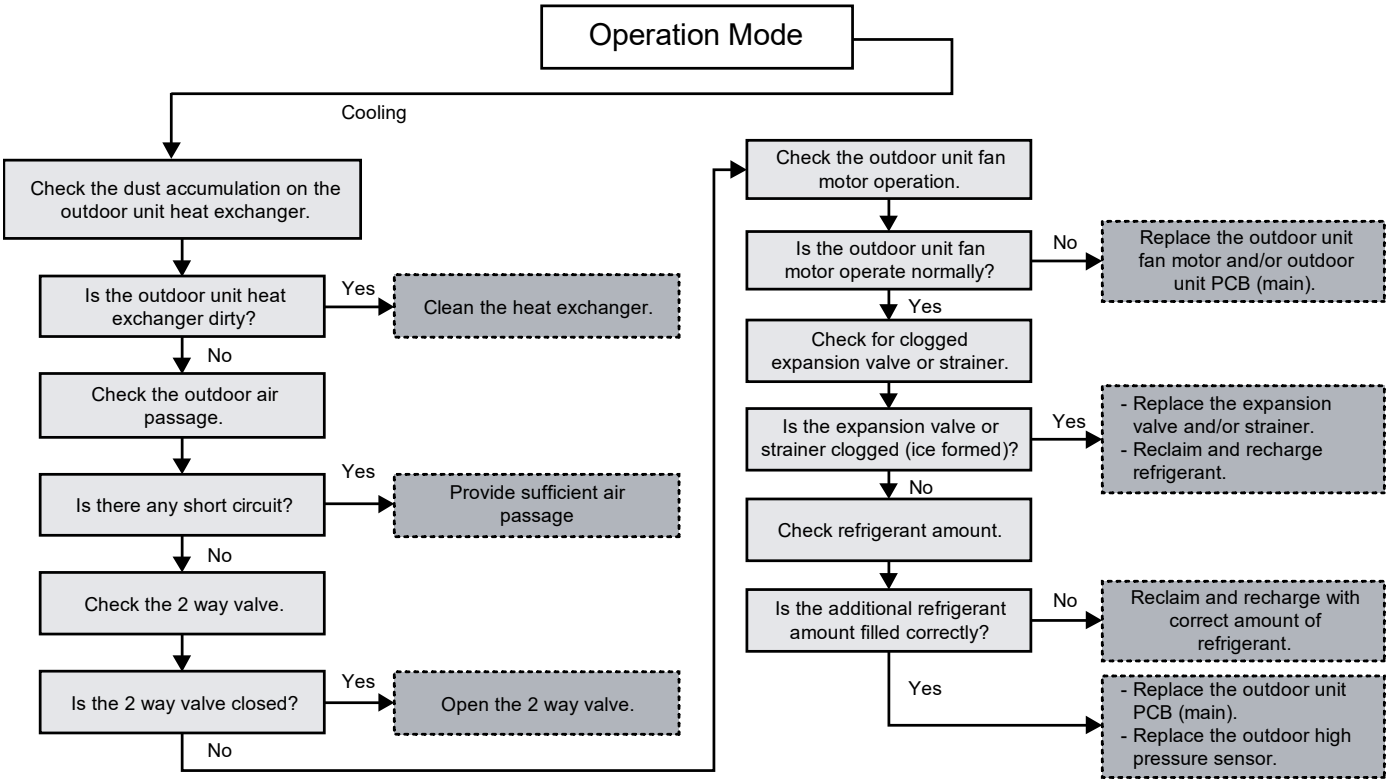
Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



19.5.26 Indoor Freeze-up Protection (H99)

Malfunction Decision Conditions:

During anti-freezing control in cooling operation, when the indoor refrigerant liquid temperature < 0°C.

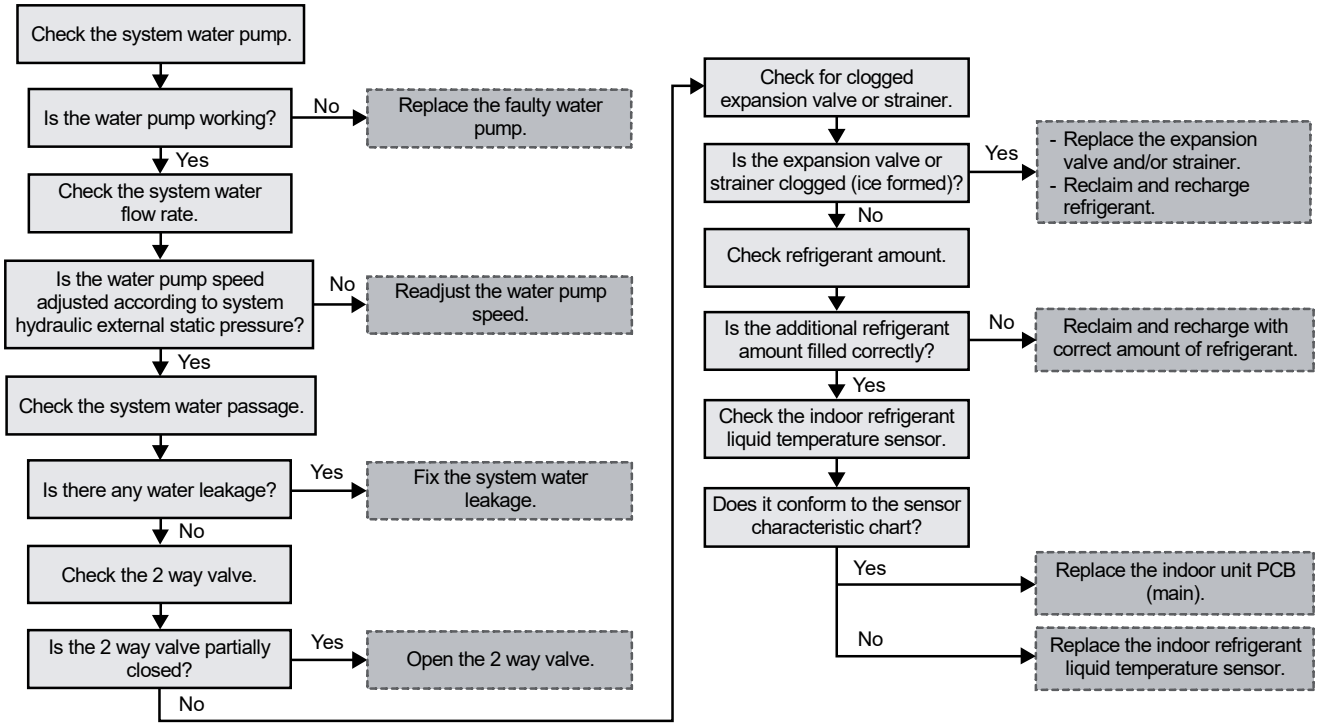
Malfunction Caused:

1 Faulty water pump.
2 Insufficient water flow rate in system.
3 Water leak in system.
4 2 way valve partially closed.
5 Clogged expansion valve or strainer.
6 Refrigerant shortage (refrigerant leakage).
7 Faulty indoor refrigerant liquid temperature sensor.
8 Faulty indoor unit PCB (main).

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



19.5.27 Outdoor High Pressure Switch Activate (F12)

Malfunction Decision Conditions:

During operation of cooling and heating, when pressure 4.5 MPa and above is detected by outdoor high pressure switch.

Malfunction Caused:

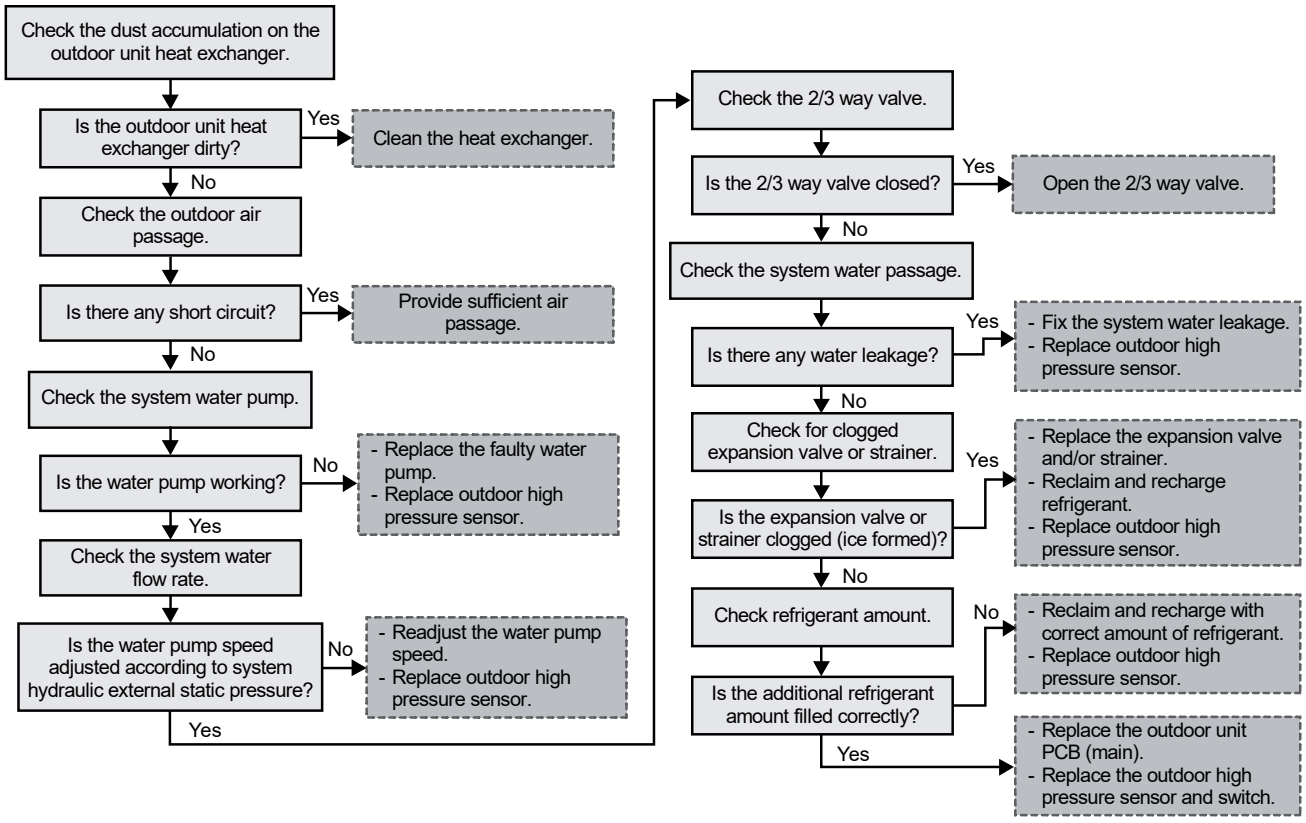
1 Dust accumulation on the outdoor unit heat exchanger.
2 Air short circuit at outdoor unit.
3 Faulty water pump.
4 Insufficient water flow rate in system.
5 Water leak in system.
6 2/3 way valve closed.
7 Clogged expansion valve or strainer.
8 Excessive refrigerant.
9 Faulty outdoor high pressure sensor and switch.
10 Faulty outdoor unit PCB.

Abnormality Judgment:

Continue 4 times in 20 minutes.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.28 Compressor Rotation Failure (F14)

Malfunction Decision Conditions:

A compressor rotation failure is detected by checking the compressor running condition through the position detection circuit.

Malfunction Caused:

1 Compressor terminal disconnect.
2 Faulty outdoor unit PCB (main).
3 Faulty compressor.

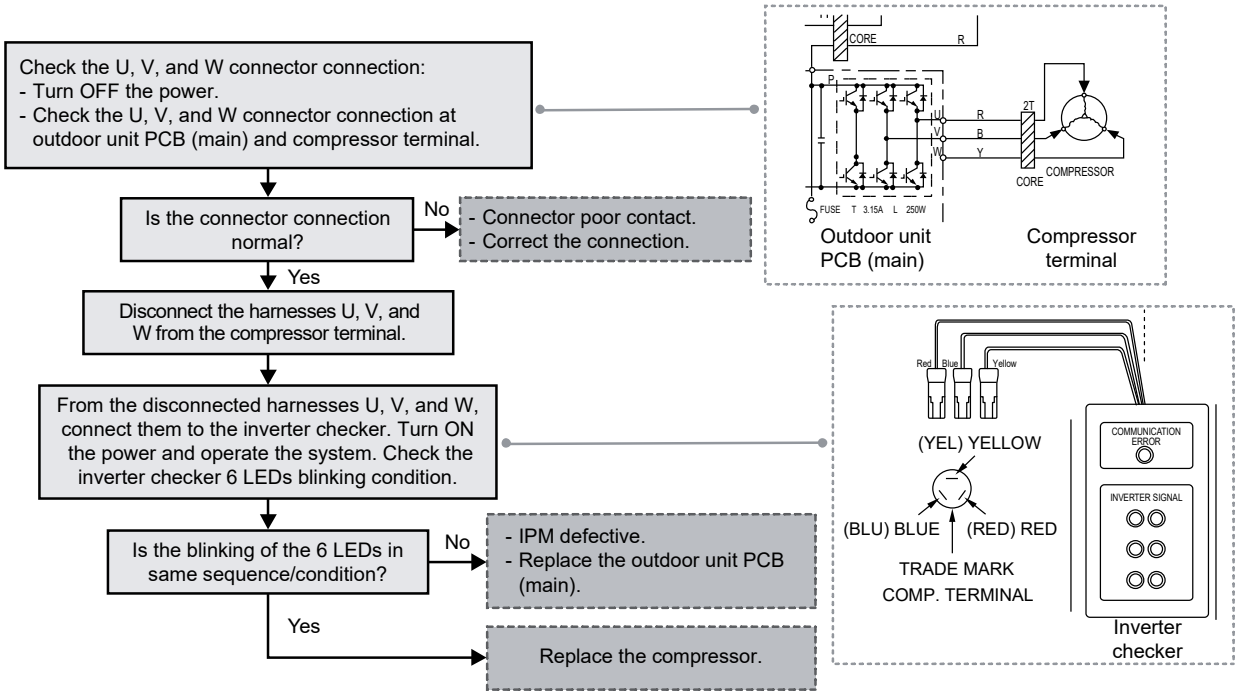
Abnormality Judgment:

Continue 4 times in 20 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



19.5.29 Outdoor Fan Motor (DC Motor) Mechanism Locked (F15)

Malfunction Decision Conditions:

The rotation speed detected by the Hall IC of the fan motor during fan motor operation is used to determine abnormal fan motor (feedback of rotation > 2550 rpm or < 50 rpm).

Malfunction Caused:

1 Operation stop due to short circuit inside the fan motor winding.
2 Operation stop due to breaking of wire inside the fan motor.
3 Operation stop due to breaking of fan motor lead wires.
4 Operation stop due to fan motor Hall IC malfunction.
5 Operation error due to faulty outdoor unit PCB.

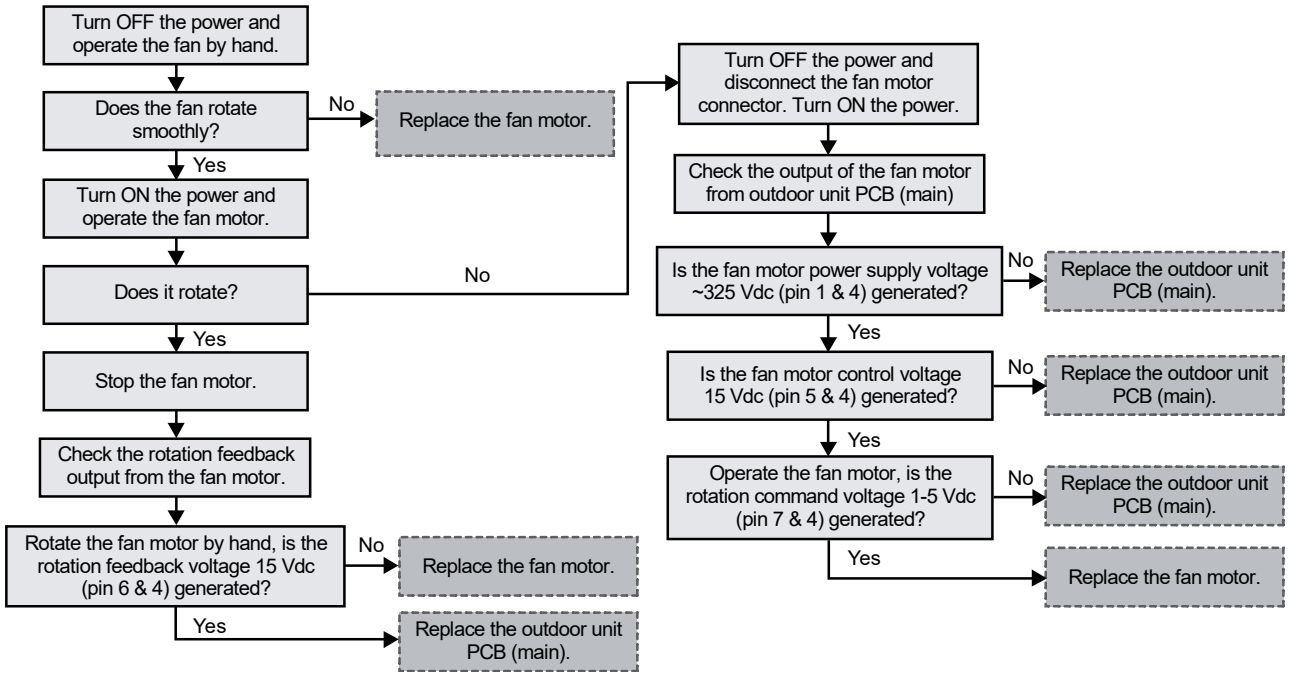
Abnormality Judgment:

Continue 2 times in 30 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



19.5.30 Input Over Current Detection (F16)

Malfunction Decision Conditions:

During operation of cooling and heating, when outdoor current above UDZ03 / 05KE5 14.6A & UDZ07 / 09KE5 16.9A is detected by the current transformer (CT) in the outdoor unit PCB.

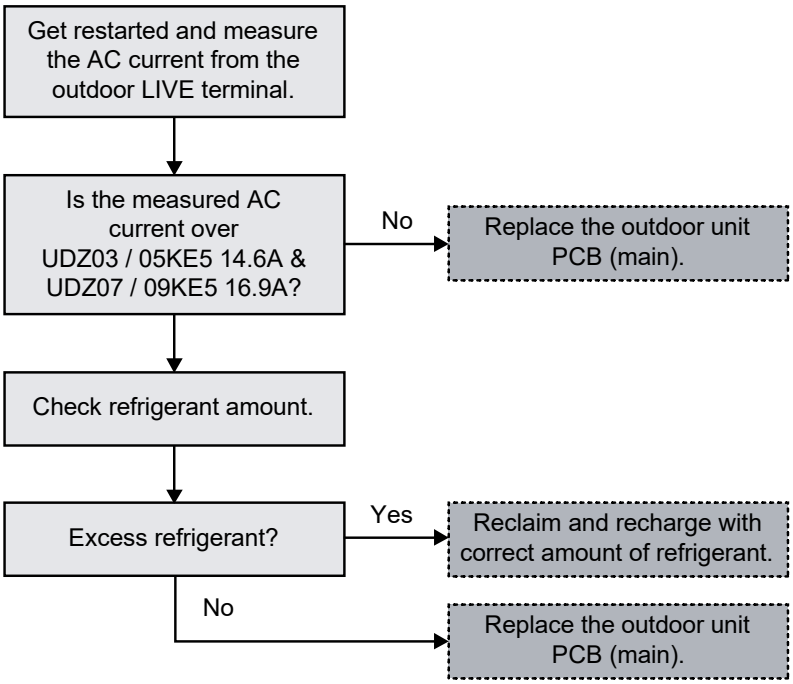
Malfunction Caused:

1 Excessive refrigerant.
2 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue 3 times in 20 minutes.

Troubleshooting: **Caution** For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.31 Compressor Overheating (F20)

Malfunction Decision Conditions:

During operation of cooling and heating, when temperature above 112°C is detected by the compressor tank temperature sensor.

Malfunction Caused:

1 Faulty compressor tank temperature sensor.
2 2/3 way valve closed.
3 Refrigerant shortage (refrigerant leakage).
4 Clogged expansion valve or strainer.
5 Faulty outdoor unit PCB (main).
6 Faulty compressor.

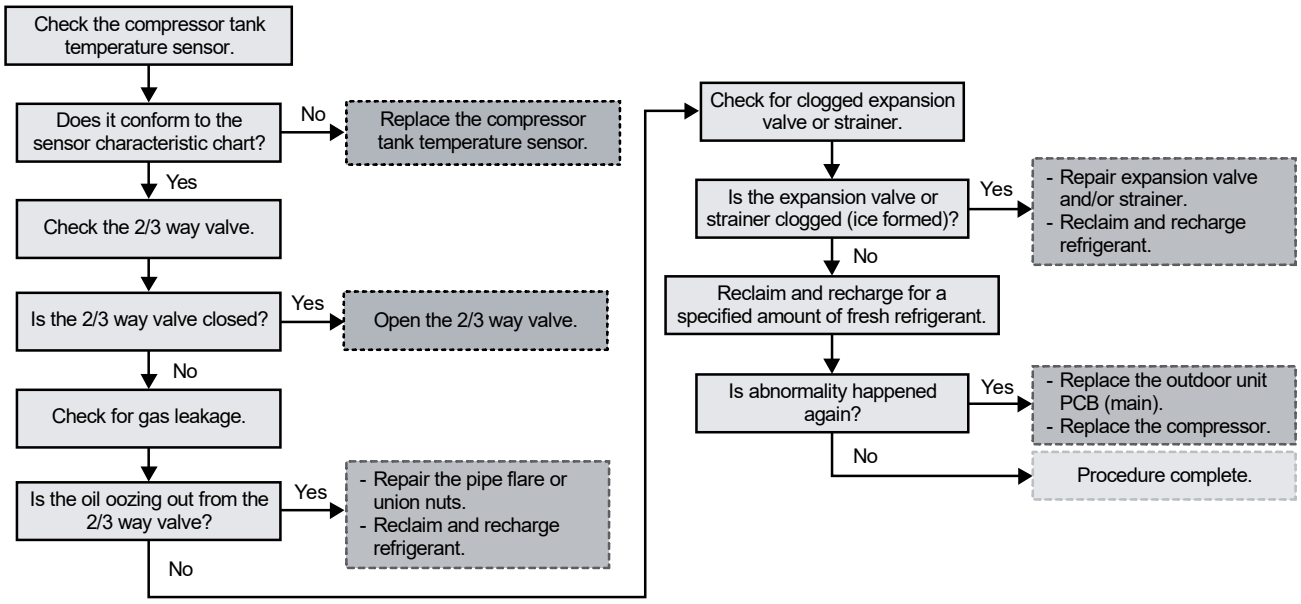
Abnormality Judgment:

Continue 4 times in 30 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



19.5.32 IPM Overheating (F22)

Malfunction Decision Conditions:

During operation of cooling and heating, when temperature 95°C is detected by the outdoor IPM temperature sensor.

Malfunction Caused:

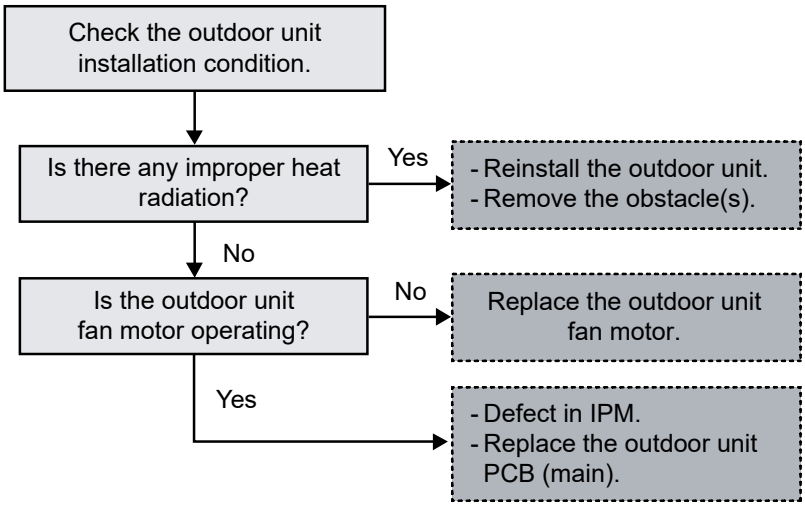
1 Faulty outdoor unit fan motor.
2 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue 3 times in 30 minutes.

Troubleshooting: Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



19.5.33 Output Over Current Detection (F23)

Malfunction Decision Conditions:

During operation of cooling and heating, when outdoor DC current is above set value is detected by the IPM DC Peak sensing circuitry in the outdoor unit PCB (main).

Malfunction Caused:

1 Faulty outdoor unit PCB (main).
2 Faulty compressor.

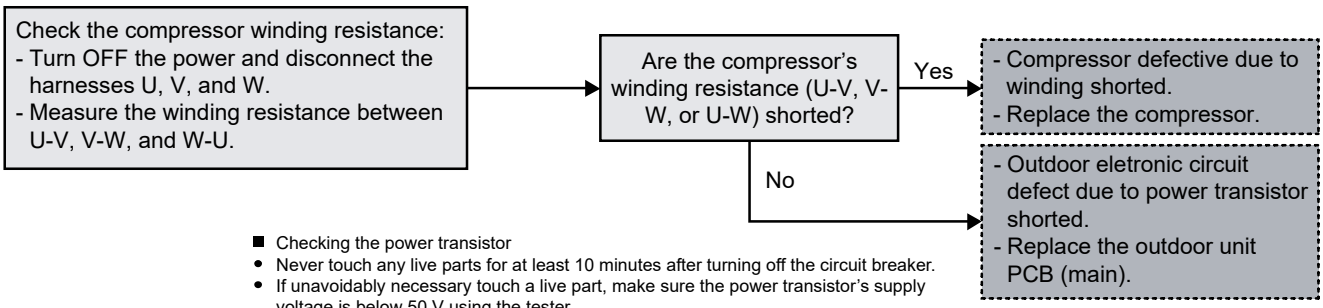
Abnormality Judgment:

Continue for 7 times.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



■ Checking the power transistor

- Never touch any live parts for at least 10 minutes after turning off the circuit breaker.
- If unavoidably necessary touch a live part, make sure the power transistor's supply voltage is below 50 V using the tester.
- For the UVW, make measurements at the Faston terminal on the board or the relay connector.

Tester's negative terminal	Power transistor (+)	UVW	Power transistor (-)	UVW
Tester's positive terminal	UVW	Power transistor (+)	UVW	Power transistor (-)
Normal resistance	Several kohms to several Mohms			
Abnormal resistance	0 or ∞			

19.5.34 Refrigeration Cycle Abnormality (F24)

Malfunction Decision Conditions:

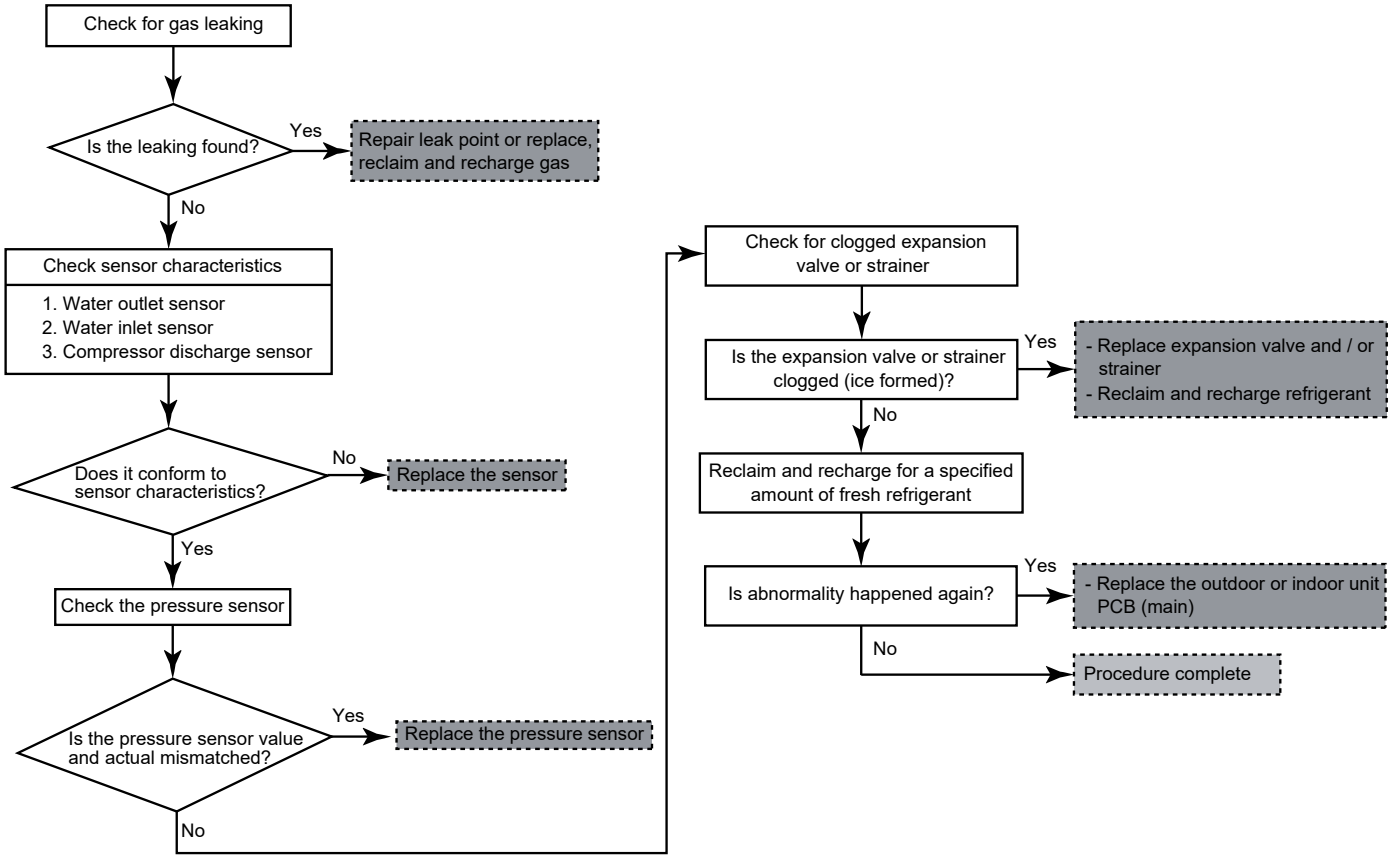
1 During compressor running (heating / cooling) for more than 10 minutes except deice, pumpdown and test mode.
2 During heating / cooling, water outlet and water inlet difference is less than 1°C.
3 During heating / cooling, high pressure < 1MPa (143 Psi) for more than 10 minutes or < 0.2 MPa (28 Psi) for more than 5 minutes.
4 During heating / cooling, discharge temperature - saturation temperature of high pressure ≥ 60°C.

Malfunction Caused:

1 Refrigerant shortage (refrigerant leakage).
2 Faulty indoor water inlet, indoor water outlet, compressor discharge temp sensor or high pressure sensor.
3 2/3 way valve closed.
4 Clogged expansion valve or strainer.
5 Faulty indoor or outdoor PCB (main).

Abnormality Judgment:

Continue 2 times in 30 minutes.



19.5.35 Four Way Valve Abnormality (F25)

Malfunction Decision Conditions:

1 During heating operation, when the indoor pipe temperature of thermostat ON indoor unit < 0°C.
2 During cooling operation, when the indoor pipe temperature of thermostat ON indoor unit > 45°C.

Malfunction Caused:

1 Faulty sensor.
2 Faulty connector connection.
3 Faulty outdoor unit PCB (noise filter/main).
4 Faulty four way valve.

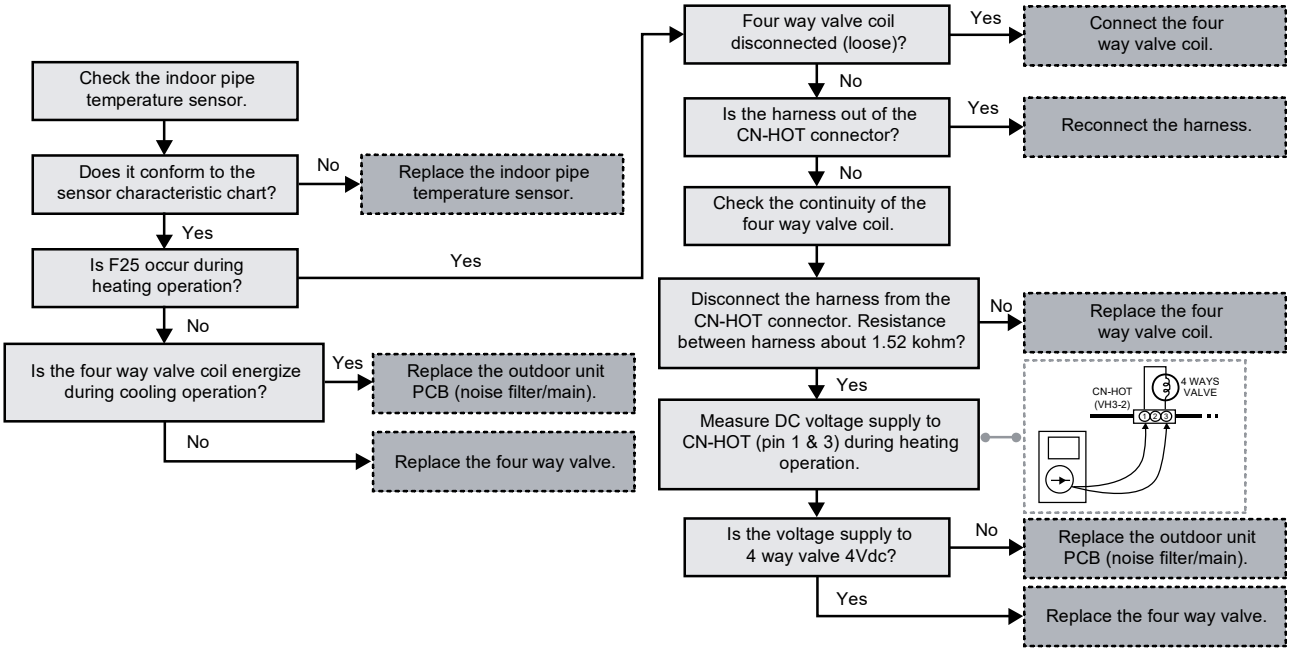
Abnormality Judgment:

Continue 4 times in 30 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



19.5.36 Outdoor High Pressure Switch Abnormal (F27)

Malfunction Decision Conditions:

During compressor stop, and outdoor high pressure switch is remain opened.

Malfunction Caused:

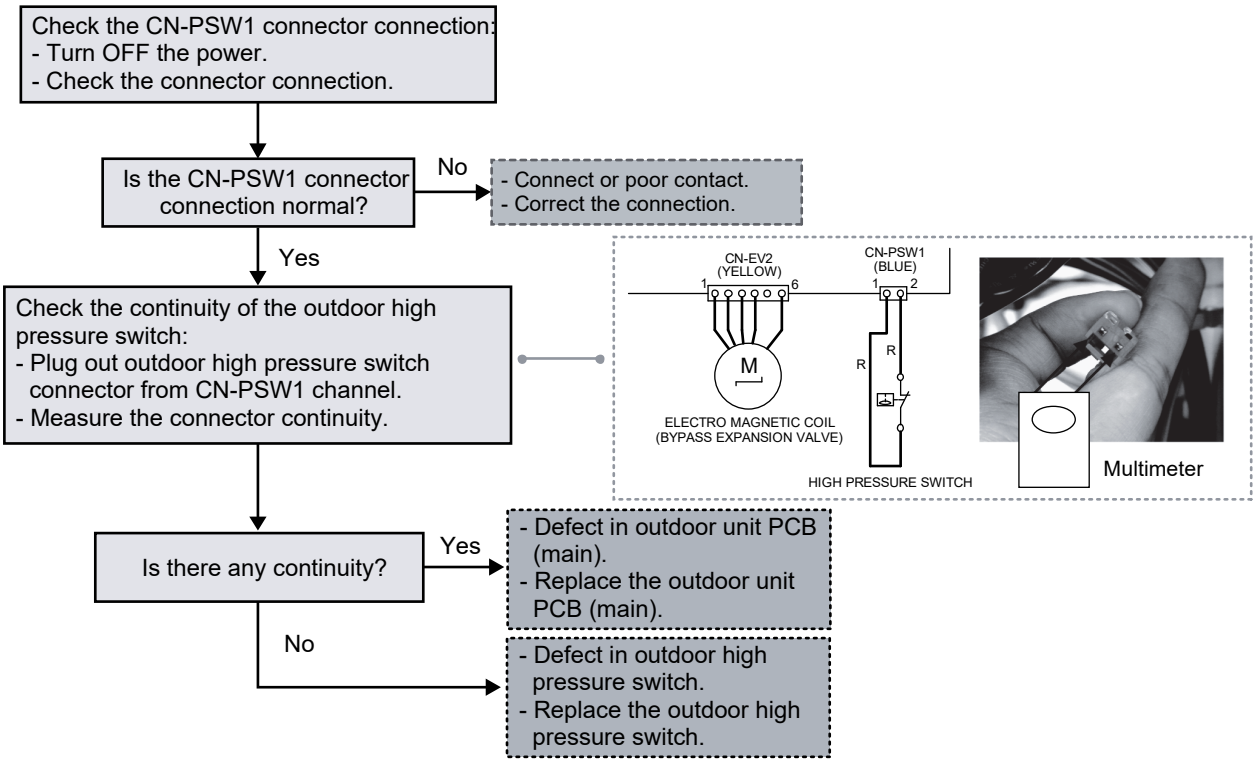
1 Faulty connector connection.
2 Faulty switch.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue for 1 minute.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component



19.5.37 Low Discharge Superheat (F29)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the outdoor pipe temperature sensor are used to determine sensor error.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).
4 Faulty High Pressure Switch
5 Refrigerant shortage (refrigerant leakage).

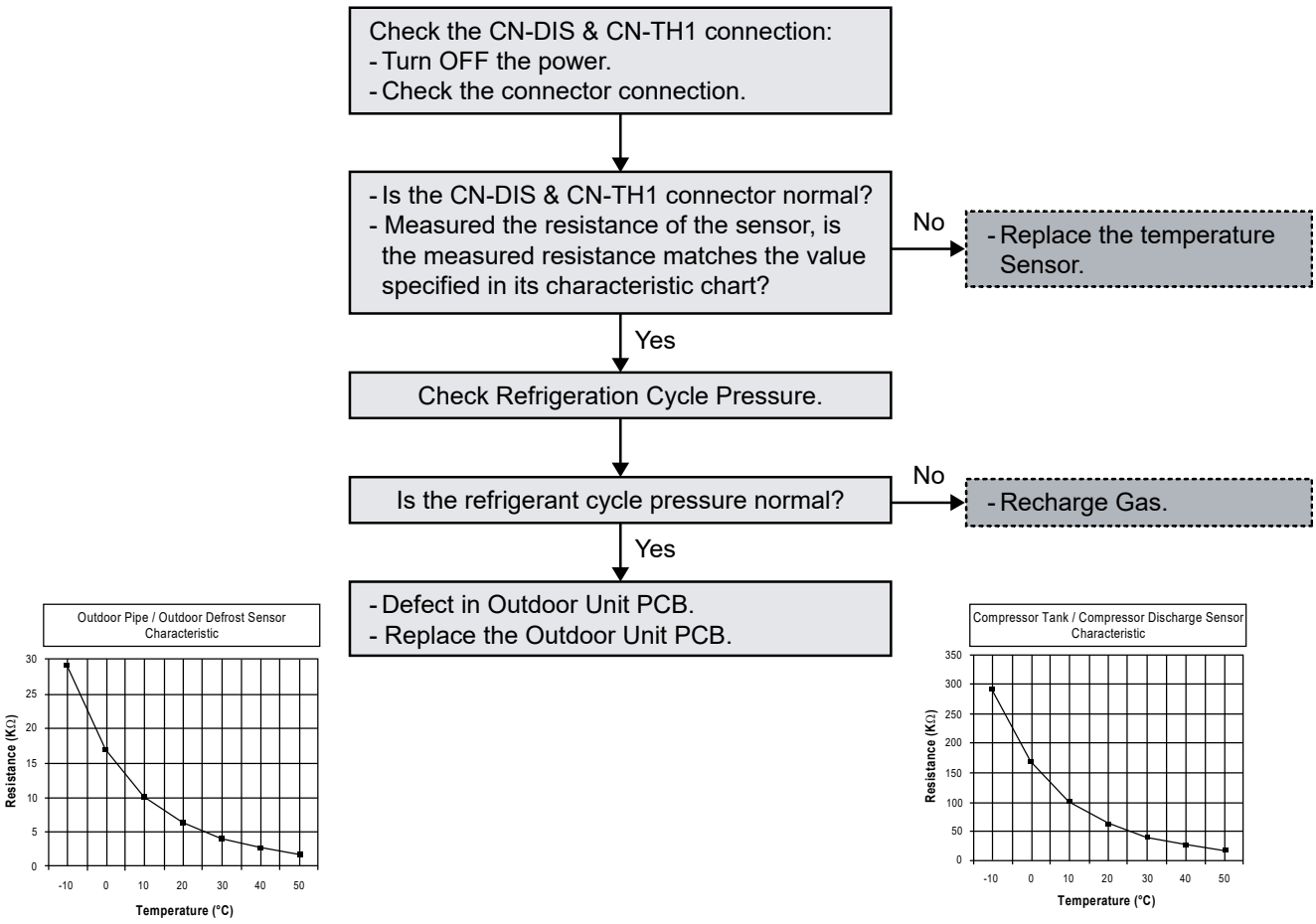
Abnormality Judgment:

1 times occurrence within 2550 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



19.5.38 Indoor Water Outlet Temperature Sensor 2 Abnormality (F30)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the indoor water outlet temperature sensor 2 are used to determine sensor error.

Malfunction Caused:

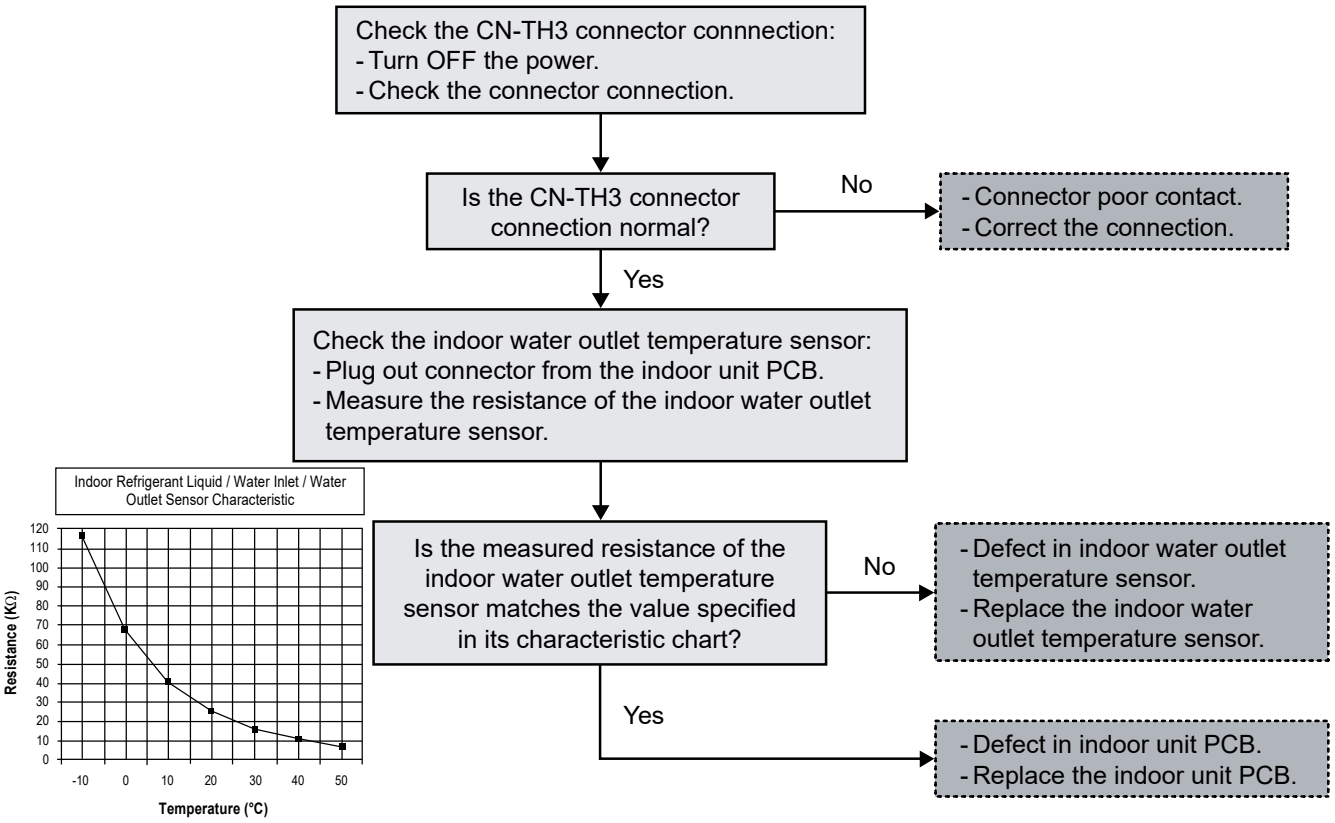
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty indoor unit PCB.

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.39 Outdoor Air Temperature Sensor Abnormality (F36)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the outdoor air temperature sensor are used to determine sensor error.

Malfunction Caused:

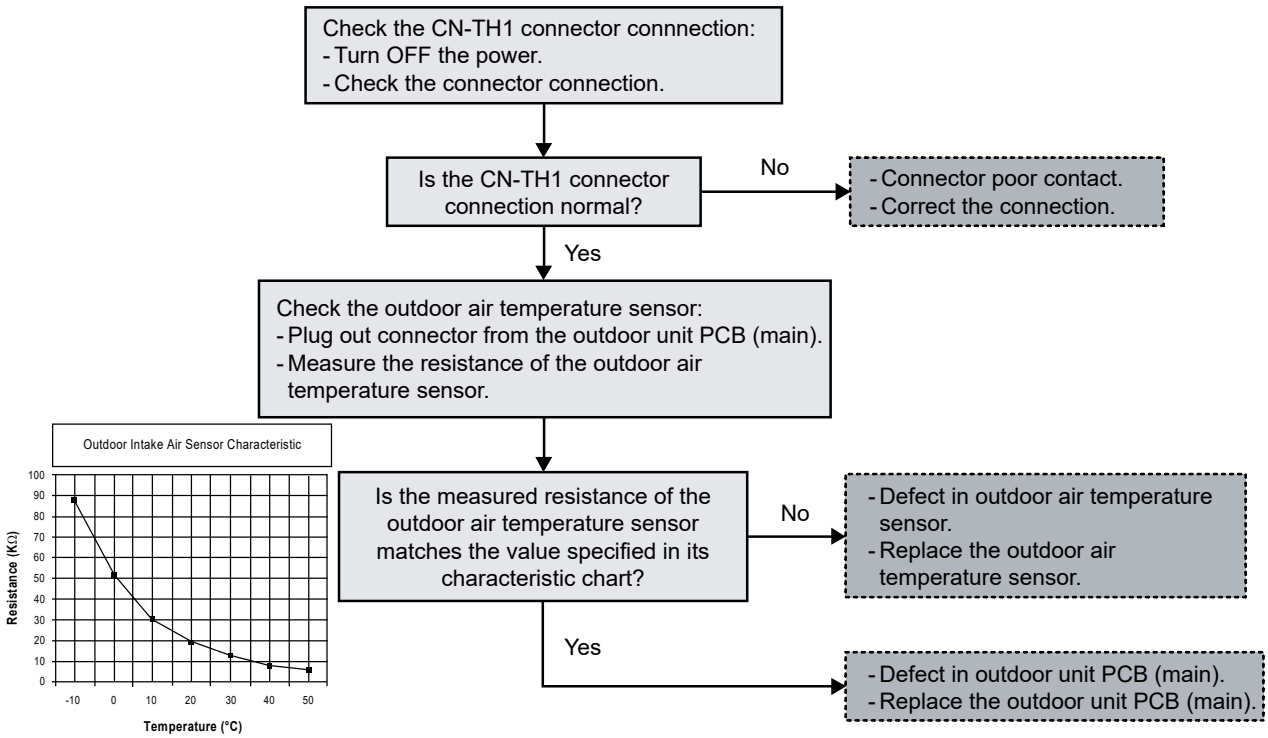
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.40 Indoor Water Inlet Temperature Sensor Abnormality (F37)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the indoor water inlet temperature sensor are used to determine sensor error.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty indoor unit PCB (main).

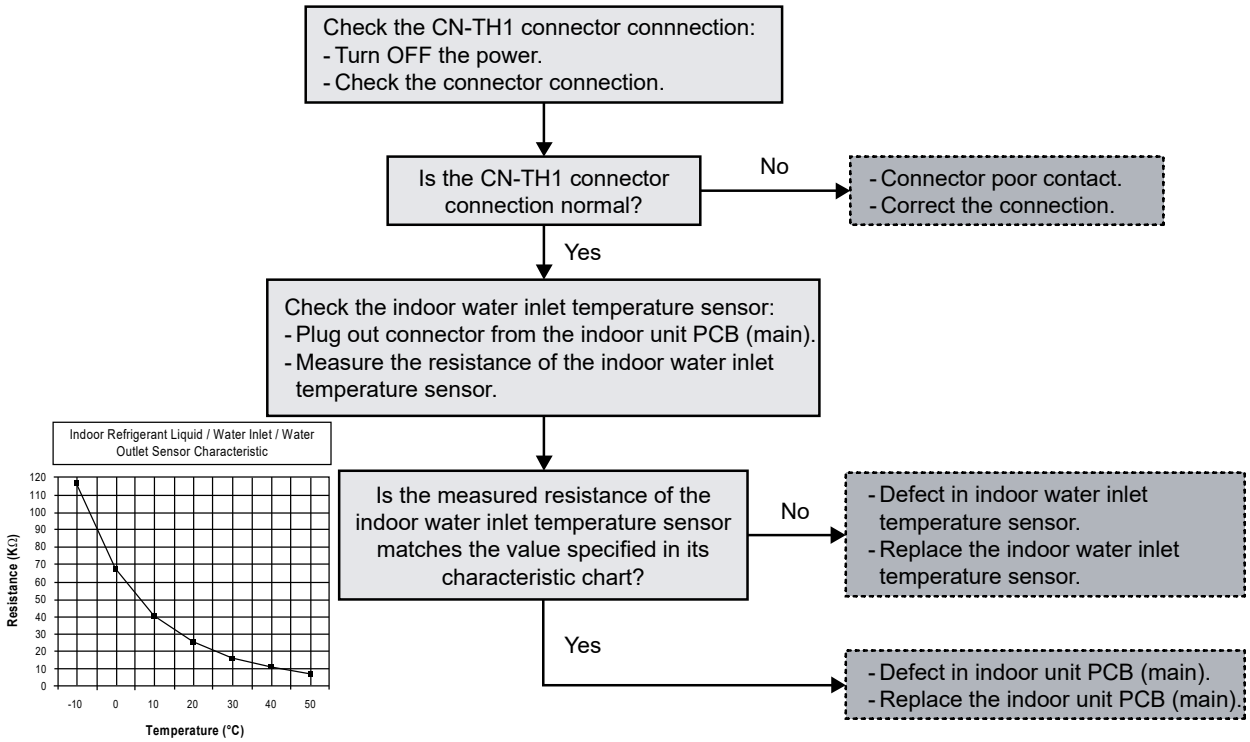
Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



19.5.41 Outdoor Discharge Pipe Temperature Sensor Abnormality (F40)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the outdoor discharge pipe temperature sensor are used to determine sensor error.

Malfunction Caused:

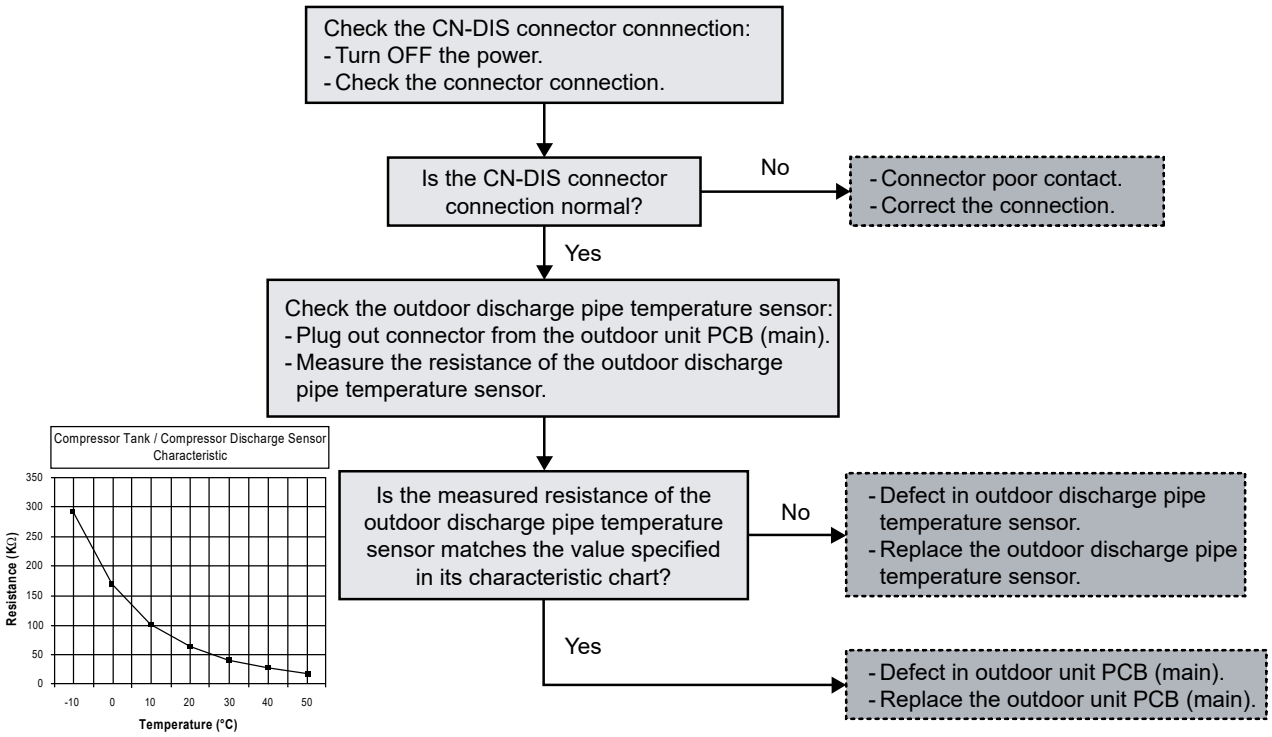
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.42 Power Factor Correction (PFC) Abnormality (F41)

Malfunction Decision Conditions:

During operation of cooling and heating, when the PFC protection circuitry in the outdoor unit PCB (main) senses abnormal high DC voltage level.

Malfunction Caused:

1 Power supply surge.
2 Compressor windings not uniform.
3 Faulty outdoor unit PCB (main).

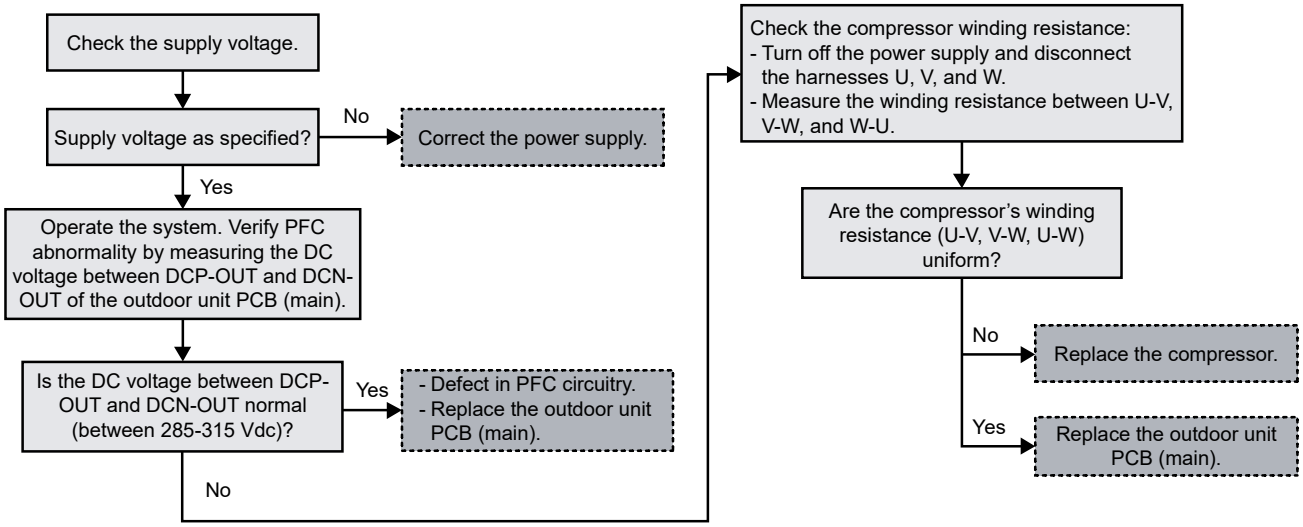
Abnormality Judgment:

Continue 4 times in 10 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.43 Outdoor Pipe Temperature Sensor Abnormality (F42)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the outdoor pipe temperature sensor are used to determine sensor error.

Malfunction Caused:

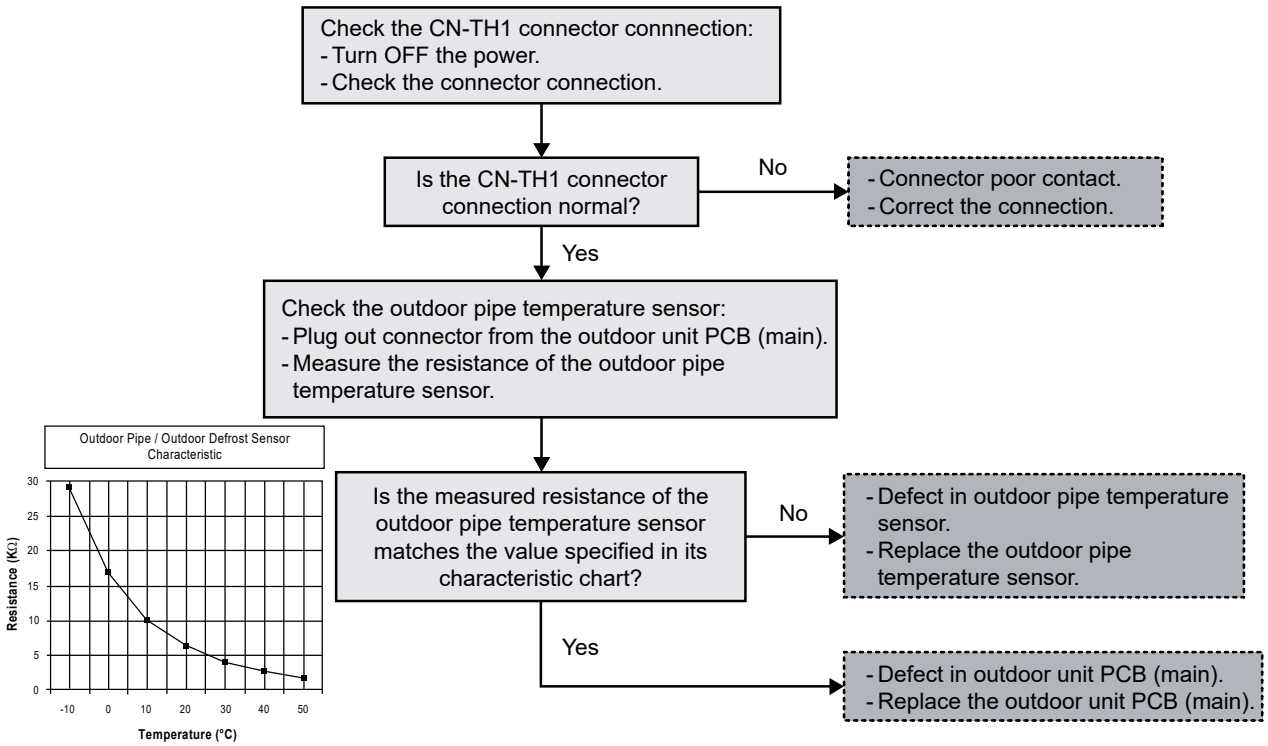
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.44 Outdoor Defrost Temperature Sensor Abnormality (F43)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the outdoor defrost temperature sensor are used to determine sensor error.

Malfunction Caused:

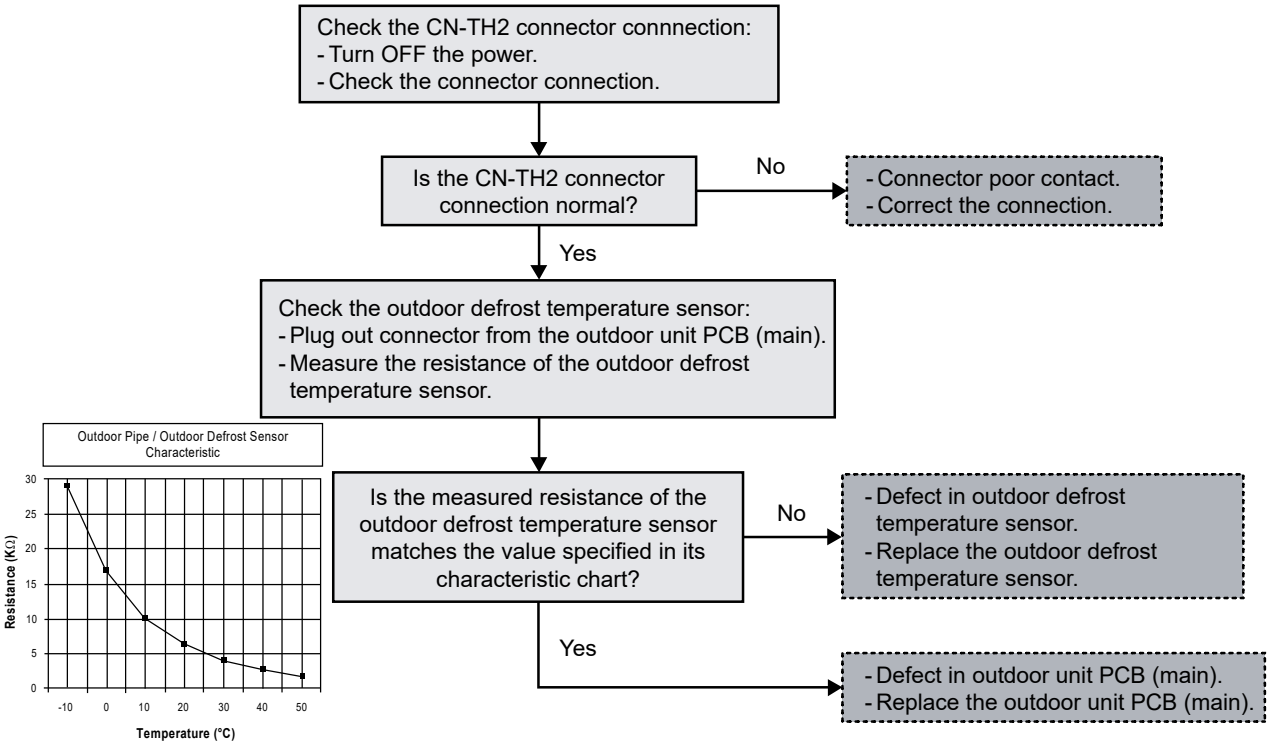
1 Faulty connector connection.
2 Faulty sensor.
3 Faulty outdoor unit PCB (main).

Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.



19.5.45 Indoor Water Outlet Temperature Sensor Abnormality (F45)

Malfunction Decision Conditions:

During startup and operation of cooling and heating, the temperatures detected by the indoor water outlet temperature sensor are used to determine sensor errors.

Malfunction Caused:

1 Faulty connector connection.
2 Faulty sensor.
3 Faulty indoor unit PCB (main).

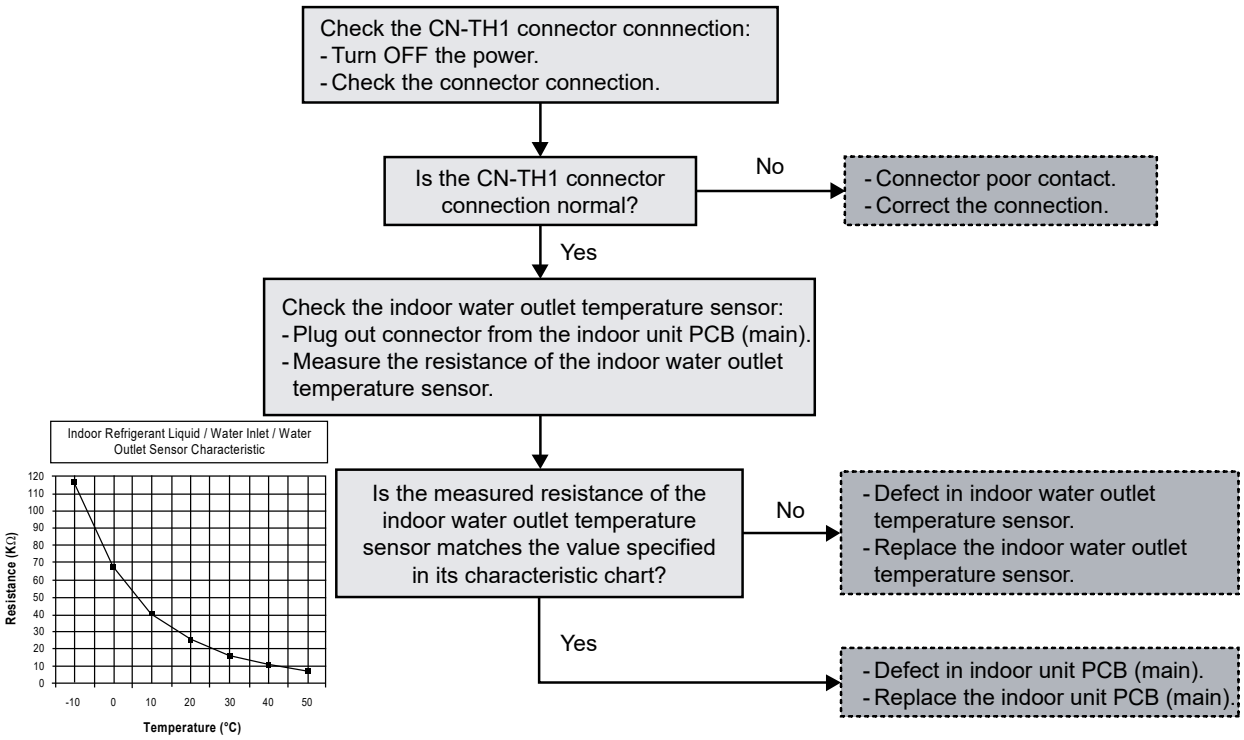
Abnormality Judgment:

Continue for 5 seconds.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



19.5.46 Outdoor Current Transformer Open Circuit (F46)

Malfunction Decision Conditions:

A current transformer (CT) open circuit is detected by checking the compressor running frequency (⩾ rated frequency) and CT detected input current (< 0.65 A) for continuously 20 seconds.

Malfunction Caused:

1 CT defective.
2 Faulty outdoor unit PCB (main).
3 Compressor defective (low compression).

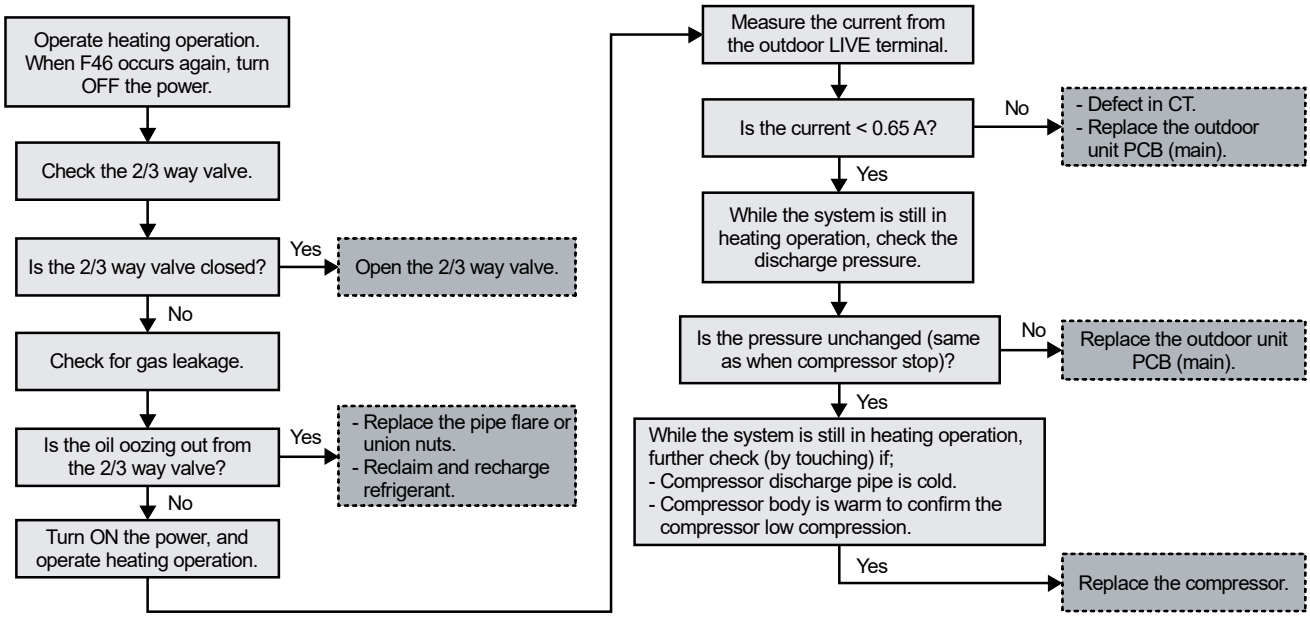
Abnormality Judgment:

Continue 3 times in 20 minutes.

Troubleshooting:

Caution

For safety reason and to prevent component breakdown,
always switch off the power before remove and connect the component.



19.5.47 Outdoor Bypass Outlet Temperature Sensor (F49)

Malfunction Decision Conditions:

During start up and operation of cooling and heating, the temperature detected by outdoor bypass sensor is used to determine sensor error.

Malfunction Caused:

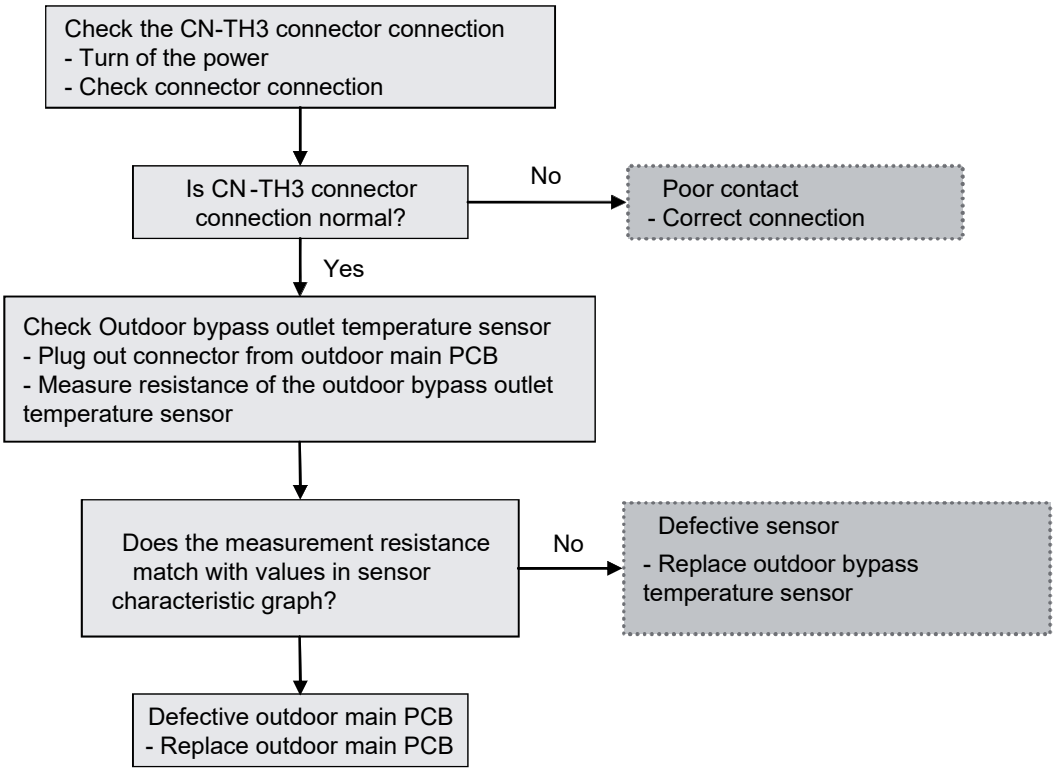
1 Faulty connector connection
2 Faulty sensor
3 Faulty outdoor unit PCB (main)

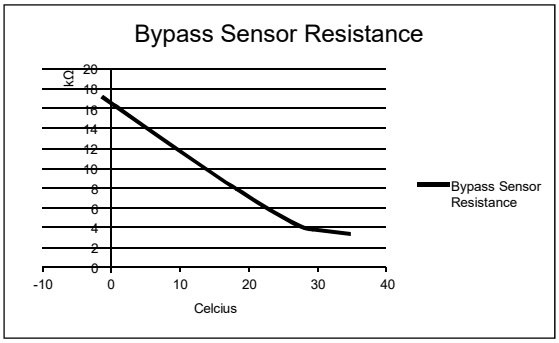
Abnormality Judgment:

Continuous for 5 seconds

Troubleshooting:

Caution For safety reason and to prevent component breakdown, always switch off the power before remove and connect the component.

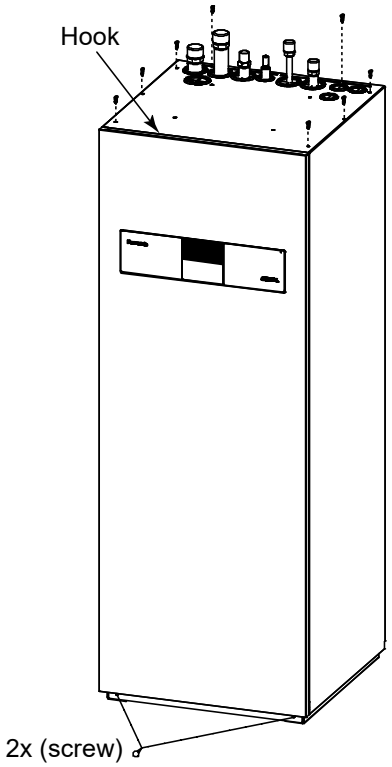




20. Disassembly and Assembly Instructions

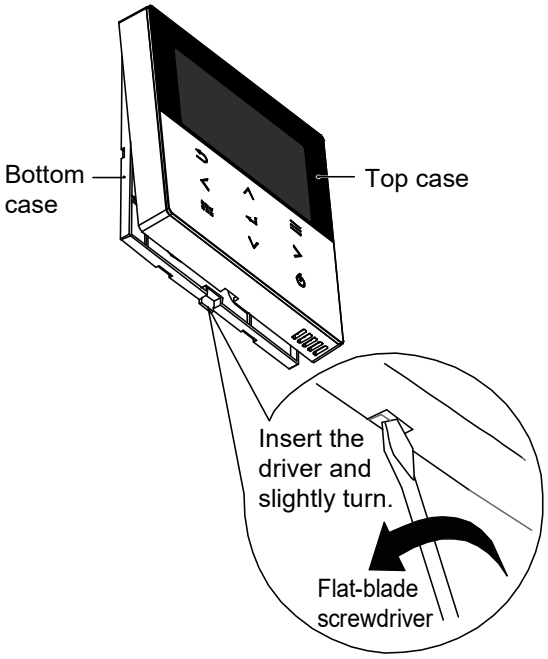
> ⚠ **WARNING**
>
> High Voltage are generated in the electrical parts area by the capacitor. Ensure that the capacitor has discharged sufficiently before proceeding with repair work. Failure to heed this caution may result in electric shocks.

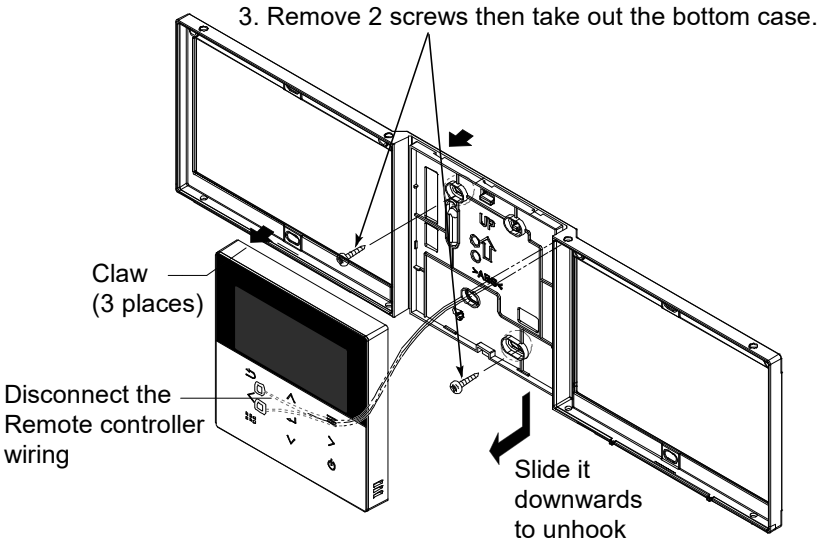
20.1 To Remove Front Plate and Top Plate



1. Remove 2 screws at the bottom to remove the Front Plate
2. Remove 8 screws at the top to remove the Top Plate

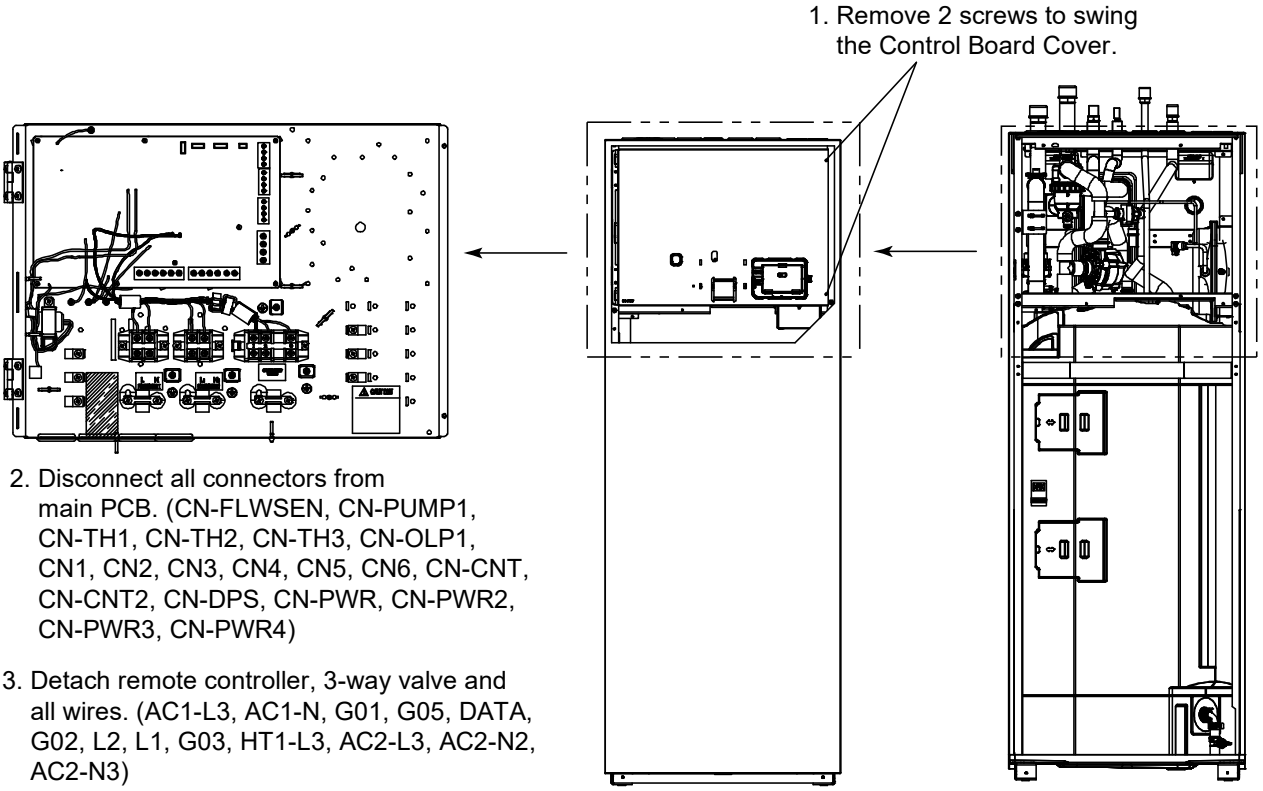
20.2 To Remove Remote Control





1. Remove the top case from the bottom case.
2. Disconnect the Remote Controller wiring.

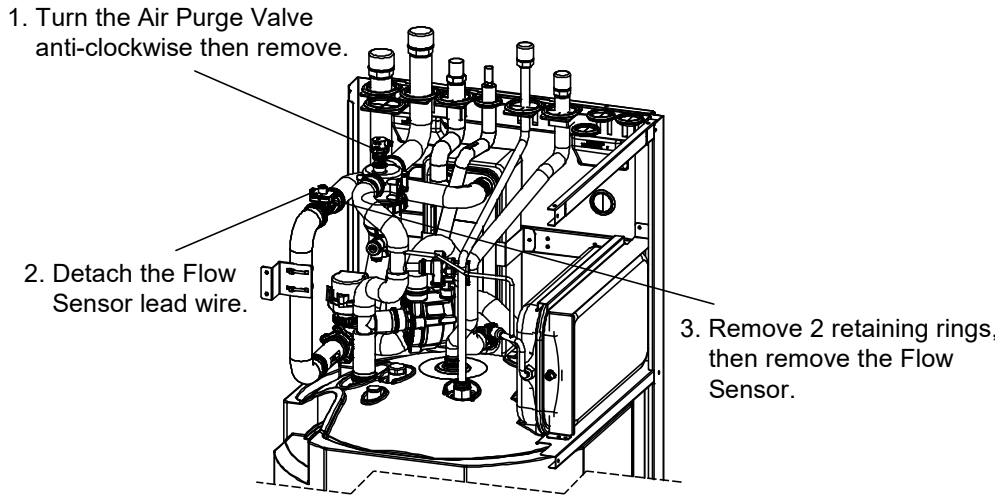
20.3 To Remove Electronic Controller



2. Disconnect all connectors from main PCB. (CN-FLWSEN, CN-PUMP1, CN-TH1, CN-TH2, CN-TH3, CN-OLP1, CN1, CN2, CN3, CN4, CN5, CN6, CN-CNT, CN-CNT2, CN-DPS, CN-PWR, CN-PWR2, CN-PWR3, CN-PWR4)

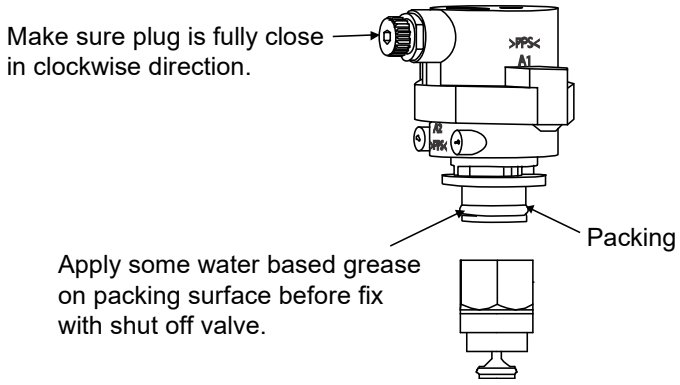
3. Detach remote controller, 3-way valve and all wires. (AC1-L3, AC1-N, G01, G05, DATA, G02, L2, L1, G03, HT1-L3, AC2-L3, AC2-N2, AC2-N3)

20.4 To Remove Flow Sensor and Air Purge Valve

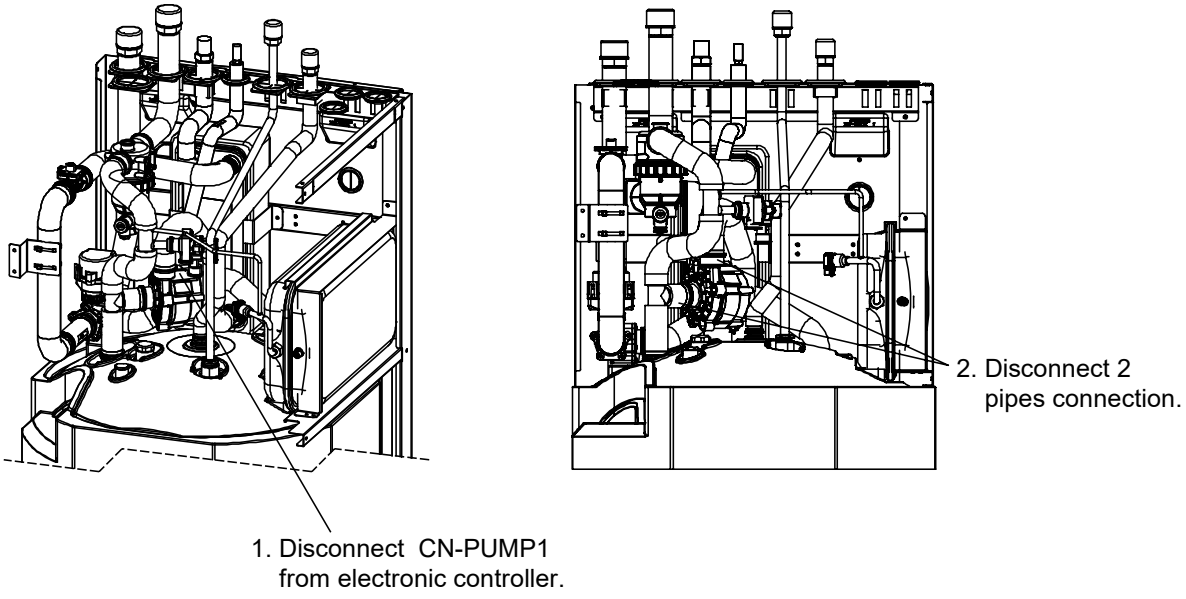


⚠ When reinstall Flow Switch, ensure the arrow on the flow switch is parallel with the pipe shaft and is facing in the direction of flow.

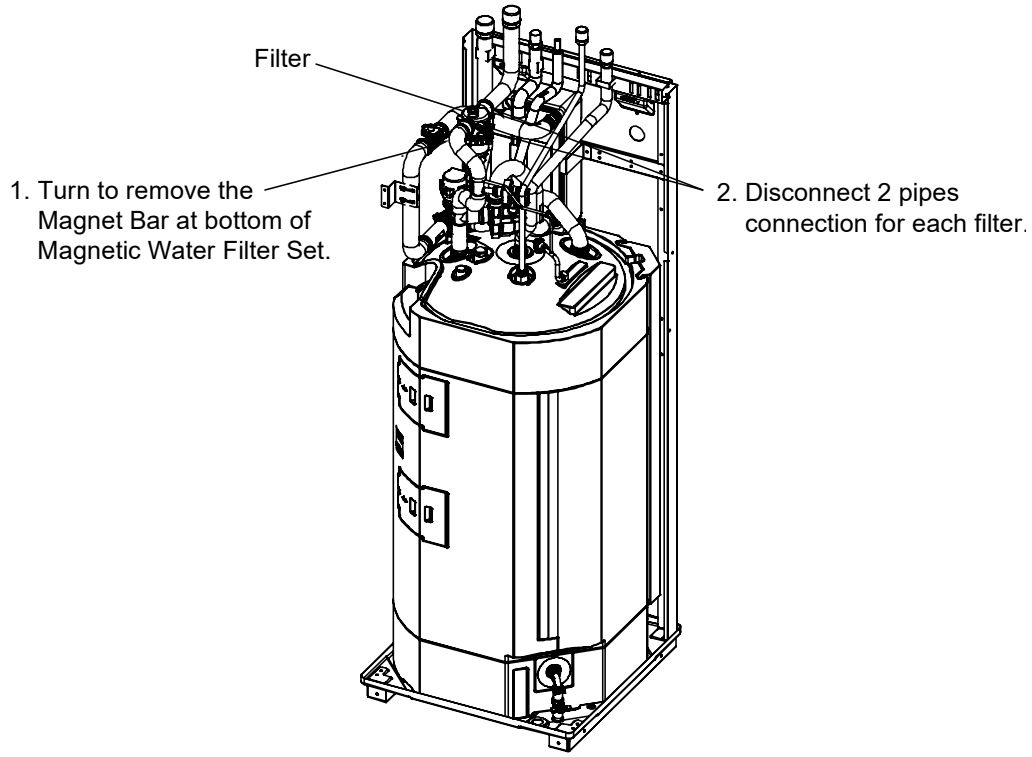
⚠ During reinstall Air Purge Valve.



20.5 To Remove Water Pump



20.6 To Remove Water Filter



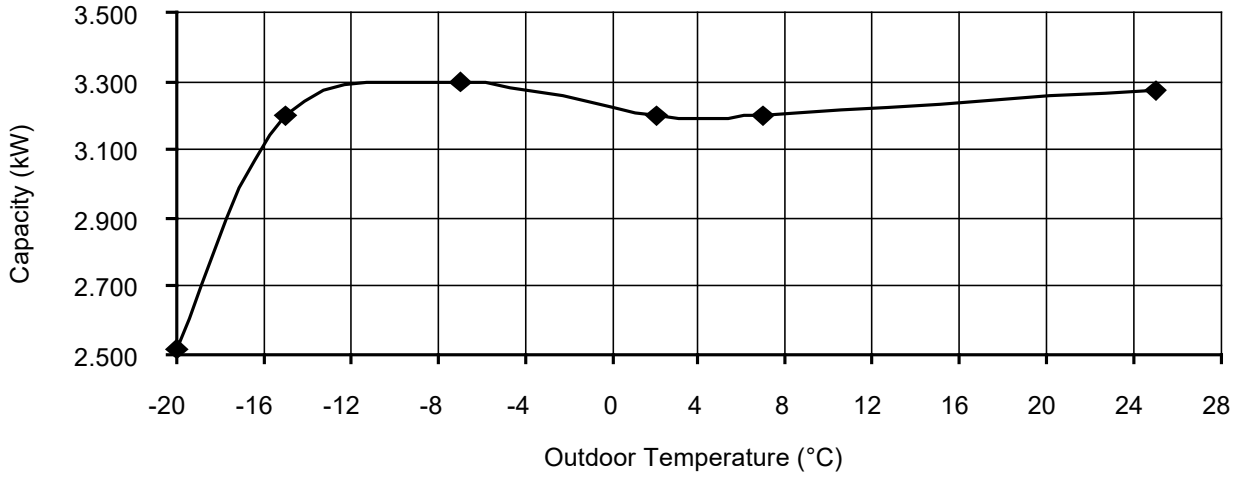
21. Technical Data

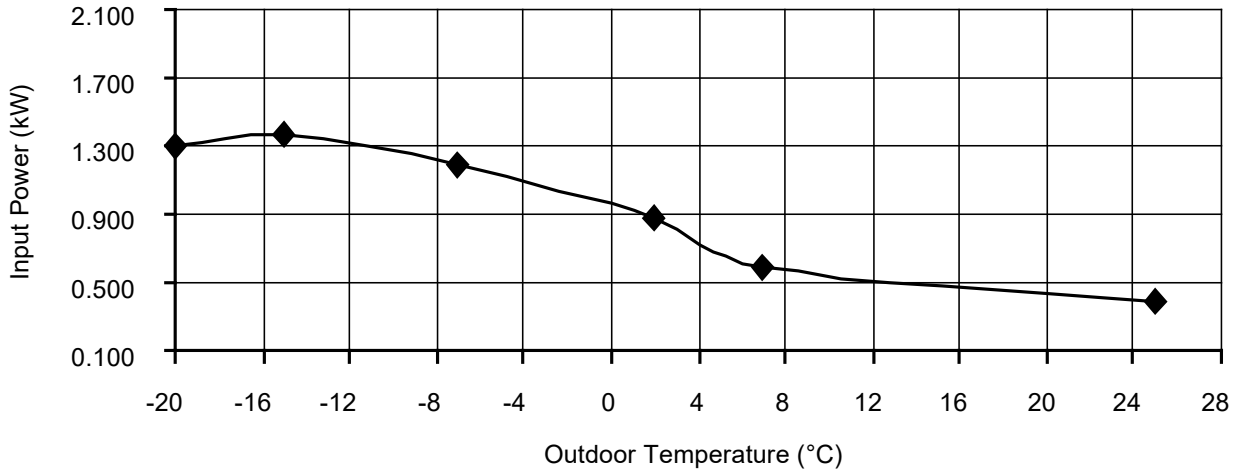
21.1 Operation Characteristics

21.1.1 WH-UDZ03KE5

Heating Characteristics at Different Outdoor Air Temperature

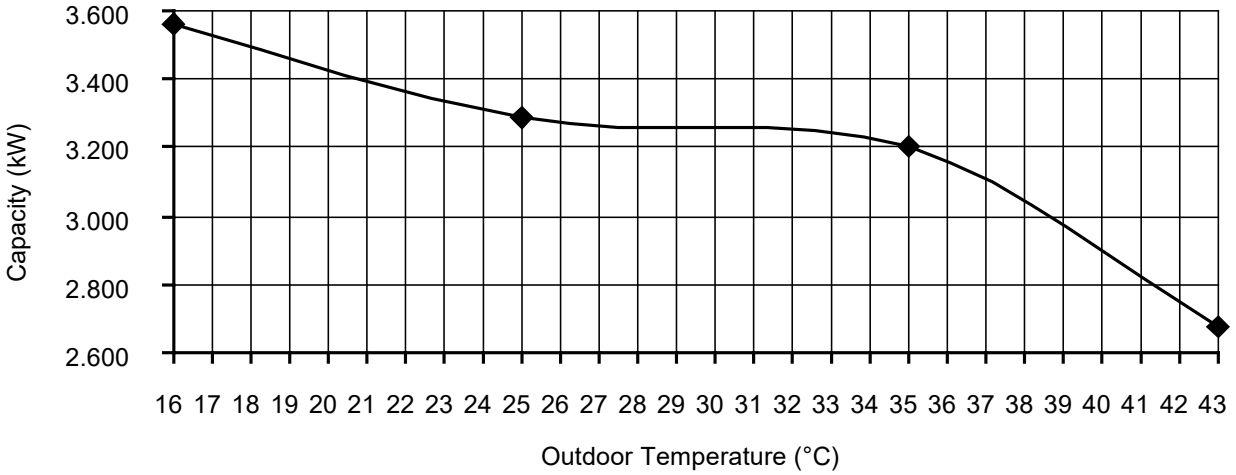
Condition
Outdoor air temperature : 7°C (DBT), 6°C (WBT)
Indoor water inlet temperature : 30°C
Indoor water outlet temperature : 35°C
Piping length : 7 m

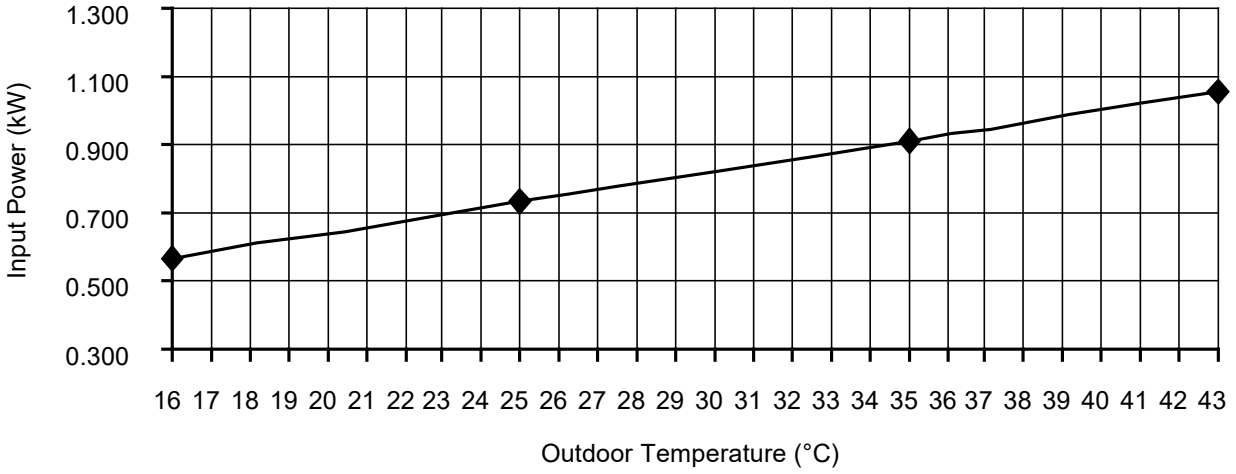




Cooling Characteristics at Different Outdoor Air Temperature

Condition
Outdoor air temperature : 35°C (DBT), -°C (WBT)
Indoor water inlet temperature : 12°C
Indoor water outlet temperature : 7°C
Piping length : 7 m





Cooling Characteristics at Different Outdoor Air Temperature

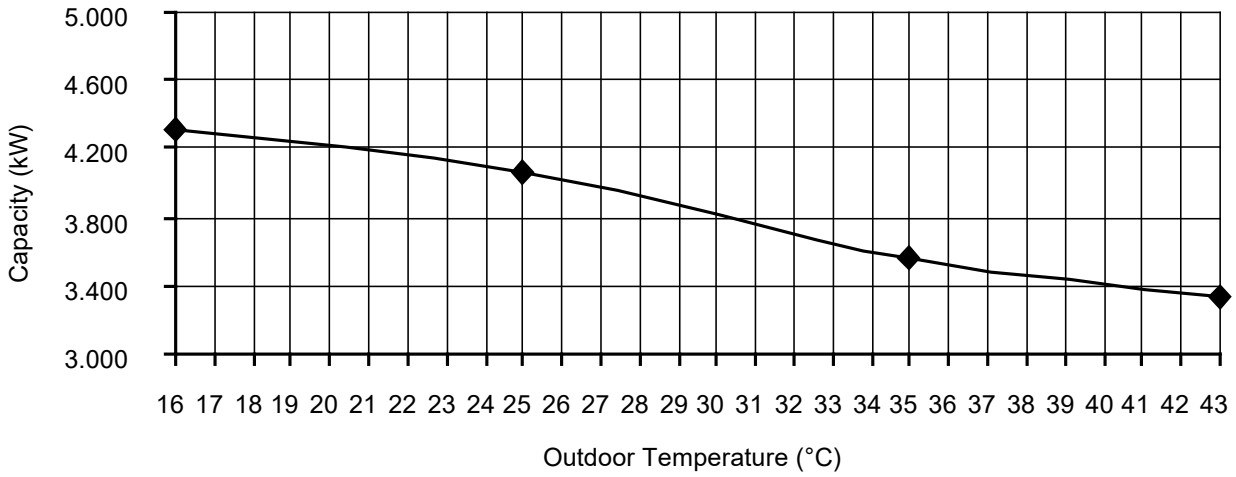
Condition

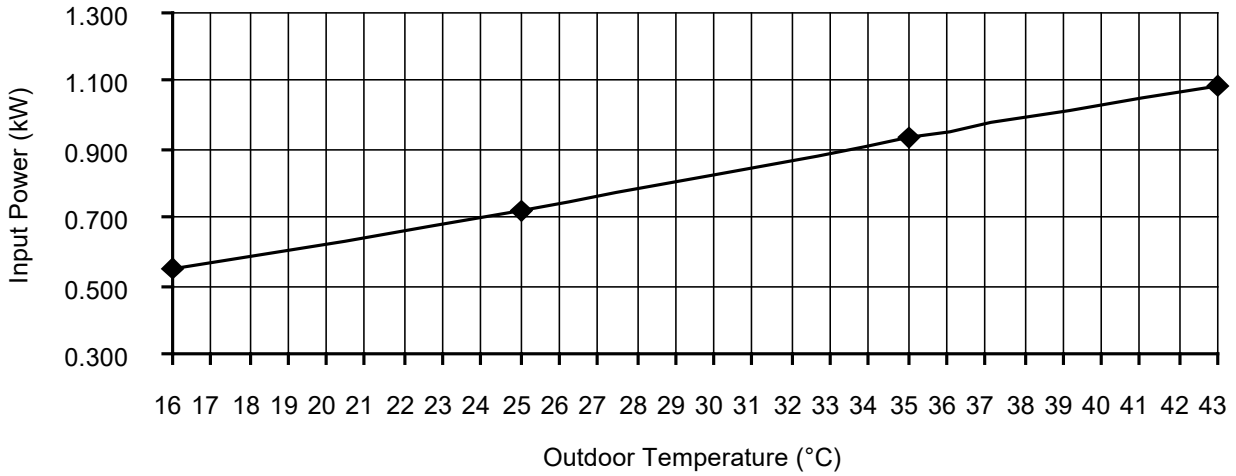
Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 19°C

Indoor water outlet temperature : 14°C

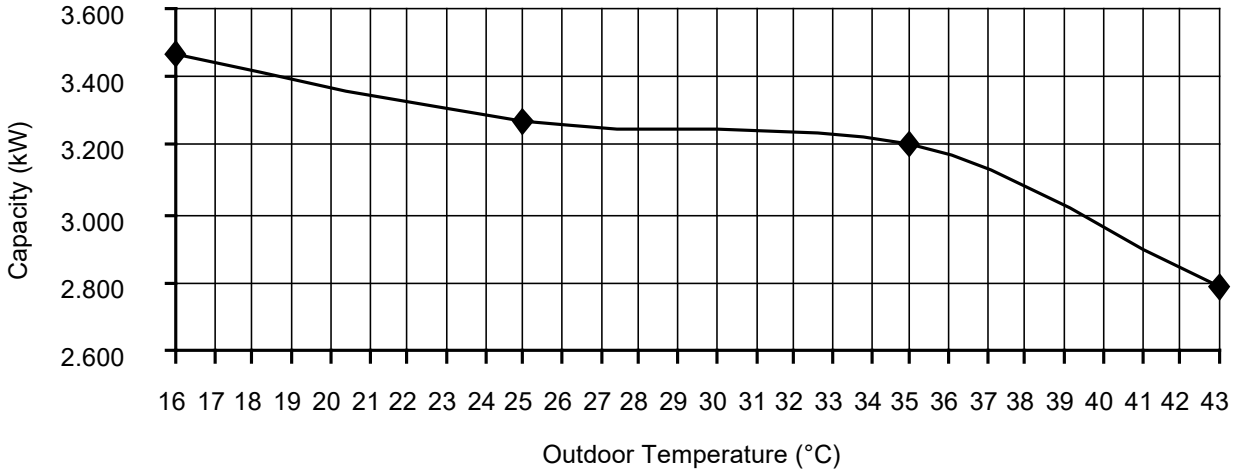
Piping length : 7 m

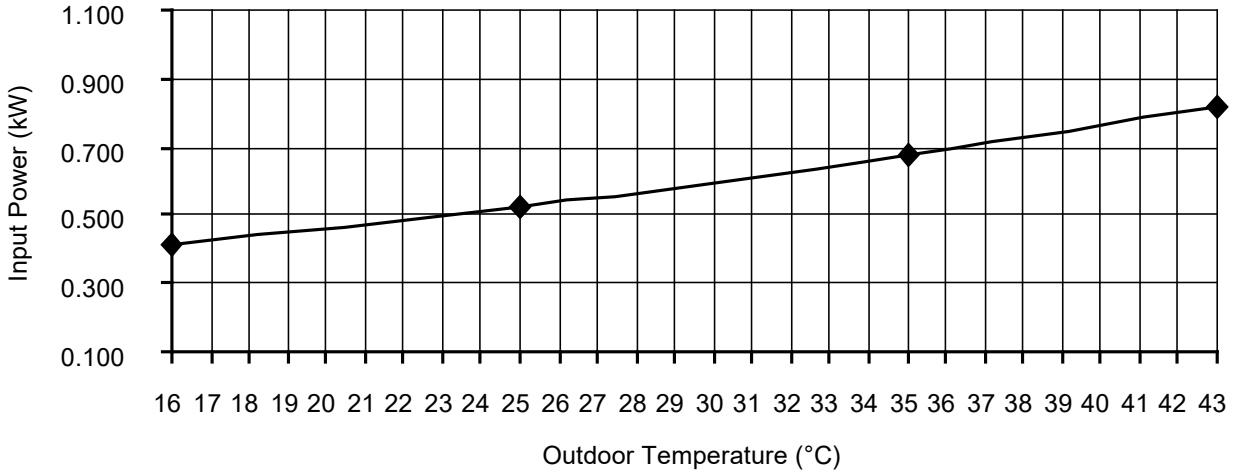




Cooling Characteristics at Different Outdoor Air Temperature

Condition
Outdoor air temperature : 35°C (DBT), -°C (WBT)
Indoor water inlet temperature : 23°C
Indoor water outlet temperature : 18°C
Piping length : 7 m





Heating Characteristics at Different Piping Length

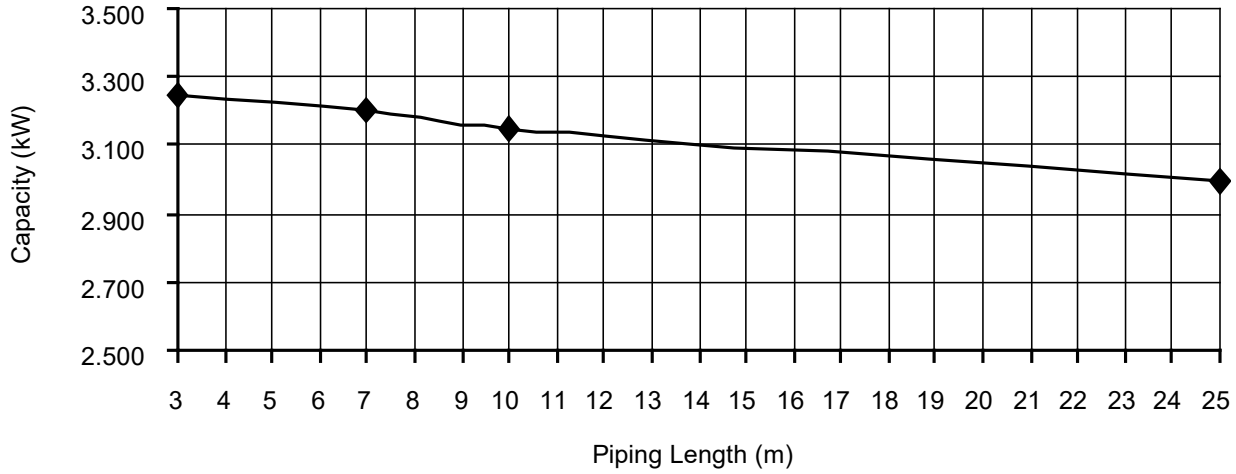
Condition

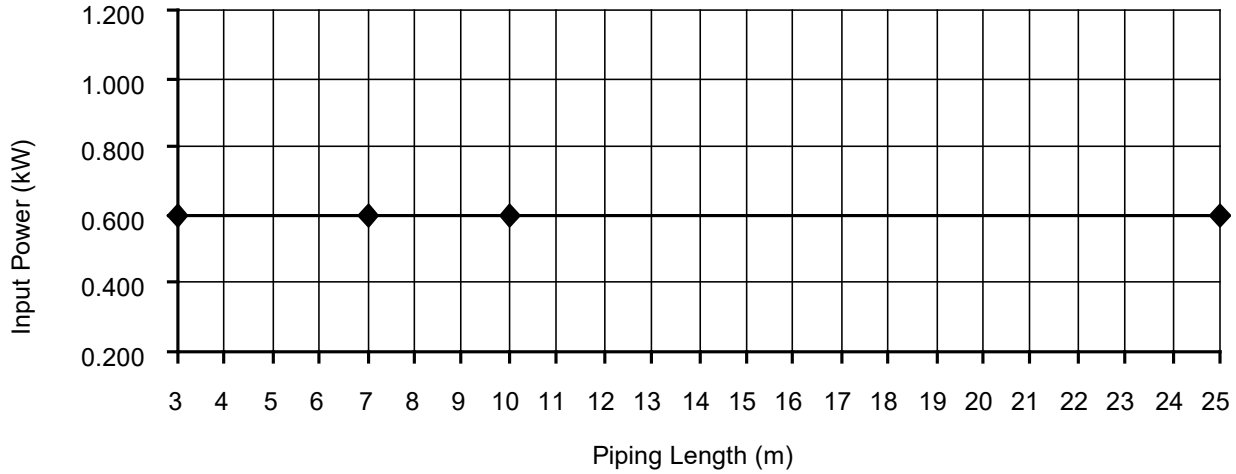
Outdoor air temperature : 7°C (DBT), 6°C (WBT)

Indoor water inlet temperature : 30°C

Indoor water outlet temperature : 35°C

Piping length : 7 m





Cooling Characteristics at Different Piping Length

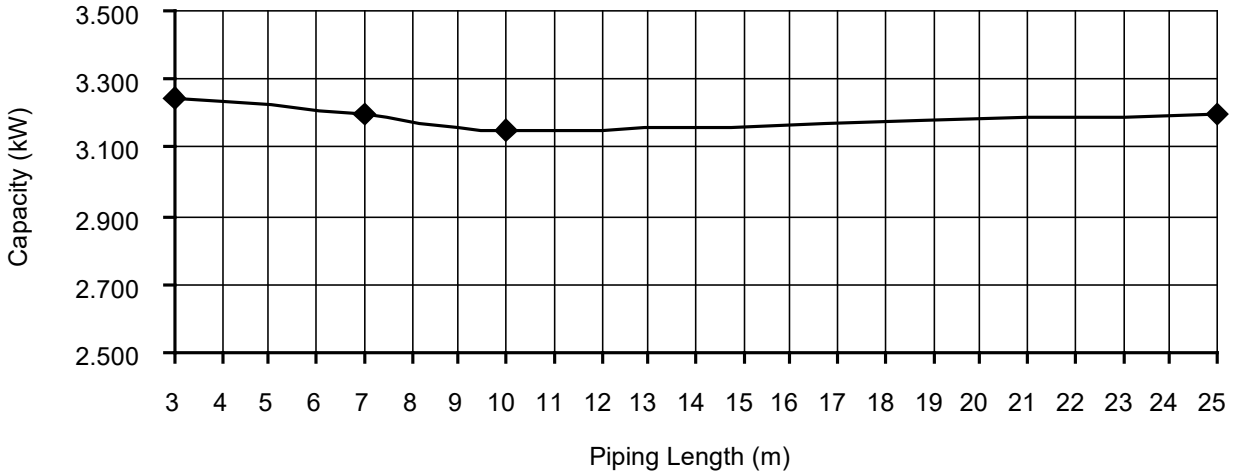
Condition

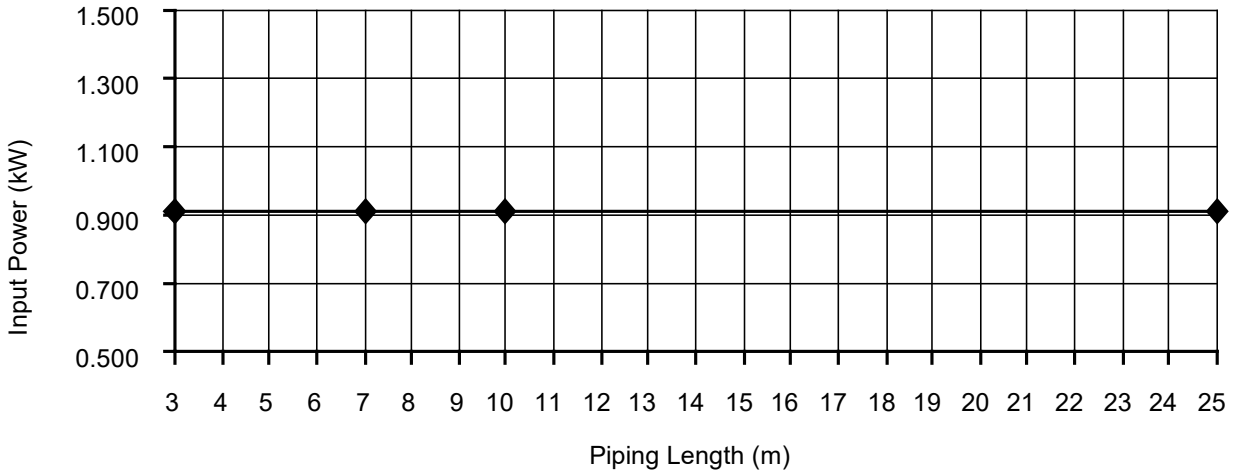
Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 12°C

Indoor water outlet temperature : 7°C

Piping length : 7 m

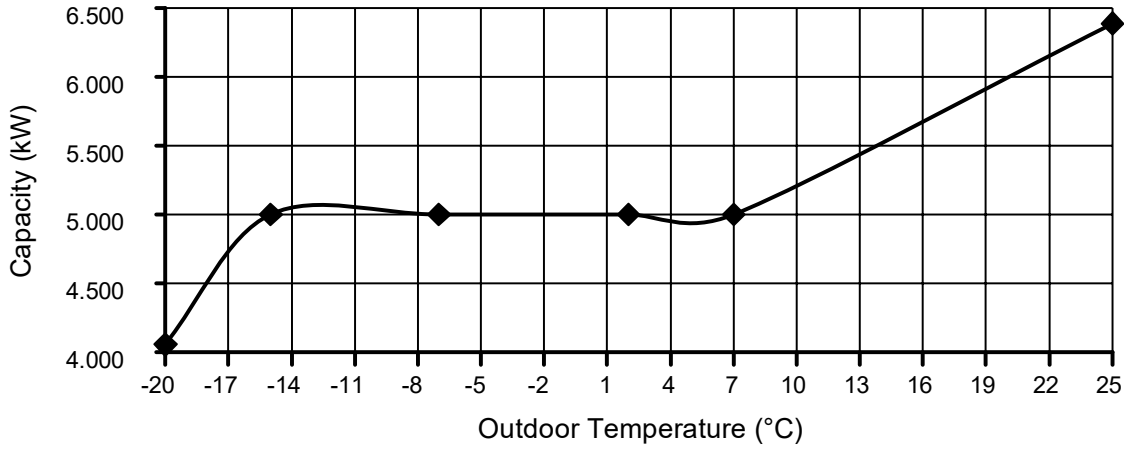


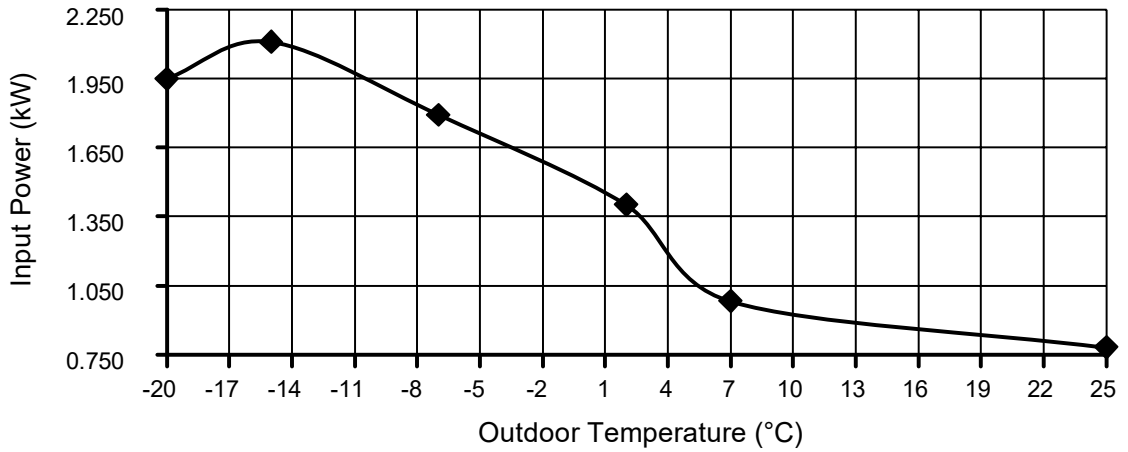


21.1.2 WH-UDZ05KE5

Heating Characteristics at Different Outdoor Air Temperature

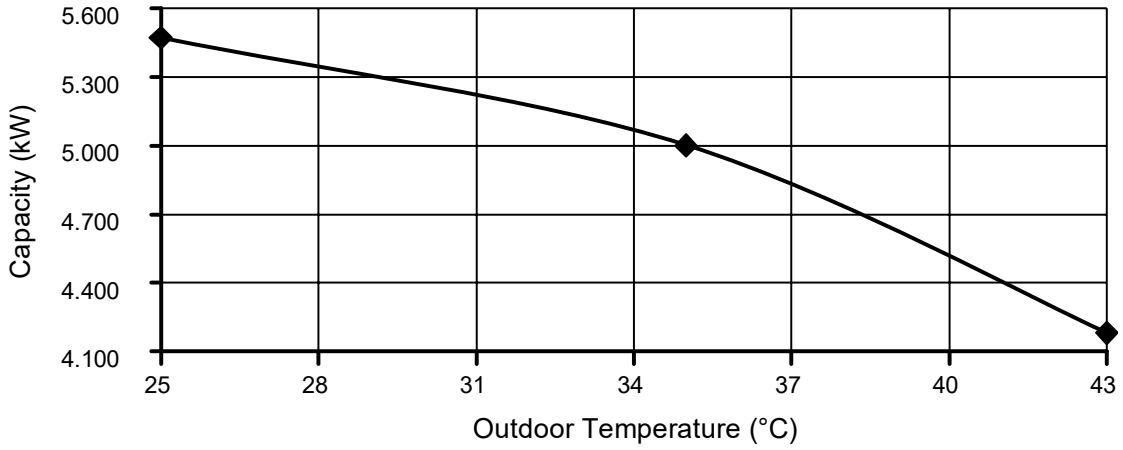
Condition
Outdoor air temperature : 7°C (DBT), 6°C (WBT)
Indoor water inlet temperature : 30°C
Indoor water outlet temperature : 35°C
Piping length : 7 m

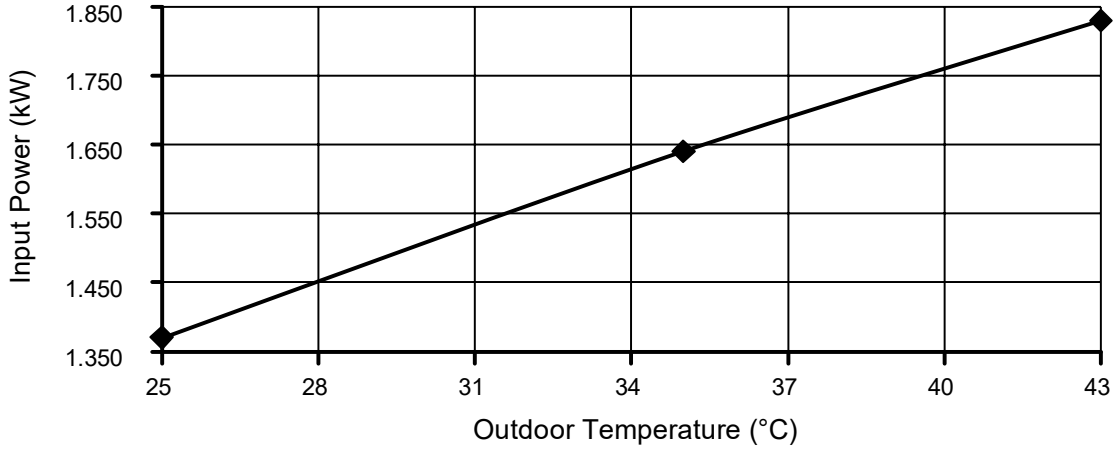




Cooling Characteristics at Different Outdoor Air Temperature

Condition
Outdoor air temperature : 35°C (DBT), -°C (WBT)
Indoor water inlet temperature : 12°C
Indoor water outlet temperature : 7°C
Piping length : 7 m





Cooling Characteristics at Different Outdoor Air Temperature

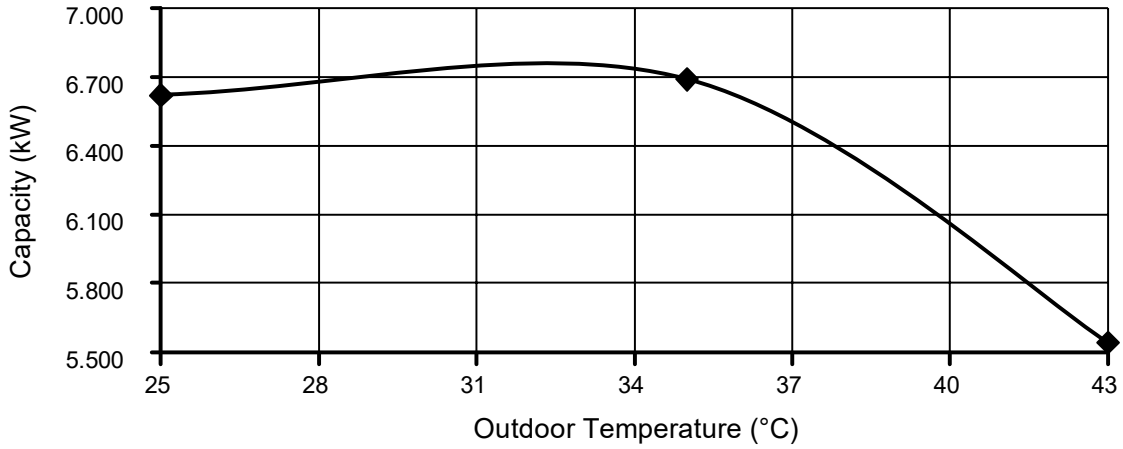
Condition

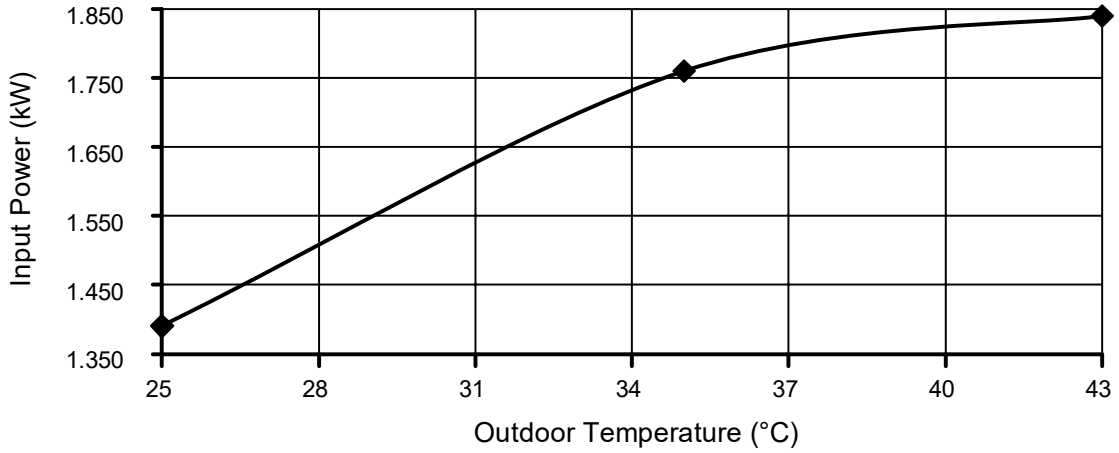
Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 19°C

Indoor water outlet temperature : 14°C

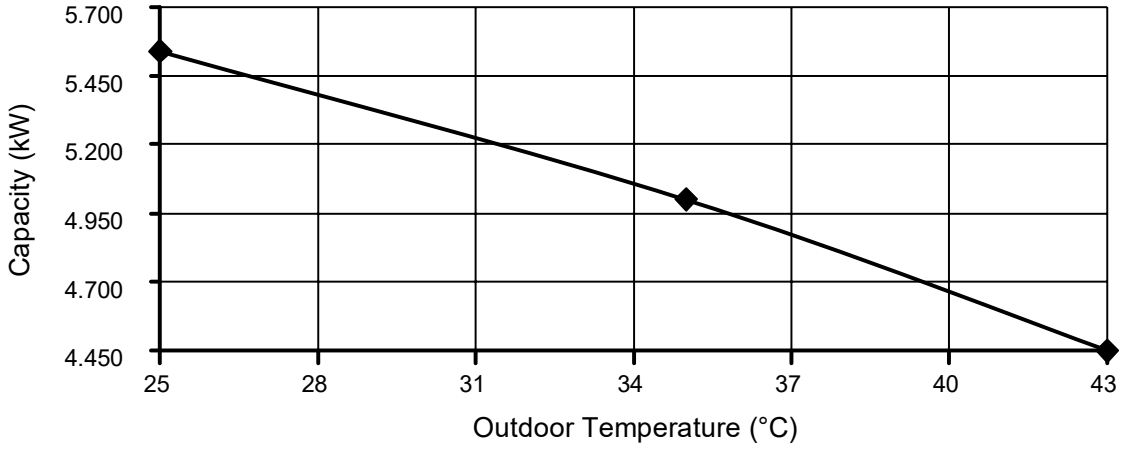
Piping length : 7 m

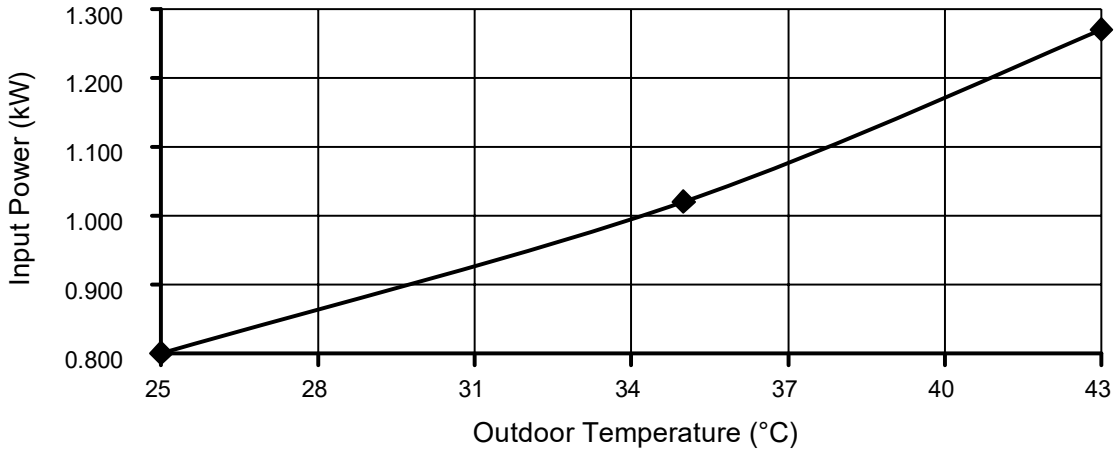




Cooling Characteristics at Different Outdoor Air Temperature

Condition
Outdoor air temperature : 35°C (DBT), -°C (WBT)
Indoor water inlet temperature : 23°C
Indoor water outlet temperature : 18°C
Piping length : 7 m





Heating Characteristics at Different Piping Length

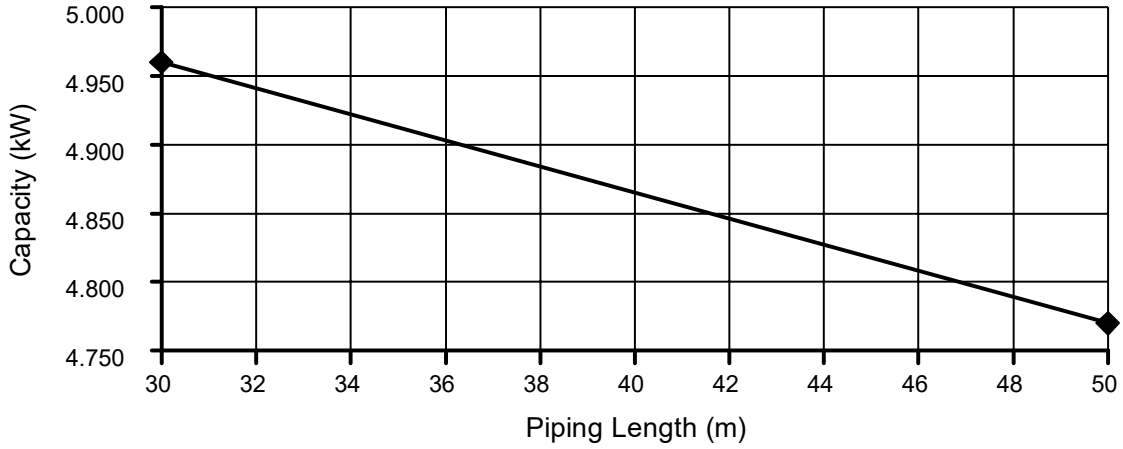
Condition

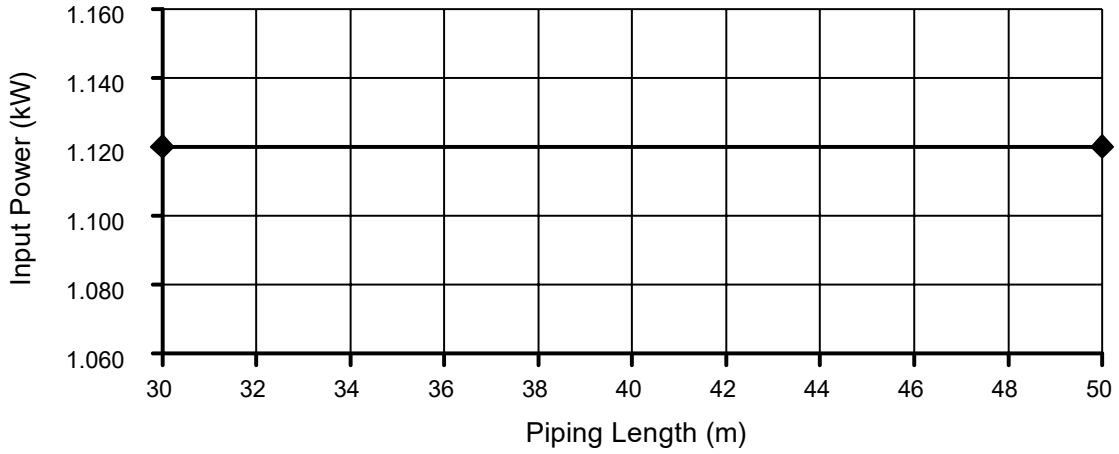
Outdoor air temperature : 7°C (DBT), 6°C (WBT)

Indoor water inlet temperature : 30°C

Indoor water outlet temperature : 35°C

Piping length : 30 m/50 m





Cooling Characteristics at Different Piping Length

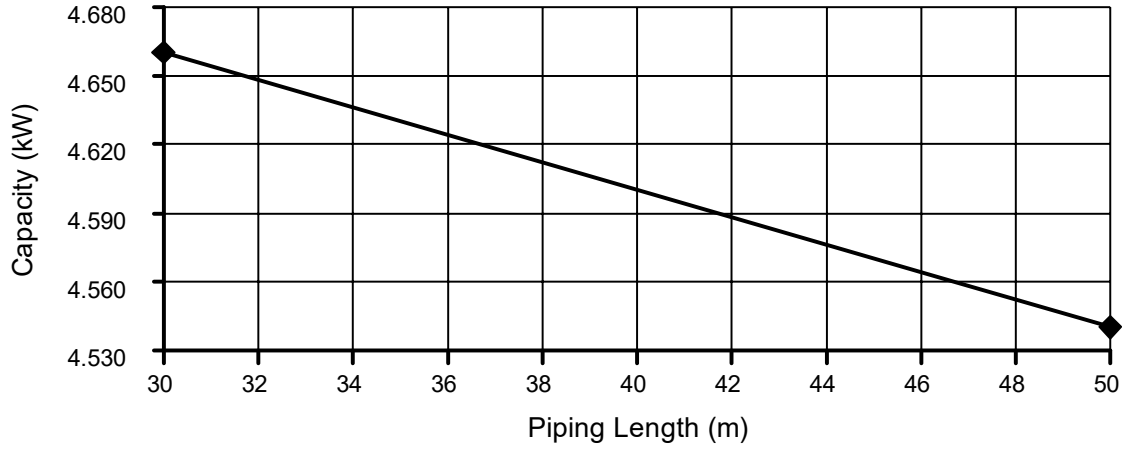
Condition

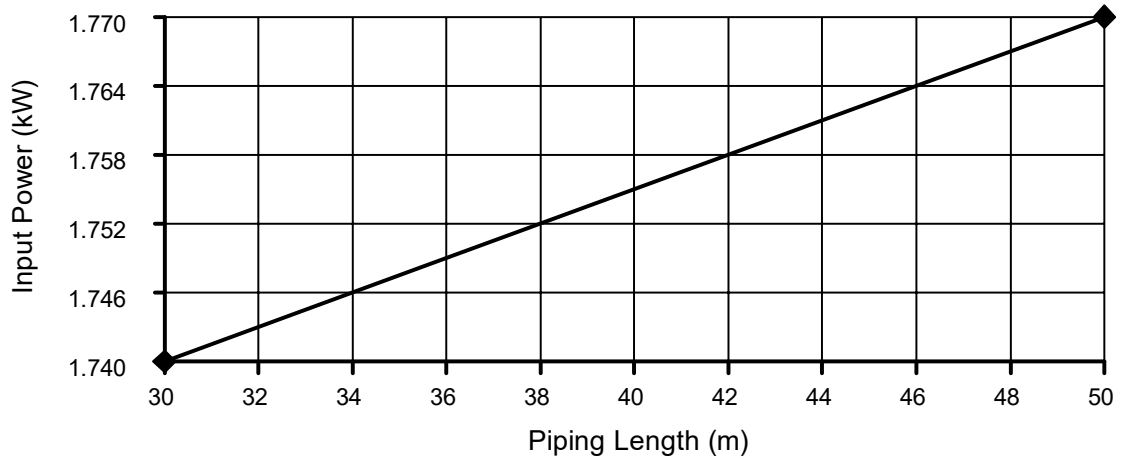
Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 12°C

Indoor water outlet temperature : 7°C

Piping length : 30 m/50 m

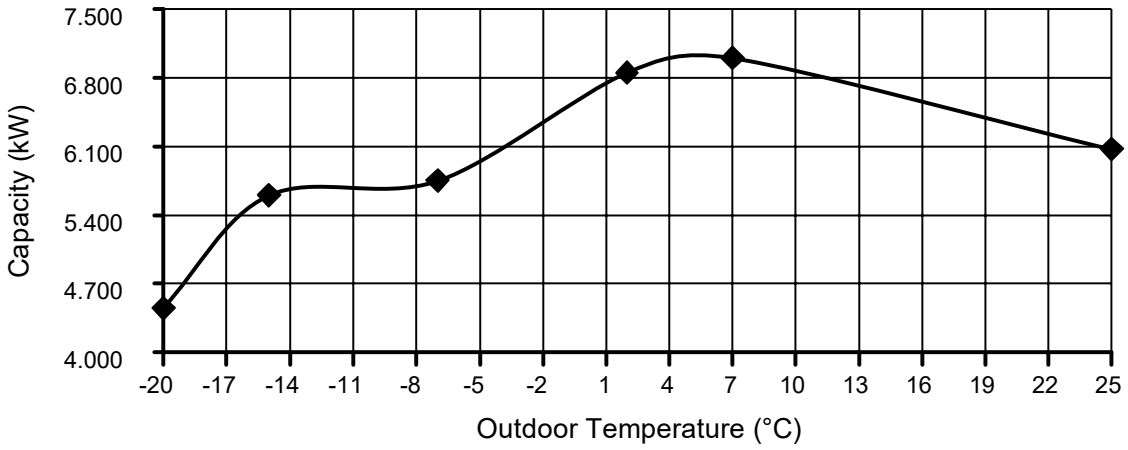


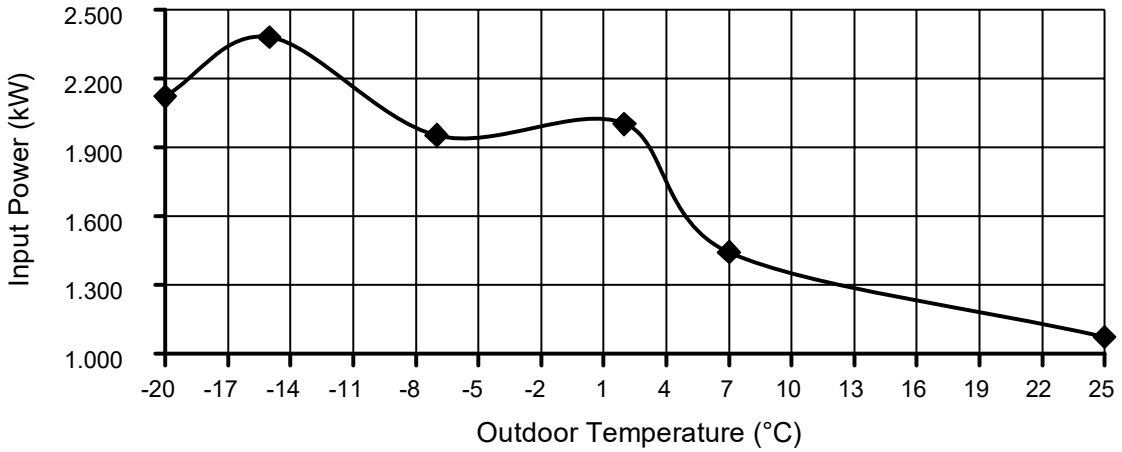


21.1.3 WH-UDZ07KE5

Heating Characteristics at Different Outdoor Air Temperature

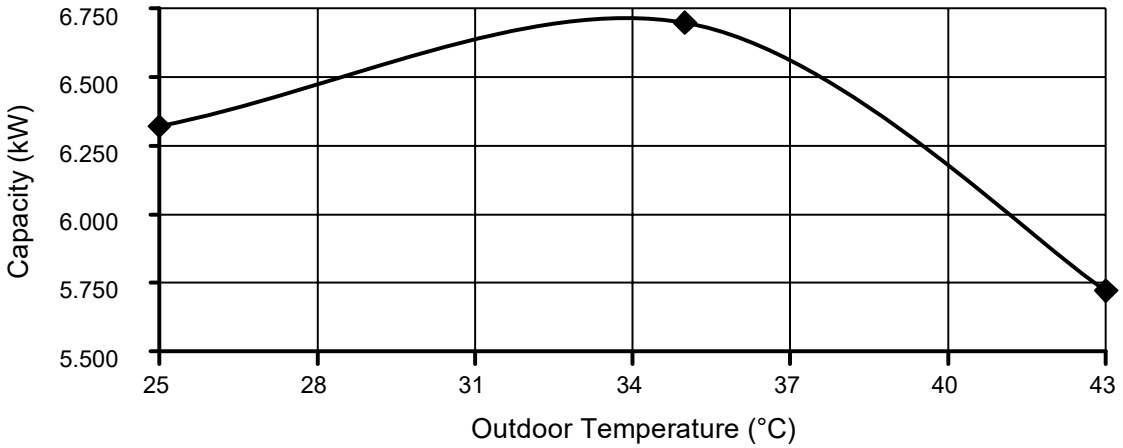
Condition
Outdoor air temperature : 7°C (DBT), 6°C (WBT)
Indoor water inlet temperature : 30°C
Indoor water outlet temperature : 35°C
Piping length : 7 m

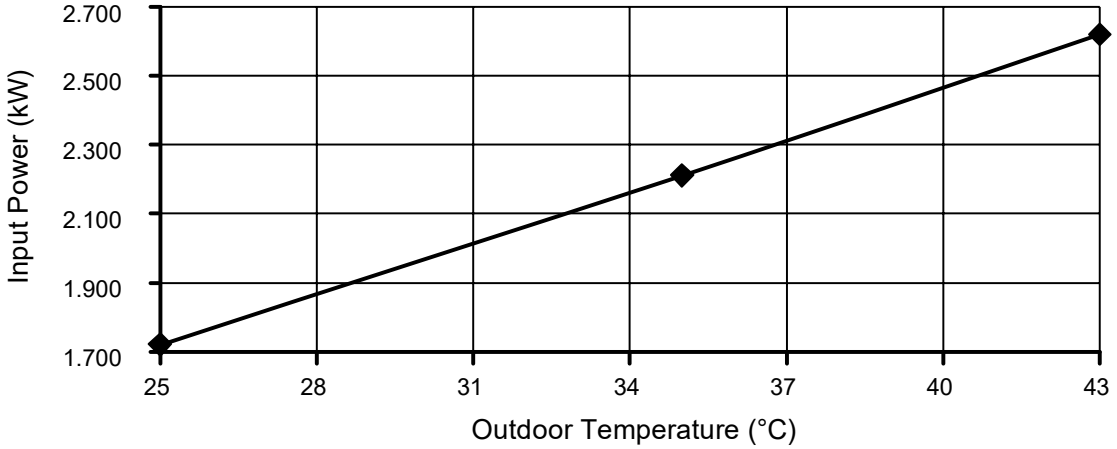




Cooling Characteristics at Different Outdoor Air Temperature

Condition
Outdoor air temperature : 35°C (DBT), -°C (WBT)
Indoor water inlet temperature : 12°C
Indoor water outlet temperature : 7°C
Piping length : 7 m





Cooling Characteristics at Different Outdoor Air Temperature

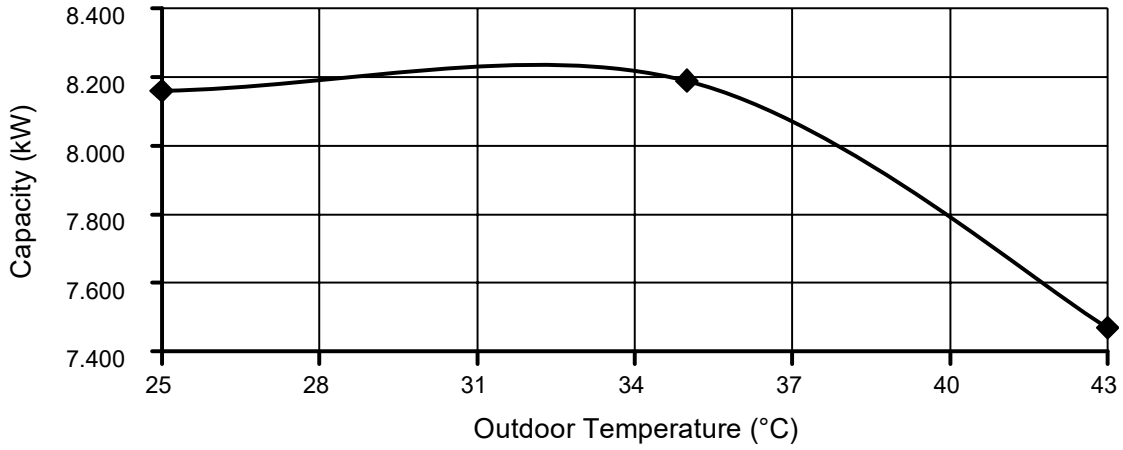
Condition

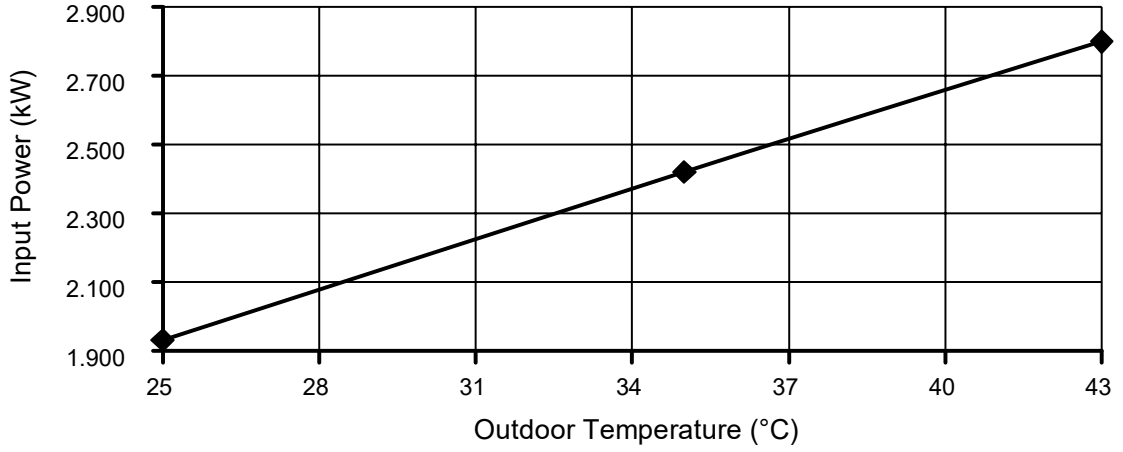
Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 19°C

Indoor water outlet temperature : 14°C

Piping length : 7 m





Cooling Characteristics at Different Outdoor Air Temperature

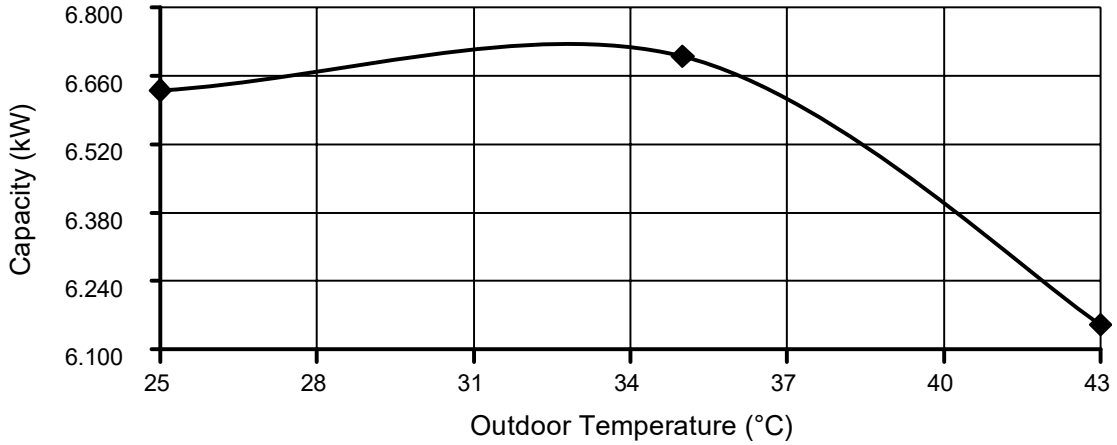
Condition

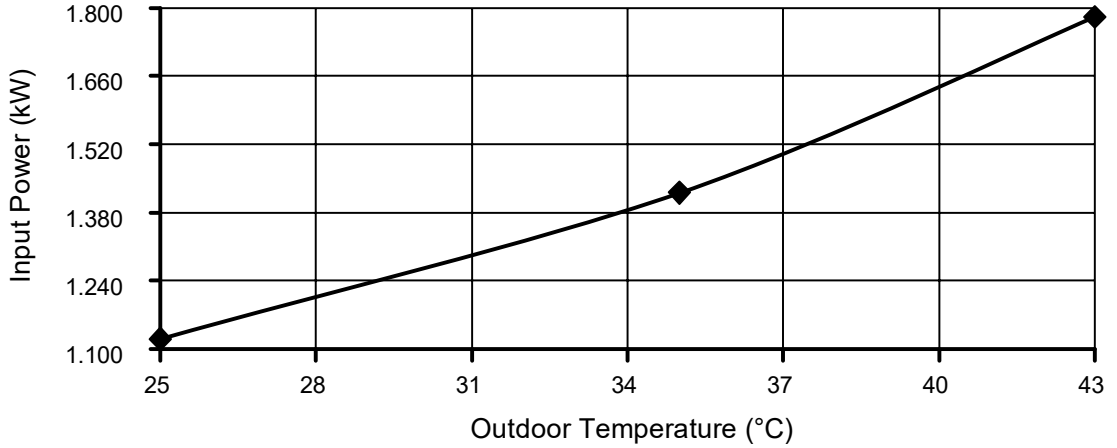
Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 23°C

Indoor water outlet temperature : 18°C

Piping length : 7 m





Heating Characteristics at Different Piping Length

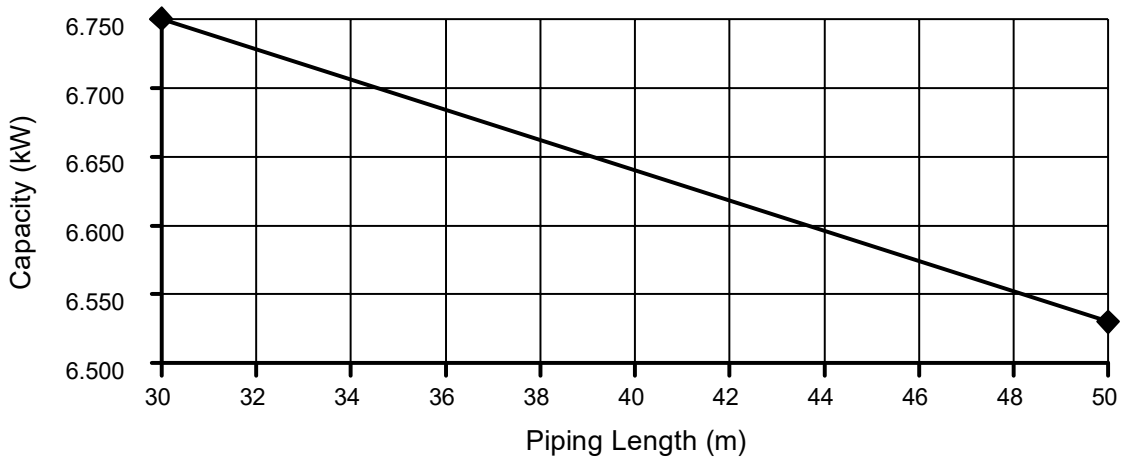
Condition

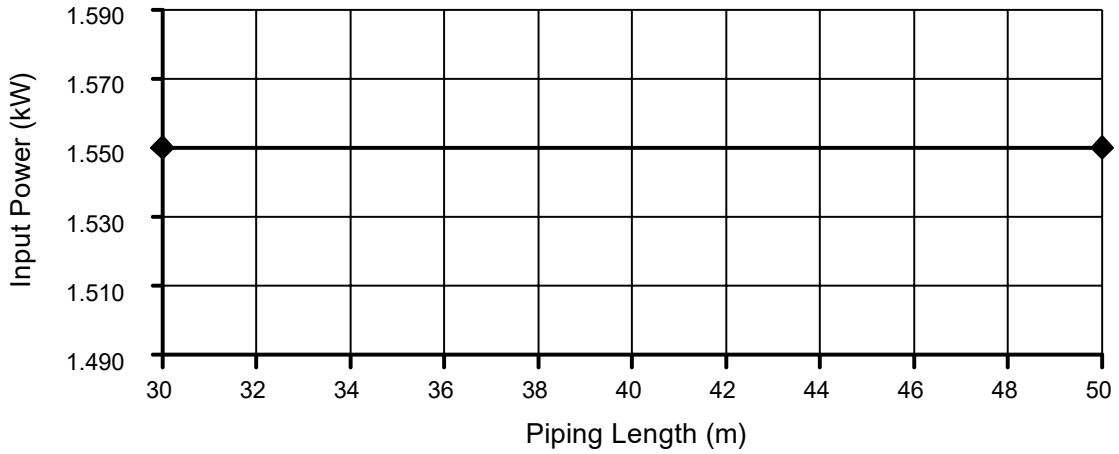
Outdoor air temperature : 7°C (DBT), 6°C (WBT)

Indoor water inlet temperature : 30°C

Indoor water outlet temperature : 35°C

Piping length : 30 m/50 m





Cooling Characteristics at Different Piping Length

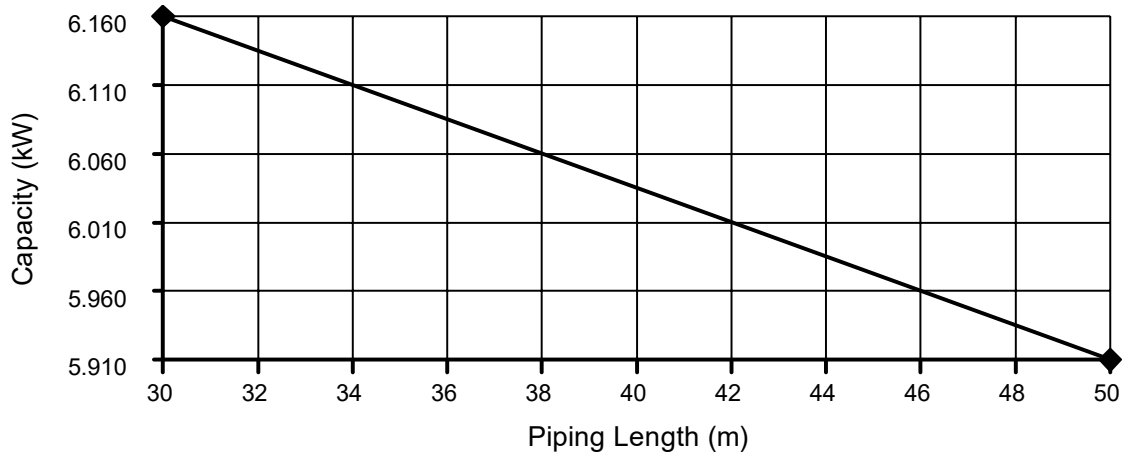
Condition

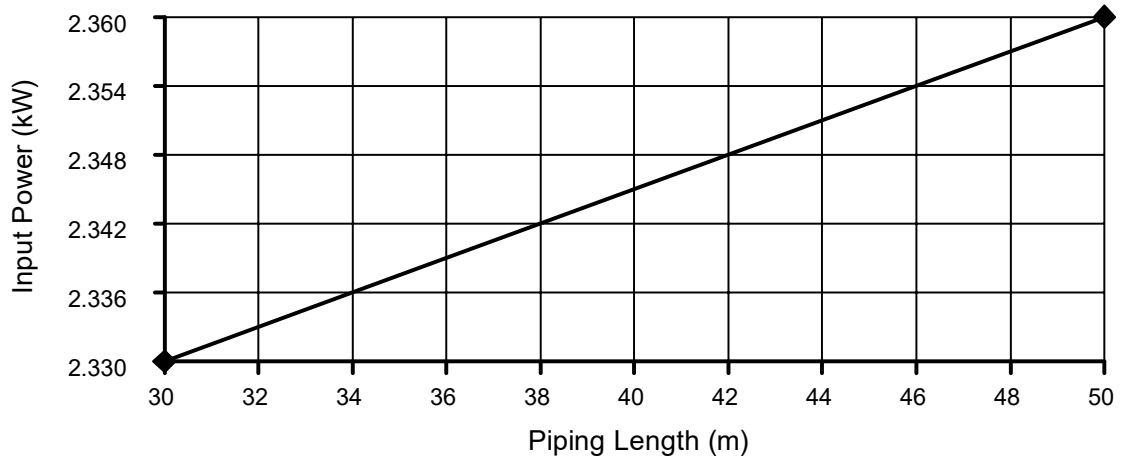
Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 12°C

Indoor water outlet temperature : 7°C

Piping length : 30 m/50 m

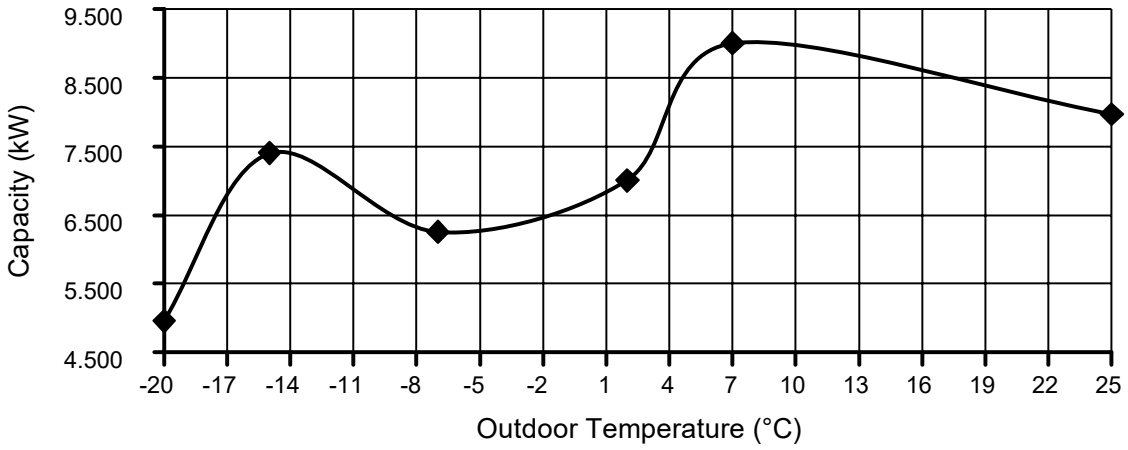


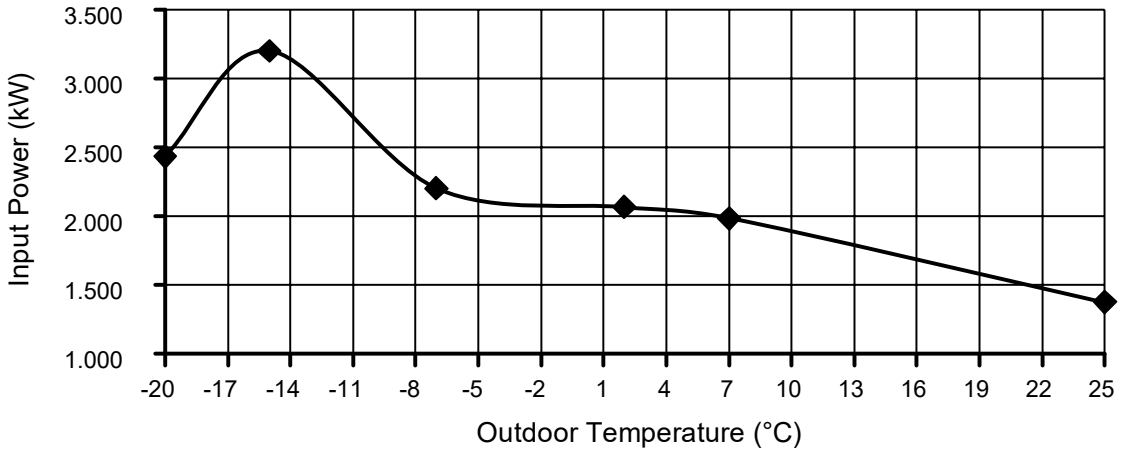


21.1.4 WH-UDZ09KE5

Heating Characteristics at Different Outdoor Air Temperature

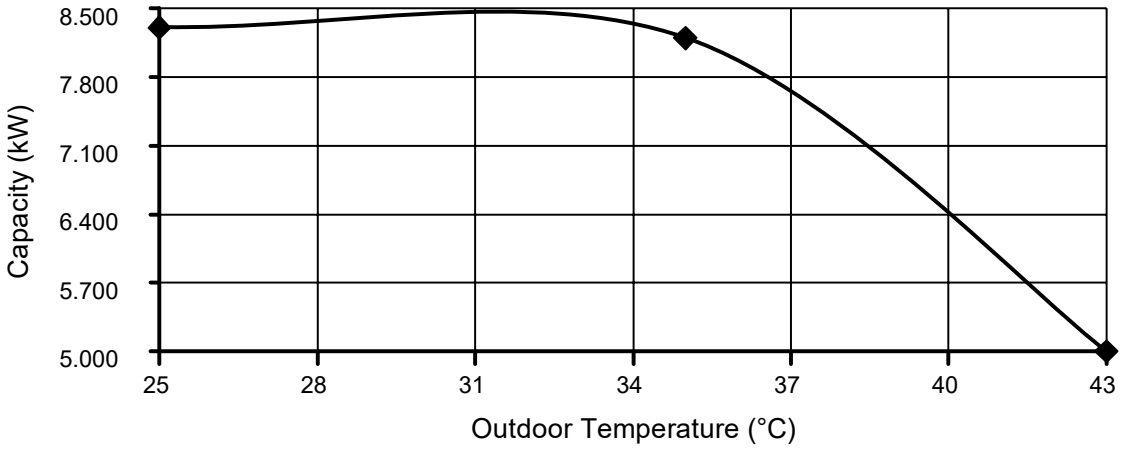
Condition
Outdoor air temperature : 7°C (DBT), 6°C (WBT)
Indoor water inlet temperature : 30°C
Indoor water outlet temperature : 35°C
Piping length : 7 m

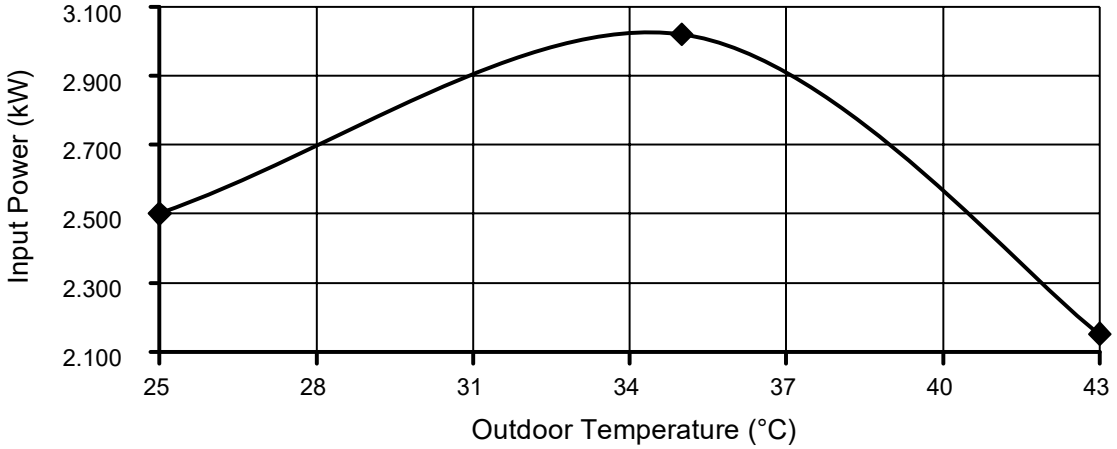




Cooling Characteristics at Different Outdoor Air Temperature

Condition
Outdoor air temperature : 35°C (DBT), -°C (WBT)
Indoor water inlet temperature : 12°C
Indoor water outlet temperature : 7°C
Piping length : 7 m





Cooling Characteristics at Different Outdoor Air Temperature

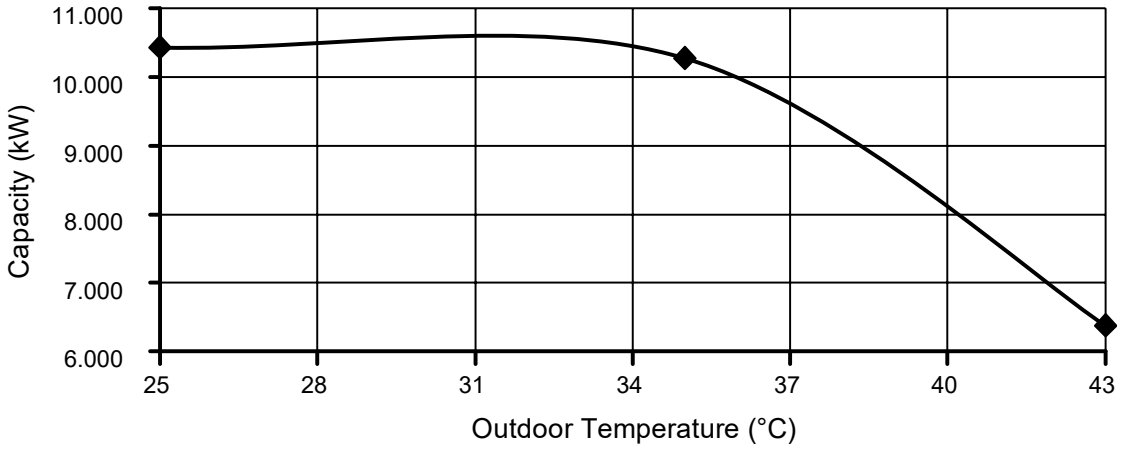
Condition

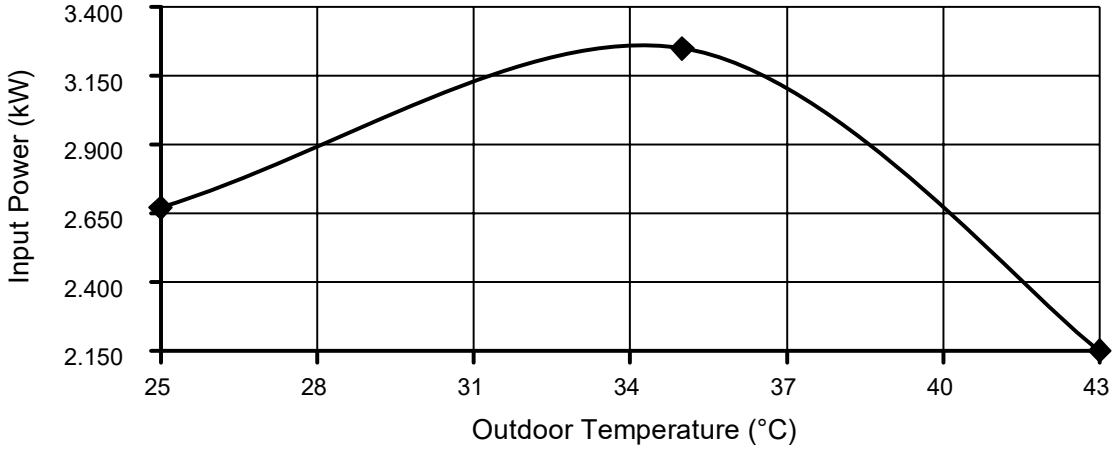
Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 19°C

Indoor water outlet temperature : 14°C

Piping length : 7 m





Cooling Characteristics at Different Outdoor Air Temperature

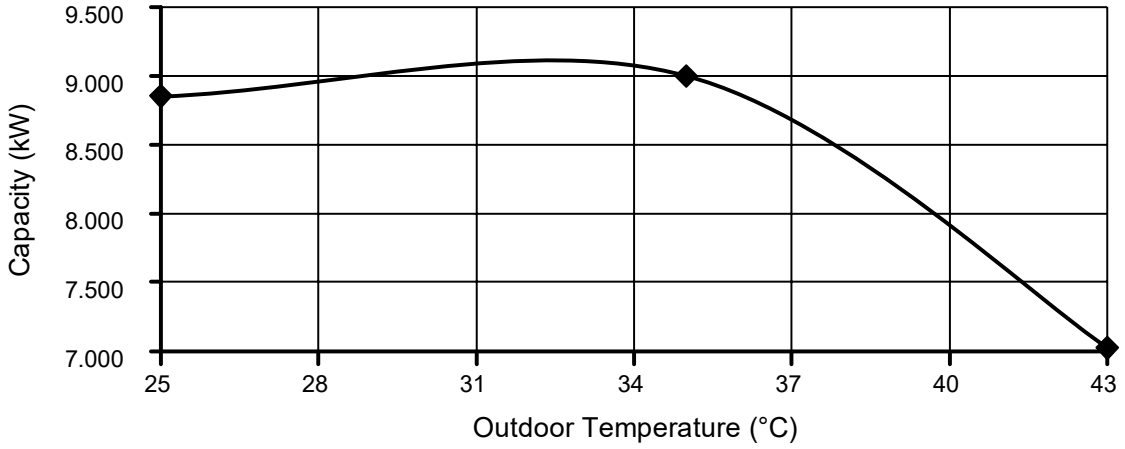
Condition

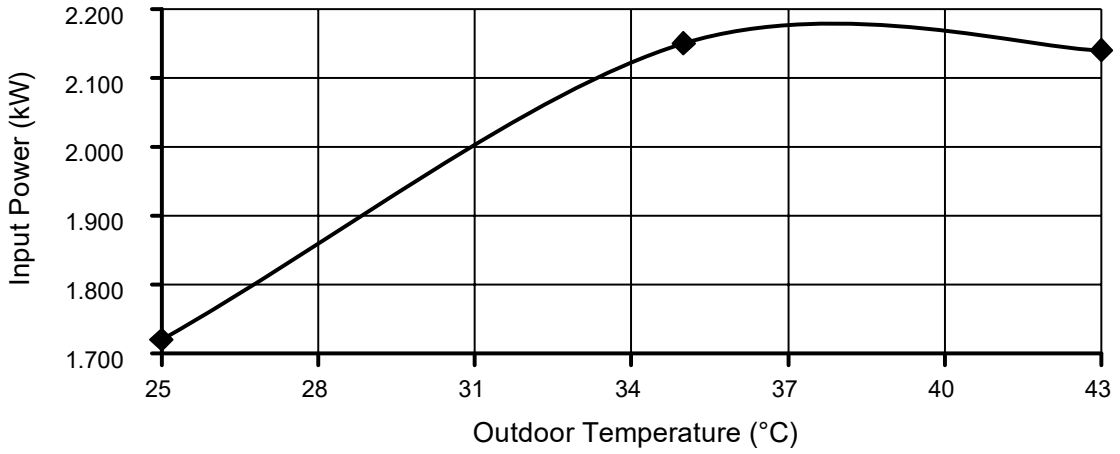
Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 23°C

Indoor water outlet temperature : 18°C

Piping length : 7 m





Heating Characteristics at Different Piping Length

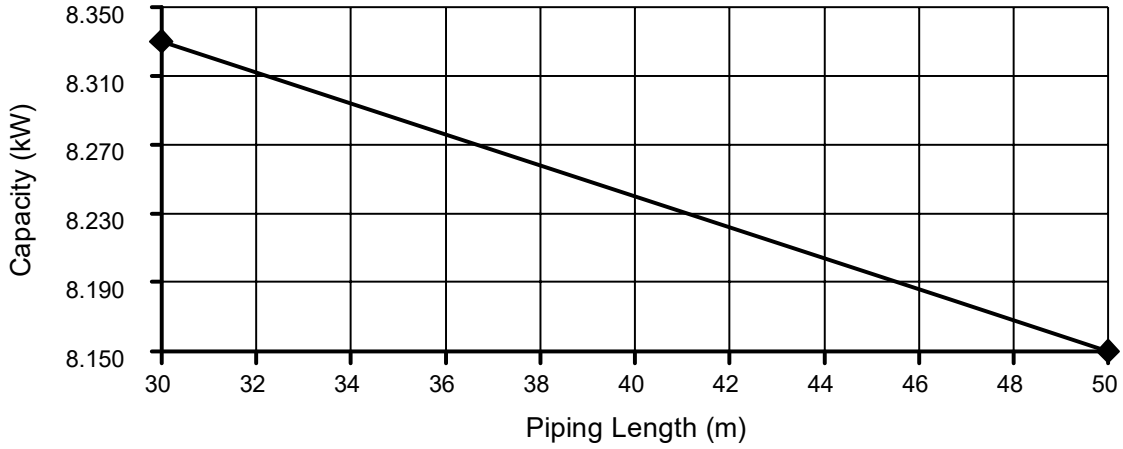
Condition

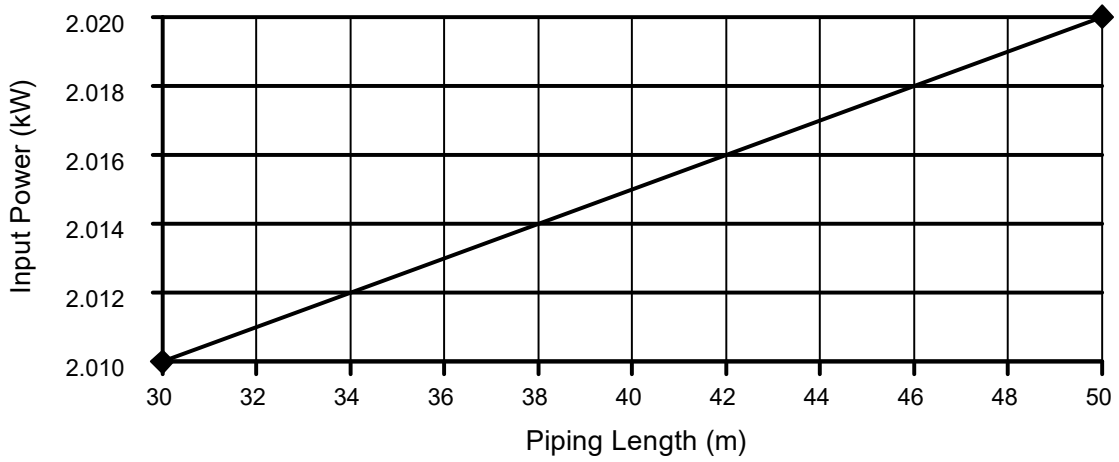
Outdoor air temperature : 7°C (DBT), 6°C (WBT)

Indoor water inlet temperature : 30°C

Indoor water outlet temperature : 35°C

Piping length : 30 m/50 m





Cooling Characteristics at Different Piping Length

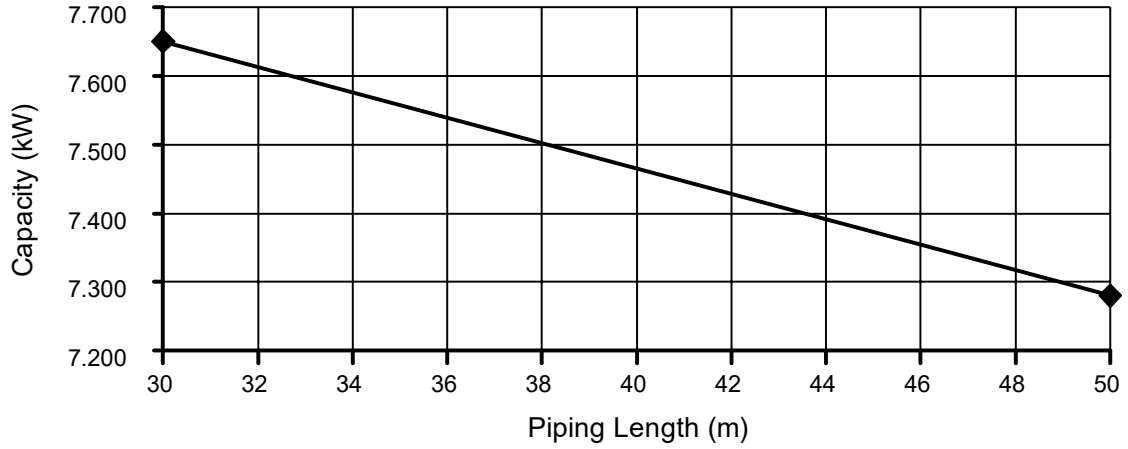
Condition

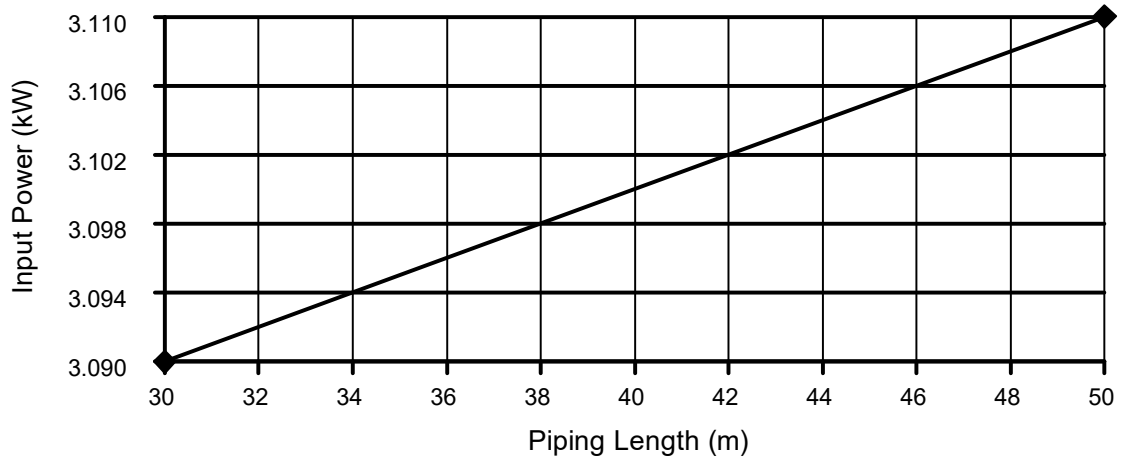
Outdoor air temperature : 35°C (DBT), -°C (WBT)

Indoor water inlet temperature : 12°C

Indoor water outlet temperature : 7°C

Piping length : 30 m/50 m





21.2 Heating Capacity Table

21.2.1 WH-UDZ03KE5

Water Out (°C)	25		35		45		55		60	
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)
-20	2500	1110	2520	1310	2240	1590	2120	1800	-	-
-15	3000	1140	3200	1370	3000	1620	2750	1920	-	-
-7	2990	910	3300	1180	3250	1470	3200	1790	3000	1880
2	2920	690	3200	880	3200	1130	3200	1460	3150	1670
7	3090	490	3200	600	3200	840	3200	1140	2950	1220
25	3270	230	3270	380	3610	630	4060	1110	4030	1140

21.2.2 WH-UDZ05KE5

Water Out (°C)	35		45		55		60	
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)
-20	4050	1950	3760	2200	3390	2480	-	-
-15	5000	2110	4750	2490	4300	2610	-	-
-7	5000	1790	5000	2140	5000	2650	4680	2710
2	5000	1400	5000	1790	5000	2180	4800	2400
7	5000	980	5000	1310	5000	1650	4580	1900
25	6390	780	6170	1190	6530	1580	5150	1330

21.2.3 WH-UDZ07KE5

Water Out (°C)	35		45		55		60	
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)
-20	4450	2120	4230	2480	3900	2850	-	-
-15	5600	2380	5300	2780	5000	3200	-	-
-7	5750	1950	5650	2300	5350	2700	4980	2900
2	6850	2000	6750	2400	6250	2800	6180	2910
7	7000	1440	7000	1920	7000	2400	6860	2730
25	6070	1070	6670	1500	7150	1770	6880	1900

21.2.4 WH-UDZ09KE5

Water Out (°C)	35		45		55		60	
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)
-20	4950	2430	4580	2700	4040	3000	-	-
-15	7400	3200	6450	3280	5400	3420	-	-
-7	6250	2200	6100	2680	5900	3060	5650	3240
2	7000	2060	6850	2500	6300	2890	7260	3310
7	9000	1980	9000	2580	8900	3040	8600	3420
25	7970	1370	8750	1930	9160	2340	6080	1720

21.3 Cooling Capacity Table

21.3.1 WH-UDZ03KE5

Water Out (°C)	7		14		18	
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)
16	3560	570	4320	550	3470	410
25	3290	730	4060	720	3270	520
35	3200	910	3560	930	3200	680
43	2680	1060	3340	1090	2790	820

21.3.2 WH-UDZ05KE5

Water Out (°C)	7		14		18	
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)
25	5470	1370	6620	1390	5540	800
35	5000	1640	6690	1760	5000	1020
43	4180	1830	5540	1840	4450	1270

21.3.3 WH-UDZ07KE5

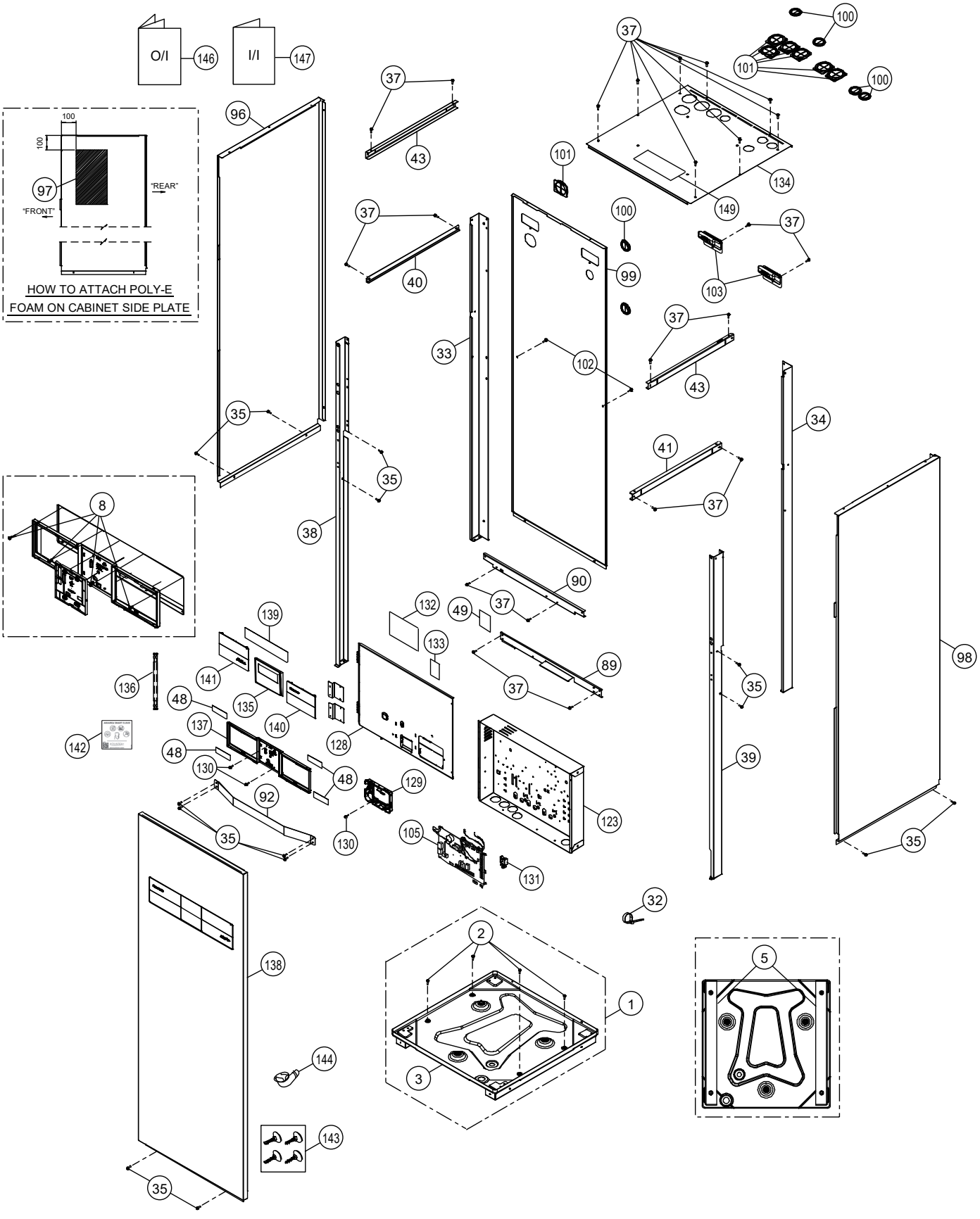
Water Out (°C)	7		14		18	
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)
25	6320	1720	8160	1930	6630	1120
35	6700	2210	8190	2420	6700	1420
43	5720	2620	7470	2800	6150	1780

21.3.4 WH-UDZ09KE5

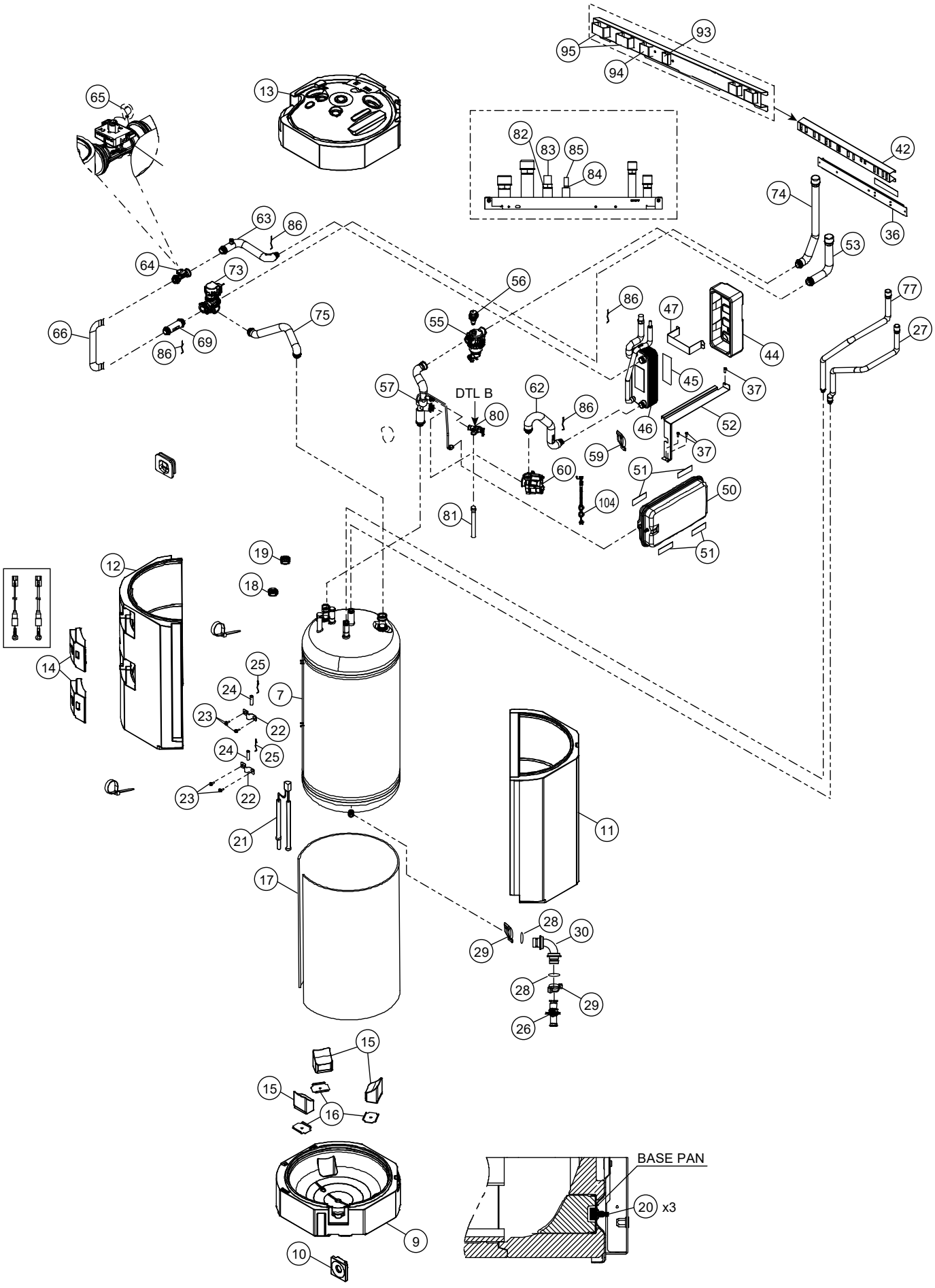
Water Out (°C)	7		14		18	
Outdoor Air (°C)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)	Capacity (W)	Input Power (W)
25	8310	2500	10430	2670	8850	1720
35	8200	3020	10280	3250	9000	2150
43	5000	2150	6380	2150	7020	2140

22. Exploded View and Replacement Parts List

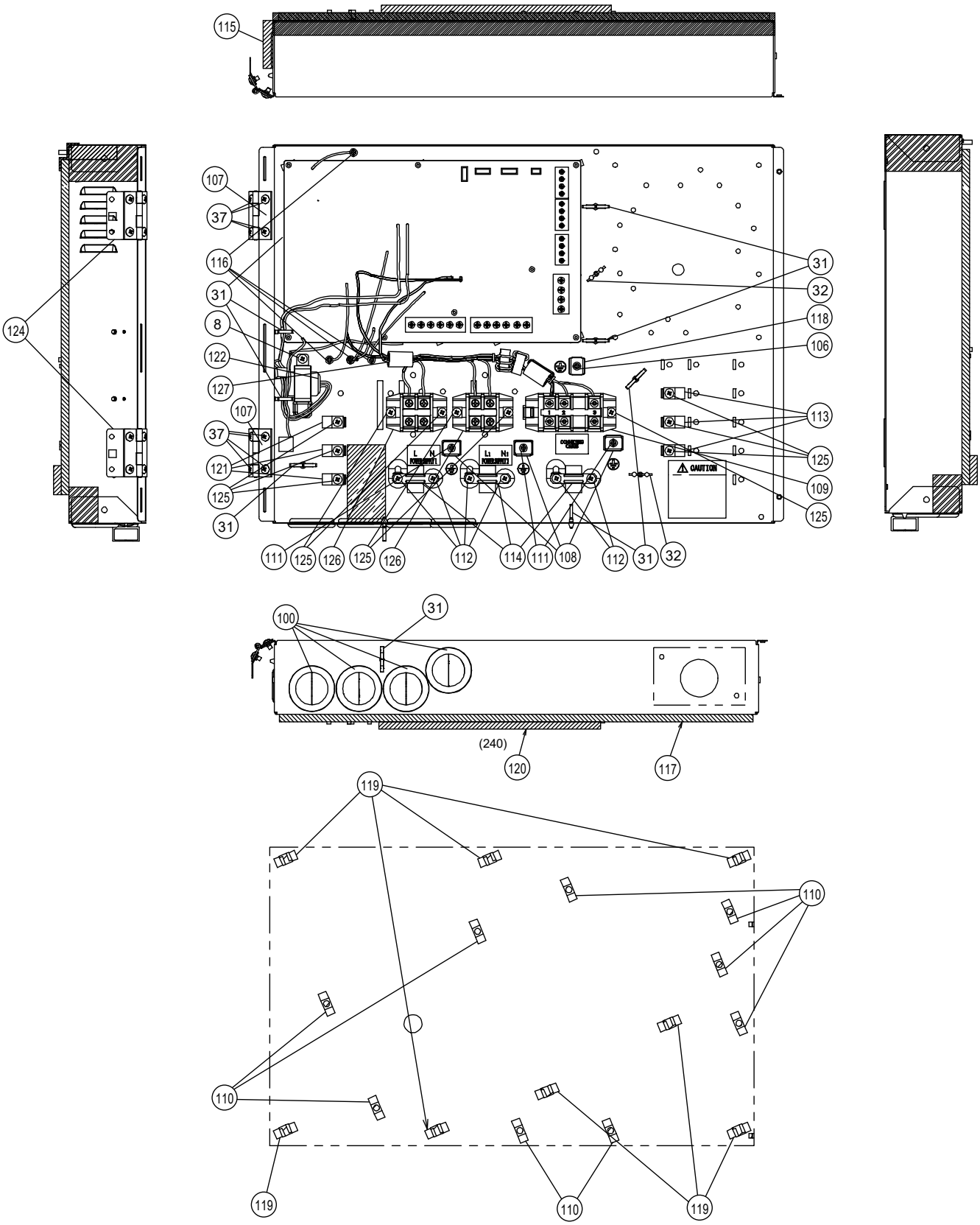
22.1 Indoor Unit



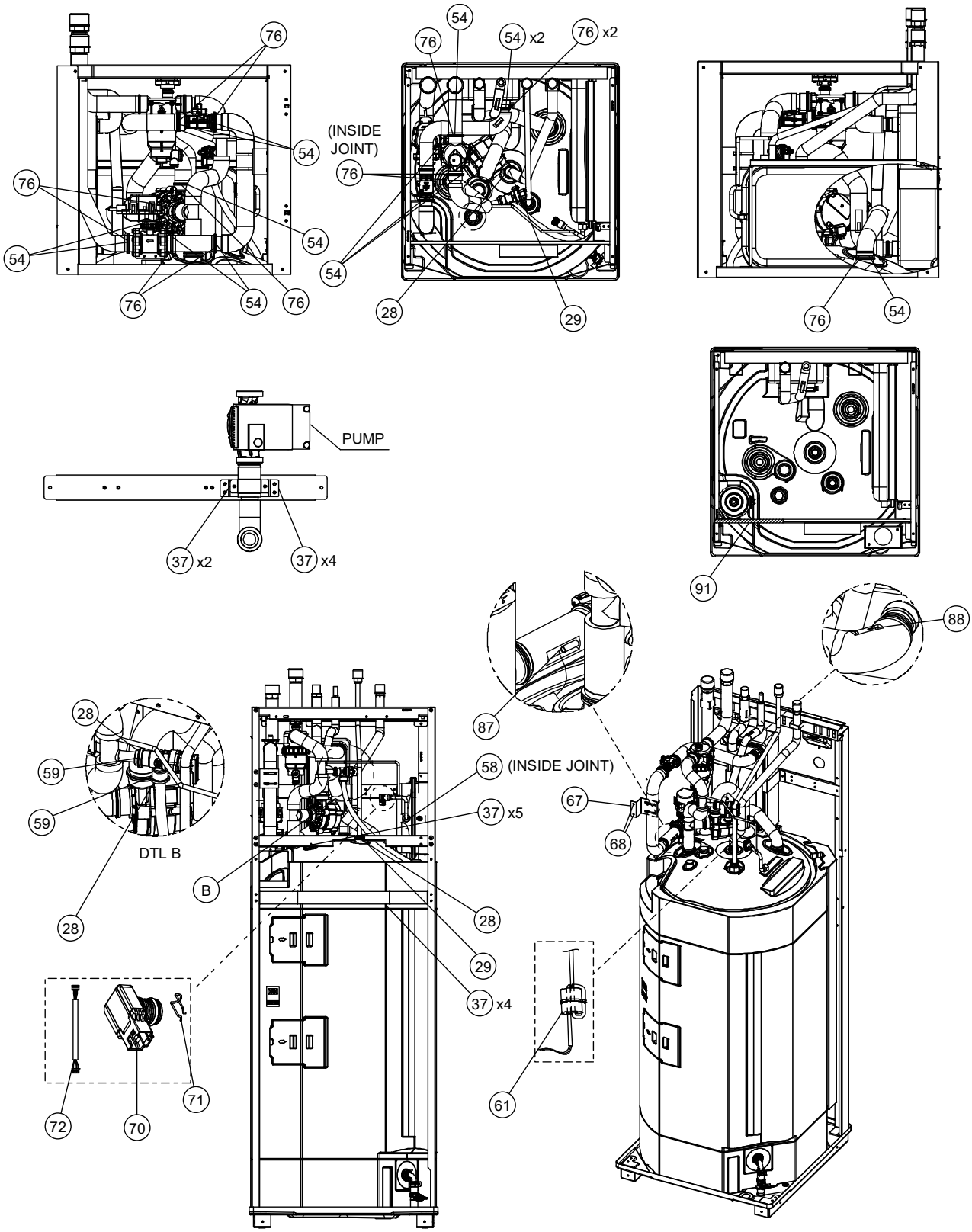
Note:
The above exploded view is for the purpose of parts disassembly and replacement.
The non-numbered parts are not kept as standard service parts.



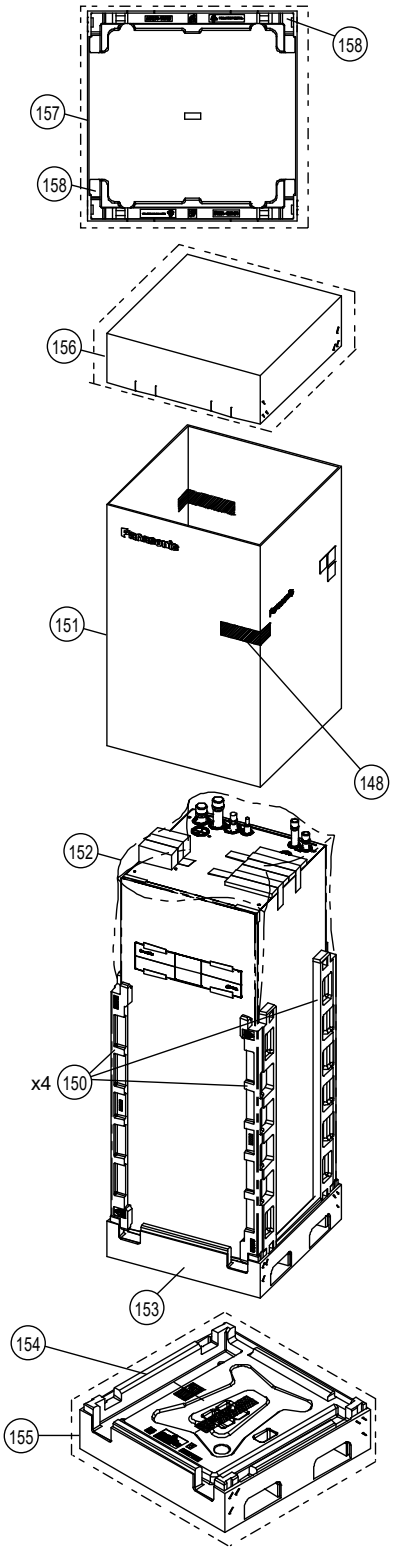
Note:
The above exploded view is for the purpose of parts disassembly and replacement.
The non-numbered parts are not kept as standard service parts.



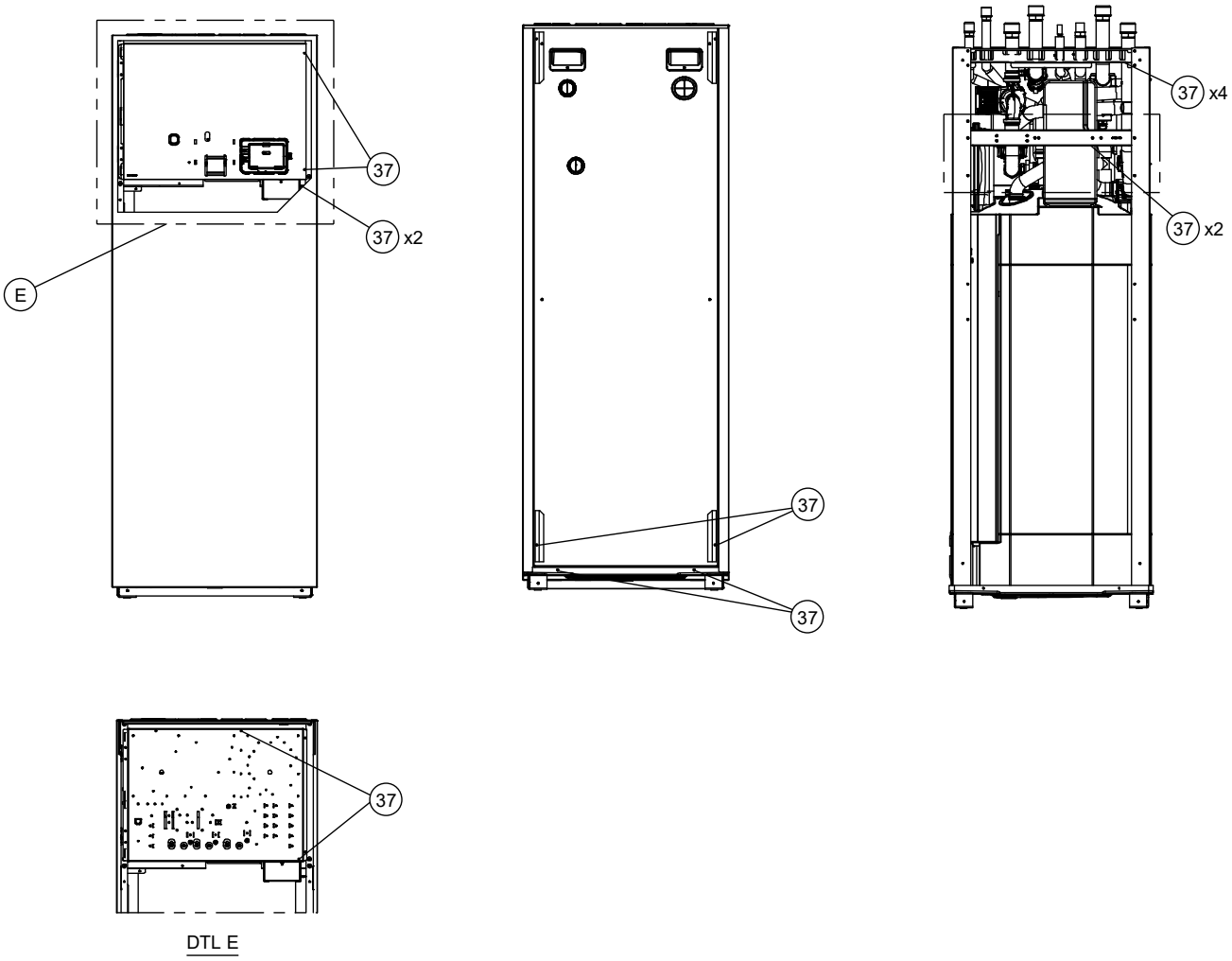
Note:
The above exploded view is for the purpose of parts disassembly and replacement.
The non-numbered parts are not kept as standard service parts.



Note:
The above exploded view is for the purpose of parts disassembly and replacement.
The non-numbered parts are not kept as standard service parts.



Note:
The above exploded view is for the purpose of parts disassembly and replacement.
The non-numbered parts are not kept as standard service parts.



DTL E

Note:
The above exploded view is for the purpose of parts disassembly and replacement.
The non-numbered parts are not kept as standard service parts.

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-ADC0309K0E5A	REMARK
	1	CHASSIS - COMPLETE	1	ACXD50C05890	
	2	SCREW	4	H55451J	
	3	BASE PAN ASSY	1	ACXD52K05960	
	5	BRACKET - UNIT MOUNTING COMPLETE	2	ACXD57C00420	
⚠	7	TANK COMPLETE	1	ACXB56C01211	
	8	SELF TAPPING SCREW	5	XTB4+8FFJ	
	9	FOAMED POLYSTYRENE	1	ACXG07-07690	
	10	FOAMED POLYSTYRENE	1	ACXG07-07700	
	11	FOAMED POLYSTYRENE	1	ACXG07-07710	
	12	FOAMED POLYSTYRENE	1	ACXG07-07720	
	13	FOAMED POLYSTYRENE	1	ACXG07-07731	
	14	FOAMED POLYSTYRENE	2	ACXG07-07740	
	15	FOAMED POLYSTYRENE	3	ACXG07-07750	
	16	U-SHAPED PLATE	3	ACXD62-02690	
	17	NON - COMBUSTIBLE FOAM	1	ACXG05-00540	
	18	CAP	1	H521306	
	19	CAP	1	ACXH52-04200	
	20	SCREW	3	H551029J	
⚠	21	SENSOR - COMPLETE	1	ACXA50C18700	O
	22	PARTICULAR PIECE	2	D934023	
	23	NUT	4	H561116	
	24	STRAIGHT TUBE	2	T102044	
	25	PLATE SPRING	2	H711010	
⚠	26	COCK - COMPLETE	1	ACXB65C00111	O
	27	TUBE ASSY - COMPLETE	1	ACXT00C55330	
⚠	28	PACKING	6	ACXB81-00010	
	29	RETAINING RING (12.7)	4	H581007	
	30	TUBE ASSY	1	ACXT00-96890	
	31	BAND	9	4605011	
	32	BAND	3	4605009	
	33	L-SHAPED PLATE	1	ACXD60-04771	
	34	L-SHAPED PLATE	1	ACXD60-04781	
	35	SCREW	14	H551040J	
	36	CONNECTING BAR	1	ACXE26-02841	
	37	SCREW	64	H551217	
	38	U-SHAPED PLATE	1	ACXD62-02700	
	39	U-SHAPED PLATE	1	ACXD62-02712	
	40	L-SHAPED PLATE	1	ACXD60-04532	
	41	U-SHAPED PLATE	1	ACXD62-02390CZ	
	42	CONNECTING BAR	1	ACXE26-02852	
	43	U-SHAPED PLATE	2	ACXD62-02720	

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-ADC0309K0E5A	REMARK
	44	FOAMED POLYSTYRENE	1	ACXG07-07000	
	45	FOAMED POLYSTYRENE	1	G071738	
⚠	46	HOT WATER COIL - COMPLETE	1	ACXB90C01700	
	47	CONVEX PLATE	1	ACXD65-01720	
	48	POLY - E. FOAM	4	ACXE5A18-28CZ	
	49	POLY - E. FOAM	1	E20A40-40CZ	
⚠	50	RECEIVER	1	ACXB14-00610	O
	51	EPT SEAL	4	D8A40-120	
	52	PARTICULAR PLATE	1	ACXD90-26060CZ	
	53	TUBE ASSY - COMPLETE INHOUSE	1	ACXT00C55270	
	54	RETAINING RING (25.4)	14	ACXH58-00370	
	55	FILTER COMPLETE	1	ACXB51C00110	O
⚠	56	VALVE BODY (AIR PURGE)	1	ACXB62-00130	O
	57	QAC - COMPLETE INHOUSE	1	ACXT00C50151	
	58	PACKING	1	B811179	
	59	RETAINING RING (14-23)	3	H581038	
⚠	60	PUMP	1	ACXB53-00850	O
	61	V-COIL	1	A431122	O
	62	TUBE ASSY - COMPLETE INHOUSE	1	ACXT00C45990	
	63	TUBE ASSY - COMPLETE INHOUSE	1	ACXT00C46002	
⚠	64	FLOW SENSOR	1	ACXB62-00932	O
	65	LEAD WIRE - COMPLETE (VB04-FLWS)	1	ACXA60C82910	O
	66	TUBE ASSY - COMPLETE INHOUSE	1	ACXT00C54730	
	67	Z-SHAPED PLATE	1	ACXD63-01650	
	68	SCREW	2	H55293J	
	69	TUBE ASSY - COMPLETE INHOUSE	1	ACXT00C37270	
⚠	70	PRESSURE SENSOR	1	ACXB62-01110	O
	71	WIRE SPRING	1	ACXH72-00341	
	72	LEAD WIRE - COMPLETE (PRSENS-CN)	1	ACXA60C93661	O
⚠	73	VALVE BODY [3-WAY VALVE]	1	ACXB62-00092	O
	74	TUBE ASSY - COMPLETE INHOUSE	1	ACXT00C55280	
	75	TUBE ASSY - COMPLETE INHOUSE	1	ACXT00C46010	
⚠	76	PACKING	13	ACXB81-06910	
	77	TUBE ASSY	1	ACXT00-96420	
⚠	80	VALVE BODY [RELIEF VALVE]	1	ACXB62-00740	O
	81	TUBE ASSY	1	ACXT00-77260	
	82	UNION NUT	1	T251064	
	83	CAP	1	H52275	
	84	UNION NUT	1	T251063	
	85	CAP	1	H52061	
	86	PLATE SPRING	4	H711019	

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-ADC0309K0E5A	REMARK
⚠	87	SENSOR - COMPLETE	1	ACXA50C16380	O
⚠	88	SENSOR - COMPLETE	1	ACXA50C16390	O
	89	CONNECTING BAR	1	ACXE26-02860	
	90	CONNECTING BAR	1	ACXE26-02870	
	91	EPT SEAL	1	D3A10-195	
	92	CONNECTING BAR	1	ACXE26-02880	
	93	PARTICULAR PIECE	1	ACXD93-01860	
	94	PARTICULAR PIECE	1	ACXD93-01870	
	95	PARTICULAR PIECE	2	ACXD93-25030	
	96	CABINET SIDE PLATE L	1	ACXE04-13290ACZ	
	97	POLY - E. FOAM	1	ACXE3A210-370	
	98	CABINET SIDE PLATE R	1	ACXE04-13300ACZ	
	99	CABINET REAR PLATE	1	ACXE02-03400	
	100	PACKING	9	ACXB81-00030	
	101	PACKING	7	ACXB81-00040	
	102	SCREW	2	ACXH55-07220	
	103	HANDLE	2	ACXE16-00300	
	104	LEAD WIRE - COMPLETE (CON-PUMP)	1	ACXA60C73630	O
⚠	105	ELECTRONIC CONTROLLER (MAIN)	1	ACXA74C23570	
	106	MACHINE SCREW WASHER ASSY	1	XYN4DC8FJ	
	107	HINGE	2	H611006	
	108	WASHER	3	H571044	
⚠	109	TERMINAL BOARD ASSY	1	A28K1217	O
	110	SPACER	9	H54295	
	111	MACHINE SCREW WASHER ASSY	3	XYN5DC10FJ	
	112	SELF TAPPING SCREW	6	XTT4+25DFJ	
	113	HOLDER - P.S. CORD	3	H31042	
	114	HOLDER - P.S. CORD	3	H31103	
	115	POLY - E. FOAM (FLAME PROOF)	1	ACXEN15A10-50CZ	
	116	SELF TAPPING SCREW	4	XTT4+8FFJ	
	117	POLY - E. FOAM (FLAME PROOF)	1	ACXEN8A350-500	
	118	WASHER	1	H57094	
	119	SPACER	8	H54294	
	120	POLY - E. FOAM (FLAME PROOF)	1	ACXEN8A30-240CZ	
	121	A-PIECE	3	D77001	
⚠	122	PEAKING COILS	1	G0C103Z00003	O
	123	CONTROL BOARD ASSY	1	ACXH10-09861	
	124	L-PIECE	2	ACXD70-02300	
	125	SELF TAPPING SCREW	11	XTT4+12CFJ	
⚠	126	TERMINAL BOARD ASSY	2	A28K1238CZ	O
⚠	127	LEAD WIRE - COMPLETE (WIRE1)	1	ACXA61C10340	O

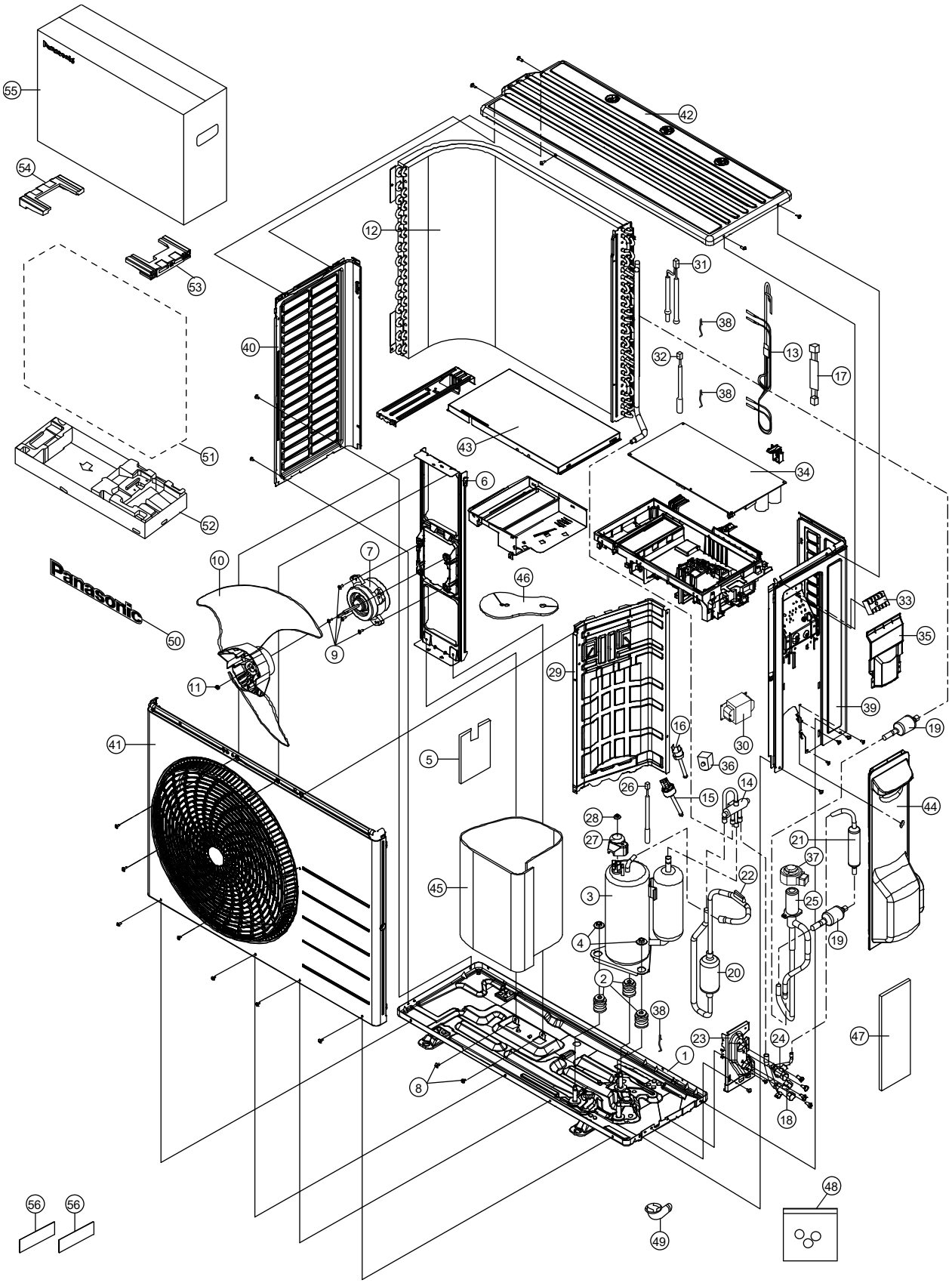
SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-ADC0309K0E5A	REMARK
	128	CONTROL BOARD COVER	1	ACXH13-09290	
	129	BOX SHAPED PLATE	1	ACXD66-03961	
	130	SELF TAPPING SCREW	3	XTB4+8CFJ	
	131	HOLDER - P.S. CORD	1	ACXH31-00210	
⚠	132	WIRING DIAGRAM	1	ACXF29-01250	
⚠	133	CAUTION LABEL	1	ACXF75-15040	
	134	CABINET TOP PLATE	1	ACXE03-04870	
⚠	135	REMOTE CONTROL SWITCH - COMPLETE	1	ACXA75C28920	O
	136	LEAD WIRE - COMPLETE (REMOTE)	1	ACXA60C99510	O
	137	BOX SHAPED PLATE (DECO)	1	ACXD66-03970	
	138	CABINET FRONT PLATE	1	ACXE06-05161ACZ	
	139	RUBBER (DECO)	1	ACXB81-07400	
	140	DECORATION BASE ASSY (PANASONIC)	1	ACXE35K03820	
	141	DECORATION BASE ASSY (AQUAREA)	1	ACXE35K03830	
	142	QR CODE	1	ACXF71-03400	
	143	ACCESSORY - ADJUSTABLE FEET	1	H82C2112	
	144	ACCESSORY - DRAIN ELBOW	1	G87C900	
⚠	146	OPERATING INSTRUCTION - COMPLETE	1	ACXF55C32390	
⚠	147	INSTALLATION INSTRUCTION COMPL	1	ACXF60C21510	
⚠	148	MODEL LABEL	1	ACXF87-54310	
⚠	149	NAME PLATE	1	ACXF09-10850	
	150	SHOCK ABSORBER	4	G713134	
⚠	151	C.C. CASE	1	ACXG50-61793	
⚠	152	BAG (UNIT)	1	ACXG86-04783	
	153	BASE BOARD - COMPLETE	1	ACXG62C02791	
	154	SHOCK ABSORBER (BOTTOM)	1	ACXG70-15201	
	155	CORRUGATED CARDBOARD (BOTTOM)	1	ACXG57-09491	
	156	TOP BOARD COMPLETE	1	ACXG60C00301	
	157	CORRUGATED CARDBOARD (TOP)	1	ACXG57-12321	
	158	SHOCK ABSORBER (TOP)	2	ACXG70-15211	

Note:

- All parts are supplied from PHVACCZ, Czech (Vendor Code: 00029407).
- “O” marked parts are recommended to be kept in stock.

22.2 Outdoor Unit

22.2.1 WH-UDZ03KE5



Note:
The above exploded view is for the purpose of parts disassembly and replacement.
The non-numbered parts are not kept as standard service parts.

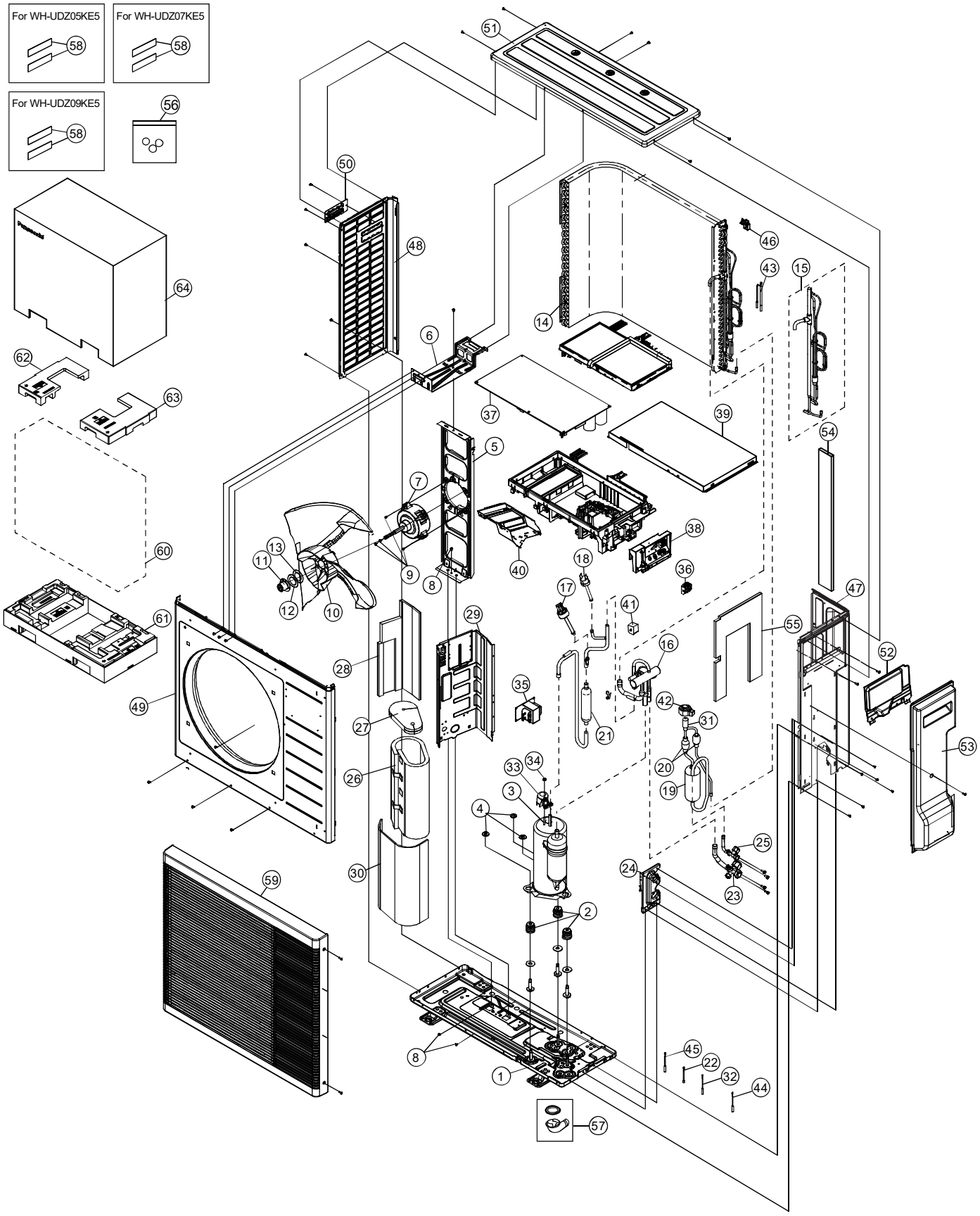
SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-UDZ03KE5	REMARK
	1	CHASSY ASS'Y	1	CWD52K1317	
	2	ANTI - VIBRATION BUSHING	3	CWH50077	
⚠	3	COMPRESSOR	1	9RD138ZAB21	O
	4	NUT - COMPRESSOR MOUNT	3	CWH561096	
	5	SOUND PROOF MATERIAL	1	CWG302762	
	6	BRACKET FAN MOTOR	1	CWD541167	
⚠	7	FAN MOTOR, DC 40W 3PH	1	L6CAYYYL0064	O
	8	SCREW - BRACKET FAN MOTOR	2	ACXH55-07140	
	9	SCREW - FAN MOTOR MOUNT	4	CWH55252J	
	10	PROPELLER FAN ASSY	1	CWH03K1066	
	11	NUT - PROPELLER FAN	1	CWH56053J	
	12	CONDENSER	1	ACXB32C26820	
	13	MANIFOLD TUBE ASS'Y (CAP TUBE)	1	CWT07K1831	
	14	4-WAYS VALVE	1	CWB001063	O
⚠	15	HIGH PRESSURE SENSOR CN-HPS	1	CWA501463	O
	16	PRESSURE SWITCH	1	ACXA10-00640	O
	17	LEAD WIRE - COMPLETE	1	CWA68C1224	O
	18	3-WAYS VALVE	1	ACXB01-04050	O
	19	STRAINER	2	CWB11094	
	20	DISCHARGE MUFFLER (1)	1	CWB121010	
	21	DISCHARGE MUFFLER (2)	1	CWB121063	
	22	HOLDER - SENSOR	1	CWH32075	
	23	HOLDER - COUPLING	1	CWH351233	
	24	2-WAYS VALVE	1	ACXB02-02680	O
	25	EXPANSION VALVE	1	CWB051029	O
⚠	26	SENSOR - CO. (DISCHARGE TEMP CN-DIS)	1	ACXA50C15550	O
	27	TERMINAL COVER	1	CWH171039A	
	28	NUT - TERMINAL COVER	1	CWH7080300J	
	29	SOUND - PROOF BOARD	1	CWH151345	
⚠	30	REACTOR	1	G0C392J00027	O
⚠	31	SENSOR - CO. (OUTDDOR AIR & PIPING TEMP CN-TH1)	1	CWA50C2893	O
⚠	32	SENSOR - CO. (EVA EXIT TEMP CN-TH3)	1	CWA50C3374	O
⚠	33	TERMINAL BOARD ASS'Y (1,2,3)	1	CWA28K1036J	O
⚠	34	ELECTRONIC CONTROLLER - MAIN	1	ACXA73C92000R	O
	35	PLATE - C.B. COVER	1	CWH131470	
⚠	36	V-COIL COMP. (4 WAY VALVE)	1	CWA43C2431	O
⚠	37	V-COIL CO. (EXPANSION VALVE)	1	ACXA43C06020	O
	38	HOLDER - SENSOR	3	CWH32143	
	39	CABINET SIDE PLATE CO.	1	ACXE04C08670	
	40	CABINET SIDE PLATE (L)	1	ACXE04-13070G	

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-UDZ03KE5	REMARK
	41	CABINET FRONT PLATE - CO.	1	ACXE06C04700	
	42	CABINET TOP PLATE	1	ACXE03-04830G	
	43	CONTROL BOARD COVER	1	CWH131473	
	44	CONTROL BOARD COVER - COMPLETE	1	ACXH13C06420	
	45	SOUND PROOF MATERIAL	1	ACXG30-08570	
	46	SOUND PROOF MATERIAL	1	CWG302630	
	47	SOUND PROOF MATERIAL	1	CWG302788	
	48	ACCESSORY COMP. (CAP)	1	ACXH82C03830	
	49	ACCESSORY COMP. (DRAIN ELBOW)	1	ACXH82C21740	
	50	PANASONIC BADGE	1	ACXE37-19200	
	51	BAG	1	CWG861078	
	52	BASE BOARD - COMPLETE	1	CWG62C1162	
	53	SHOCK ABSORBER	1	ACXG70-14440A	
	54	SHOCK ABSORBER	1	ACXG70-14450A	
	55	C.C. CASE	1	ACXG50-48673	
	56	MODEL LABEL	2	ACXF85-99600	

Note:

- All parts are supplied from PAPAMY, Malaysia (Vendor Code: 00029488).
- “O” marked parts are recommended to be kept in stock.

22.2.2 WH-UDZ05KE5 WH-UDZ07KE5 WH-UDZ09KE5



Note:
The above exploded view is for the purpose of parts disassembly and replacement.
The non-numbered parts are not kept as standard service parts.

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-UDZ05KE5	WH-UDZ07KE5	REMARK
	1	BASE PAN ASSY	1	ACXD52K05210	←	
	2	ANTI - VIBRATION BUSHING	3	CWH50077	←	
⚠	3	COMPRESSOR	1	9RD220XBC21	←	O
	4	NUT - COMPRESSOR MOUNT	3	CWH561096	←	
	5	BRACKET FAN MOTOR	1	ACXD54-04490	←	
	6	PARTICULAR PLATE	1	CWD911802	←	
⚠	7	FAN MOTOR, DC 60W 3PH	1	L6CBYYYL0422	←	O
	8	SCREW - BRACKET FAN MOTOR	3	ACXH55-07140	←	
	9	SCREW - FAN MOTOR MOUNT	4	CWH551323	←	
	10	PROPELLER FAN ASSY	1	CWH00K1006	←	
	11	NUT - PROPELLER FAN	1	CWH561092	←	
	12	WASHER	1	CWH571067	←	
	13	WASHER	1	CWH571068	←	
	14	CONDENSER	1	ACXB32C26350	←	
	15	MANIFOLD TUBE ASS'Y (CAP TUBE)	1	ACXT00C27650	←	
	16	4-WAYS VALVE	1	ACXB00-00140	←	O
⚠	17	HIGH PRESSURE SENSOR CN-HPS	1	CWA501463	←	O
	18	PRESSURE SWITCH	1	ACXA10-00640	←	O
	19	RECEIVER	1	CWB14017	←	
	20	STRAINER	2	CWB111063	←	
	21	DISCHARGE MUFFLER	1	CWB121013	←	
⚠	22	LEAD WIRE - CO. (CN-HPS)	1	ACXA60C94010	←	O
	23	3-WAYS VALVE	1	CWB011363	←	O
	24	HOLDER - COUPLING	1	CWH351228	←	
	25	2-WAYS VALVE	1	CWB021464	←	O
	26	SOUND PROOF MATERIAL	1	ACXG30-13230	←	
	27	SOUND PROOF MATERIAL	1	ACXG30-13240	←	
	28	SOUND PROOF MATERIAL	1	ACXG30-13310	←	
	29	SOUND - PROOF BOARD	1	ACXH15-03900	←	
	30	SOUND PROOF MATERIAL	1	ACXG30-13250	←	
	31	EXPANSION VALVE	1	CWB051029	←	O
⚠	32	SENSOR - CO. COMPRESSOR TEMP (CN-TANK)	1	ACXA50C17400	←	O
	33	TERMINAL COVER	1	CWH171039A	←	
	34	NUT - TERMINAL COVER	1	CWH7080300J	←	
⚠	35	REACTOR	1	G0C392J00027	←	O
⚠	36	TERMINAL BOARD ASS'Y (1,2,3)	1	CWA28K1076J	←	O
⚠	37	ELECTRONIC CONTROLLER - MAIN	1	ACXA73C91120R	ACXA73C91130R	O
	38	PLATE - C.B. COVER	1	CWH102361	←	
	39	CONTROL BOARD COVER	1	ACXH13-09170	←	
	40	CONTROL BOARD COVER - COMPLETE	1	ACXH13K00830	←	
⚠	41	V-COIL COMP. (4 WAY VALVE)	1	ACXA43C00250	←	O
⚠	42	V-COIL CO. (EXPANSION VALVE)	1	A43C2342	←	O
⚠	43	SENSOR - CO. OUTDOOR AIR & PIPE TEMP (CN-TH1)	1	CWA50C2822	←	O

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-UDZ05KE5	WH-UDZ07KE5	REMARK
⚠	44	SENSOR - CO. DISCHARGE (CN-DIS)	1	CWA50C2722	←	O
⚠	45	SENSOR - CO. EVA EXIT (CN-TH3)	1	ACXA50C14960	←	O
	46	HOLDER - SENSOR	1	CWH321096G	←	
	47	CABINET SIDE PLATE CO.	1	ACXE04-13310G	←	
	48	CABINET SIDE PLATE (L)	1	CWE041585G	←	
	49	CABINET FRONT PLATE - CO.	1	ACXE06C04610	←	
	50	HANDLE	1	CWE161010G	←	
	51	CABINET TOP PLATE	1	CWE031083G	←	
	52	CONTROL BOARD COVER	1	CWH131332	←	
	53	CONTROL BOARD COVER - COMPLETE	1	ACXH13C06350	←	
	54	SOUND PROOF MATERIAL	1	ACXG30-13260	←	
	55	SOUND PROOF MATERIAL	1	ACXG30-13270	←	
	56	ACCESSORY COMP. (CAP)	1	ACXH82C09831	←	
	57	ACCESSORY COMP. (DRAIN ELBOW)	1	ACXH82C21740	←	
	58	MODEL LABEL	2	ACXF85-99610	ACXF85-99620	
	59	DISCHARGE GRILLE - COMPLETE	1	ACXE20C07970	←	
	60	BAG	1	CWG861154	←	
	61	BASE BOARD - COMPLETE	1	ACXG62C02810	←	
	62	SHOCK ABSORBER	1	ACXG70-15180	←	
	63	SHOCK ABSORBER	1	ACXG70-15190	←	
	64	C.C. CASE	1	ACXG50-61910	←	

Note:

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SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-UDZ09KE5	REMARK
	1	BASE PAN ASSY	1	ACXD52K05210	
	2	ANTI - VIBRATION BUSHING	3	CWH50077	
⚠	3	COMPRESSOR	1	9RD220XBC21	O
	4	NUT - COMPRESSOR MOUNT	3	CWH561096	
	5	BRACKET FAN MOTOR	1	ACXD54-04490	
	6	PARTICULAR PLATE	1	CWD911802	
⚠	7	FAN MOTOR, DC 60W 3PH	1	L6CBYYYL0422	O
	8	SCREW - BRACKET FAN MOTOR	3	ACXH55-07140	
	9	SCREW - FAN MOTOR MOUNT	4	CWH551323	
	10	PROPELLER FAN ASSY	1	CWH00K1006	
	11	NUT - PROPELLER FAN	1	CWH561092	
	12	WASHER	1	CWH571067	
	13	WASHER	1	CWH571068	
	14	CONDENSER	1	ACXB32C26350	
	15	MANIFOLD TUBE ASS'Y (CAP TUBE)	1	ACXT00C27650	
	16	4-WAYS VALVE	1	ACXB00-00140	O
⚠	17	HIGH PRESSURE SENSOR CN-HPS	1	CWA501463	O
	18	PRESSURE SWITCH	1	ACXA10-00640	O
	19	RECEIVER	1	CWB14017	
	20	STRAINER	2	CWB111063	
	21	DISCHARGE MUFFLER	1	CWB121013	
⚠	22	LEAD WIRE - CO. (CN-HPS)	1	ACXA60C94010	O
	23	3-WAYS VALVE	1	CWB011363	O
	24	HOLDER - COUPLING	1	CWH351228	
	25	2-WAYS VALVE	1	CWB021464	O
	26	SOUND PROOF MATERIAL	1	ACXG30-13230	
	27	SOUND PROOF MATERIAL	1	ACXG30-13240	
	28	SOUND PROOF MATERIAL	1	ACXG30-13310	
	29	SOUND - PROOF BOARD	1	ACXH15-03900	
	30	SOUND PROOF MATERIAL	1	ACXG30-13250	
	31	EXPANSION VALVE	1	CWB051029	O
⚠	32	SENSOR - CO. COMPRESSOR TEMP (CN-TANK)	1	ACXA50C17400	O
	33	TERMINAL COVER	1	CWH171039A	
	34	NUT - TERMINAL COVER	1	CWH7080300J	
⚠	35	REACTOR	1	G0C392J00027	O
⚠	36	TERMINAL BOARD ASS'Y (1,2,3)	1	CWA28K1076J	O
⚠	37	ELECTRONIC CONTROLLER - MAIN	1	ACXA73C91140R	O
	38	PLATE - C.B. COVER	1	CWH102361	
	39	CONTROL BOARD COVER	1	ACXH13-09170	
	40	CONTROL BOARD COVER - COMPLETE	1	ACXH13K00830	
⚠	41	V-COIL COMP. (4 WAY VALVE)	1	ACXA43C00250	O
⚠	42	V-COIL CO. (EXPANSION VALVE)	1	A43C2342	O
⚠	43	SENSOR - CO. OUTDOOR AIR & PIPE TEMP (CN-TH1)	1	CWA50C2822	O

SAFETY	REF. NO.	DESCRIPTION & NAME	QTY.	WH-UDZ09KE5	REMARK
⚠	44	SENSOR - CO. DISCHARGE (CN-DIS)	1	CWA50C2722	O
⚠	45	SENSOR - CO. EVA EXIT (CN-TH3)	1	ACXA50C14960	O
	46	HOLDER - SENSOR	1	CWH321096G	
	47	CABINET SIDE PLATE CO.	1	ACXE04-13310G	
	48	CABINET SIDE PLATE (L)	1	CWE041585G	
	49	CABINET FRONT PLATE - CO.	1	ACXE06C04610	
	50	HANDLE	1	CWE161010G	
	51	CABINET TOP PLATE	1	CWE031083G	
	52	CONTROL BOARD COVER	1	CWH131332	
	53	CONTROL BOARD COVER - COMPLETE	1	ACXH13C06350	
	54	SOUND PROOF MATERIAL	1	ACXG30-13260	
	55	SOUND PROOF MATERIAL	1	ACXG30-13270	
	56	ACCESSORY COMP. (CAP)	1	ACXH82C09831	
	57	ACCESSORY COMP. (DRAIN ELBOW)	1	ACXH82C21740	
	58	MODEL LABEL	2	ACXF85-99630	
	59	DISCHARGE GRILLE - COMPLETE	1	ACXE20C07970	
	60	BAG	1	CWG861154	
	61	BASE BOARD - COMPLETE	1	ACXG62C02810	
	62	SHOCK ABSORBER	1	ACXG70-15180	
	63	SHOCK ABSORBER	1	ACXG70-15190	
	64	C.C. CASE	1	ACXG50-61910	

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